

GE Healthcare

Signa EXCITE 3T94 (E2) to HDx System Upgrade

Technical Publication



5166007
Rev 8

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

© 2006 – 2010, General Electric Company
GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3200 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	Apr 12, 2006	Initial Release.
2	Jul 18, 2006	Section 2: Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251). Added blocks for restoreINFO, ICW, and update of eLicense (MRIhc15873, MRIhc15892, MRIhc16209). Section 8: Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870).
3	Oct 26, 2006	Section 6: updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607).
4	Feb 22, 2007	Section 2: Reversed flowchart procedures <i>Software Option Install w/eLicense</i> and <i>Install and Calibration Wizard</i> . Section 4: Added note for instructions on storing Service Cable Kit (CQA 13094026). Section 6: Changed Notch Filter part number (CQA 13090674). Removed PAC2B part number. Section 8: Added information on cable disconnection from existing GOC. For installing Run 1323, added notes for routing with Run 1322 and that CAT6 cable must replace existing CAT5 cable. Added note that Humidity iButton can be installed in either direction. Section 9: Updated MGD chassis install information. Revised installation location of ICN brackets (CQA 13078452). Updated board installation instructions for upgrading to 16 channels. Added instructions for storing Service Cable Kit.
5	May 24, 2007	Section 8: Added iButton installation and Environmental Sensors Block Diagram to Section 7–6, Temperature and Humidity Sensor Kit Installation. Section 9: Updated APS Board part number from 2294300–14 to 2294300–16 (ECO 2044030). Adding existing cables connected to Term Server (CQA13112624).
6	Jun 17, 2008	Section 4: Updated cable map to show OW component and cable upgrades. Updated cable maps to show new Temperature and humidity Sensor installation and cables. Section 8: Revised entire section for replacing the PC computer in the GOC, and other components, if required. Also revised temperature/humidity sensor installation to reflect the new design.
7	Sep 18, 2008	Section 1: Added HDxt (Release 15.x) information to Introduction. Section 9: Revised section to incorporate the new SUN ICN modules for upgrades.
8	Jul 27, 2010	Section 2: Updated eLicense model type from Signa to MR HDx (per Direction 2397845). Section 4: Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Section 8: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC (ECO 2096124). Section 9: Removed second set of ICN brackets in cabinet because new Sun 4170 ICN replaces two ICN's installed in previous upgrades (ECO 2101939).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	8	2–2	2	Section 5		6–24	4	Section 9	
Damage in Trans.	–	2–3	1	PENETRATION PANEL		6–25 to 6–26	1	SYSTEMCABINET	
Direction 2128126	1*	2–4	8	2164904–5	–			2164904–9	–
A	8			5–1 to 5–2	1	Section 7		9–1 to 9–6	7
i	1					CRADLE RELEASE		9–7 to 9–9	8
Section 1		Section 3		Section 6		UPGRADE		9–10 to 9–12	7
GETTING STARTED		POWER OFF		MAGNET ROOM		2164904–7	–	Section 10	
2164904–1	–	2164904–3	–	2164904–6	–	7–1 to 7–3	1	APPENDIX	
1–1 to 1–6	7	3–1 to 3–1	1	6–1 to 6–6	1	Section 8		2164904–10	–
Section 2				6–7	3	OPERATOR		10–1 to 10–2	1
UPGRADE		Section 4		6–8 to 6–14	1	WORKSPACE			
INSTALLATION		SYSTEM CABLES		6–15	4	2164904–8	–		
PROCESS		2164904–4	–	6–16 to 6–23	1	8–1 to 8–23	8		
2164904–2	–	4–1 to 4–4	4						
2–1	1	4–5 to 4–12	8						

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
SECTION 1 – GETTING STARTED	1–1
SECTION 2 – UPGRADE INSTALLATION PROCESS	2–1
SECTION 3 – POWER OFF	3–1
SECTION 4 – SYSTEM CABLES	4–1
SECTION 5 – PENETRATION PANEL	5–1
SECTION 6 – MAGNET ENCLOSURE	6–1
SECTION 7 – CRADLE RELEASE UPGRADE	7–1
SECTION 8 – OPERATOR WORKSPACE	8–1
SECTION 9 – SYSTEM CABINET	9–1
SECTION 10 – COMPLETION	10–1

SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 3.0T with EXCITE Technology (3T/94) to Signa HDx (Release 14.x) or HDxt (Release 15.x). Refer to illustration 1-1 for pictorial overview of upgraded system.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 3T94 HDx system are listed in Direction 5159905, *Signa HDx 3.0T Upgrades Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 3.0T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD (Release 12.x) upgrade to GoldSeal which includes the following and other components not listed:

- GOC and Operator Workspace components
- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Bridge
- Exciter / Receiver Boards

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997

Fax: (414)–747–6855

Field ASPEN: 1–800–525–1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41

Fax: (+33) 1–60–86–12–31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

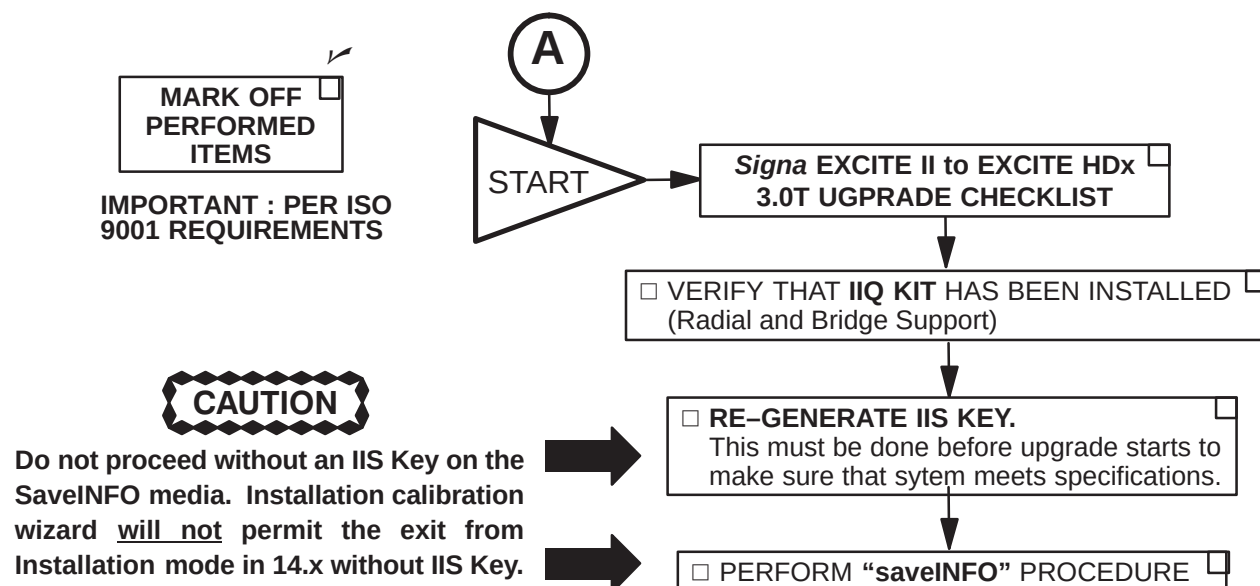
2-2 UPGRADE PROCEDURE

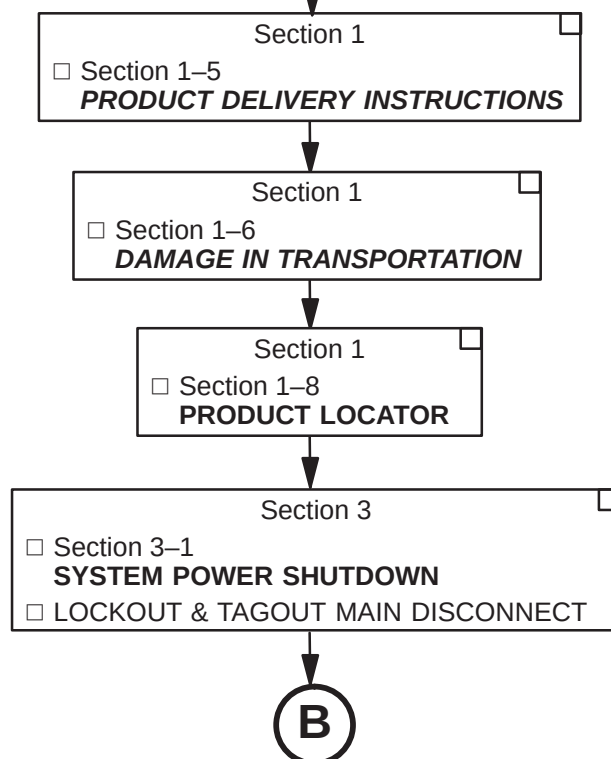
The following upgrade flow charts contain information for upgrading to Signa 3T94 HDx (Release 14.x).

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa 3T94 HDx equipment has been delivered together. Ensure that every part required is available before starting.

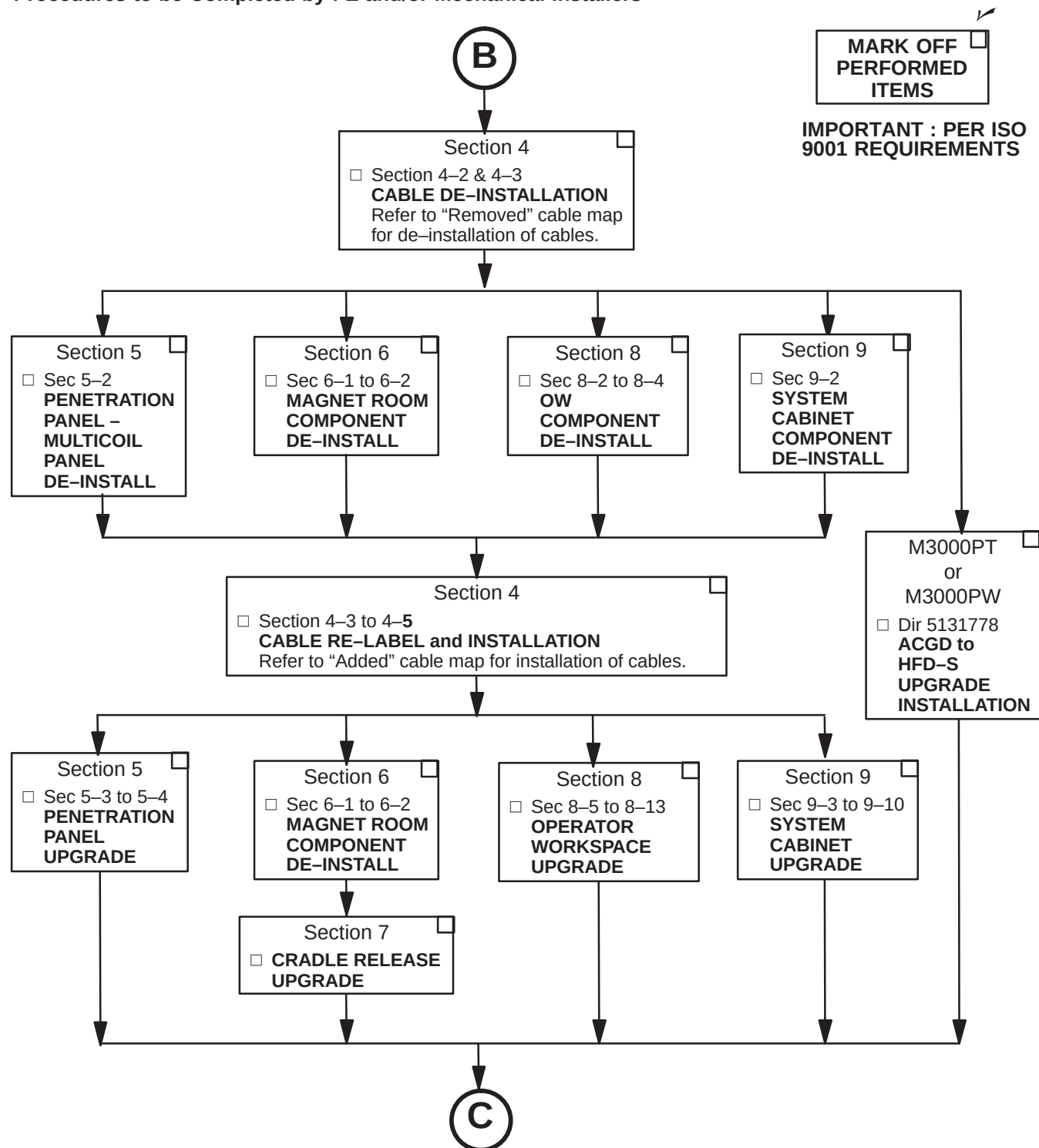
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

 Procedures to be Completed by Field Engineer


 Procedures to be Completed by FE and/or Mechanical Installers


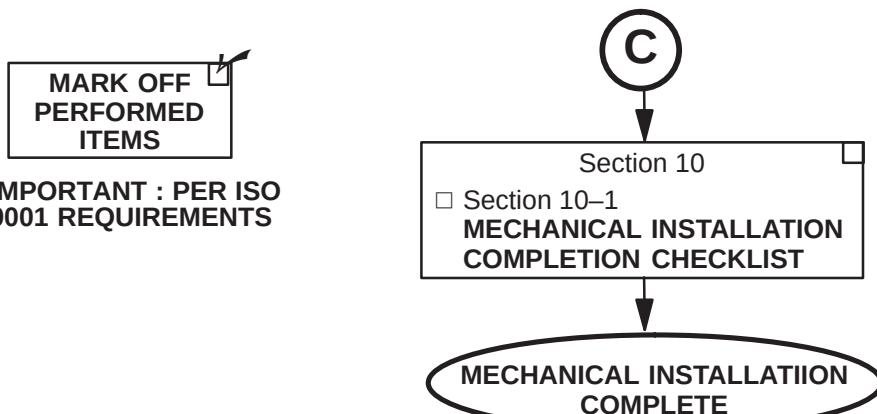
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

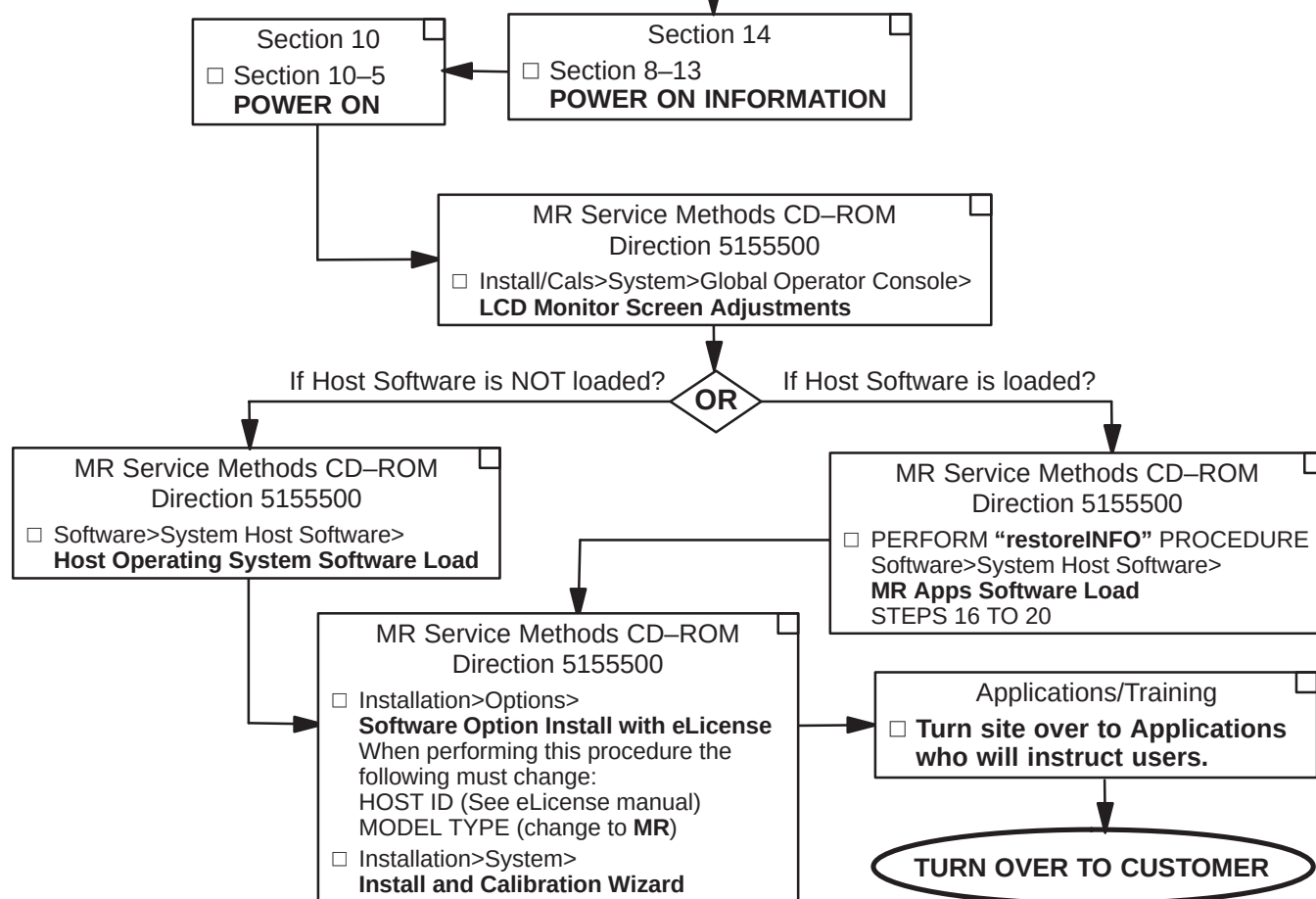


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring.

SECTION 4 – SYSTEM UPGRADE CABLES

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
4-1 INTRODUCTION	4-2
4-2 CABLES DE-INSTALLED FOR UPGRADING TO EXCITE HD	4-2
4-3 CABLE RE-LABELING	4-3
4-4 SORTING AND ROUTING NEW CABLES	4-3
4-5 CABLE MAPS	4-4

4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa 3.0T (3T/94) HDxt, Release 15.x.

4-2 CABLES DE-INSTALLED FOR UPGRADING TO EXCITE HD

Table 4-1 lists all cables to be removed from 3T/94 with EXCITE (11.x) system configurations upgrading to a Signa HDxt (15.x) system. Use table along with "Affected Cables" interconnect map for removal information.

TABLE 4-1
3T/94 with EXCITE CABLES REMOVED FOR UPGRADE TO 3T94 HDx

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
231	MR2-A11-J2	System Cabinet	PP1-J8	Pen Panel
258	MG3-A17-J1	Rear Pedestal	PP1-J4	Pen Panel
262	MG3-A3-A5-OUTPUT	BodyCoil Preamp	PP1-J8	Pen Panel
404	MG3-A17-J1	Rear Pedestal	MG2-A1- A5-J6	LPCA
488	PP1-J80	Pen Panel	MR2-A11-J8	System Cabinet
489	PP1-J81	Pen Panel	MR2-A11-J9	System Cabinet
490	PP1-J82	Pen Panel	MR2-A11-J10	System Cabinet
491	PP1-J80	Pen Panel	MG3-A36-J7	8 Channel Sw
492	PP1-J81	Pen Panel	MG3-A36-J8	8 Channel Sw
493	PP1-J82	Pen Panel	MG3-A36-J9	8 Channel Sw
499	PP1-A14-J84	Pen Panel	MR2-A11-J5	System Cabinet
728	MG3-A36-J11	8 Channel Sw	MG2-A16-OUT-J1	LPCA
729	MG3-A36-J12	8 Channel Sw	MG2-A16-OUT-J2	LPCA
730	MG3-A36-J13	8 Channel Sw	MG2-A16-OUT-J3	LPCA
731	MG3-A36-J14	8 Channel Sw	MG2-A16-OUT-J4	LPCA
732	MG3-A36-J15	8 Channel Sw	MG2-A16-OUT-J5	LPCA
733	MG3-A36-J16	8 Channel Sw	MG2-A16-OUT-J6	LPCA
734	MG3-A36-J17	8 Channel Sw	MG2-A16-OUT-J7	LPCA
735	MG3-A36-J18	8 Channel Sw	MG2-A16-OUT-J8	LPCA
749	MG2-A33-J10	SRI Module	MG2-A36	Temp Sensor
1011	MG2-A33-J5	SRI Module	MG2-A4-J4	Rt Oper Cntrl
1035	MG3-A36-J21	8 Channel Sw	MG2-A16-A3/A5-J4	LPCA
1042	PP1-A16-J77	Pen Panel	MG3-A36-J5	8 Channel Sw
1049	MG3-A36-J19	8 Channel Sw	MG2-A16-A8-J9	LPCA
1053	MR2-A11-J54	System Cabinet	PP1-J85	Pen Panel
1054	MG3-A36-J20	8 Channel Sw	MG2-A33-J15	SRI Module
1057	PP1-J85	PenPanel	MG3-A8-J1	UTNS Antenna
1062	OW-A20	OW SCIM	OW1-A21-J6	OW WIM
1063	OW-A15-VIDEO	OW Host	OW1-A6	OW Monitor

4-2 CABLES DE-INSTALLED FOR UPGRADING TO SIGNA EXCITE HD (Continued)

TABLE 4-1 (Continued)
3T/94 with EXCITE CABLES REMOVED FOR UPGRADE TO 3T EXCITE HD

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
1065	OW-A11-J2	OW Power	OW1-A6	OW Monitor
1068	OW1-A15-VIDEO2	OW Host	OW-A3	OW LCD
1069	OW-A11-J8	OW Power	OW-A3	OW LCD
1078	MG3-A36-J29,J30	8 Channel Sw	MG2-A33-J18,J19	SRI Module
1082	MR2-A11-J63	System Cabinet	OW1-A15-ETH	OW
1083	MR2-A11-J22	System Cabinet	OW1-A15-XMT1	OW
1153	MG3-A17-J2	Rear Pedestal	MG2-A16-A3 OUT	LPCA
1155	MG3-A36-J28	8 Channel Sw	MG2-A16-J2	LPCA
1168	MR2-A11-J40	System Cabinet	PP1-J6	PenPanel
1169	MR2-A11-J41	System Cabinet	PP1-J7	PenPanel
1170	MR2-A11-J42	System Cabinet	PP1-J58	PenPanel
1171	MR2-A11-J43	System Cabinet	PP1-J59	PenPanel
1172	MG3-A36-J1	8 Channel Sw	PP1-J6	PenPanel
1173	MG3-A36-J2	8 Channel Sw	PP1-J7	PenPanel
1174	MG3-A36-J3	8 Channel Sw	PP1-J58	PenPanel
1175	MG3-A36-J4	8 Channel Sw	PP1-J59	PenPanel
1176	MG3-A36-J6	8 Channel Sw	PP1-J84	PenPanel

4-3 CABLE RE-LABELING

Relabel existing cables as described below:

- The new GOC upgrades the WIM to a GOCAA. This requires that Run 1084 be relabeled as Run 1255. Attach blank label to each end of cable and mark with information below.
- Swap and relabel existing equipment room cable Run 1030 as instructed below.

ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1084 MR1-A11-J26 2354601	RUN #1255 MR2-A11-J26 5120763-3	RUN #1084 OW1-J21-J6 VIA RUN #1062 2354601	RUN #1255 OW1-A21-J18 5120763-3
RUN #1030 PP1-J78 2329624	RUN #1329 PP1-J78 5160765	RUN #1030 MR2-A11-J61 2329624	RUN #1329 MR2-DVR-J10 5160765

4-4 SORTING AND ROUTING NEW CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

4-4 SORTING AND ROUTING NEW CABLES (Continued)

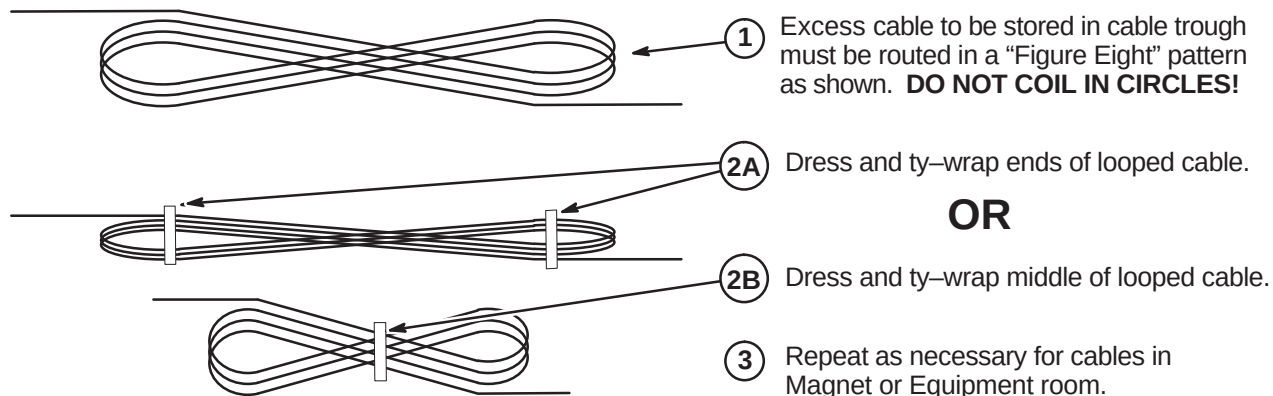
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

Note: The Service Cable Kit (5160683) is a collection of service cables delivered to site in a plastic bag. Refer to Section 9–10 for instructions on storage of this kit within System cabinet.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4–1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4–1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4-5 CABLE MAPS

A double set of “Removed” and “Added” cables maps is provided. One set for the equipment room and the other for the magnet room. Tape maps in a convenient location to check off affected “Removed” and “Added” cables.

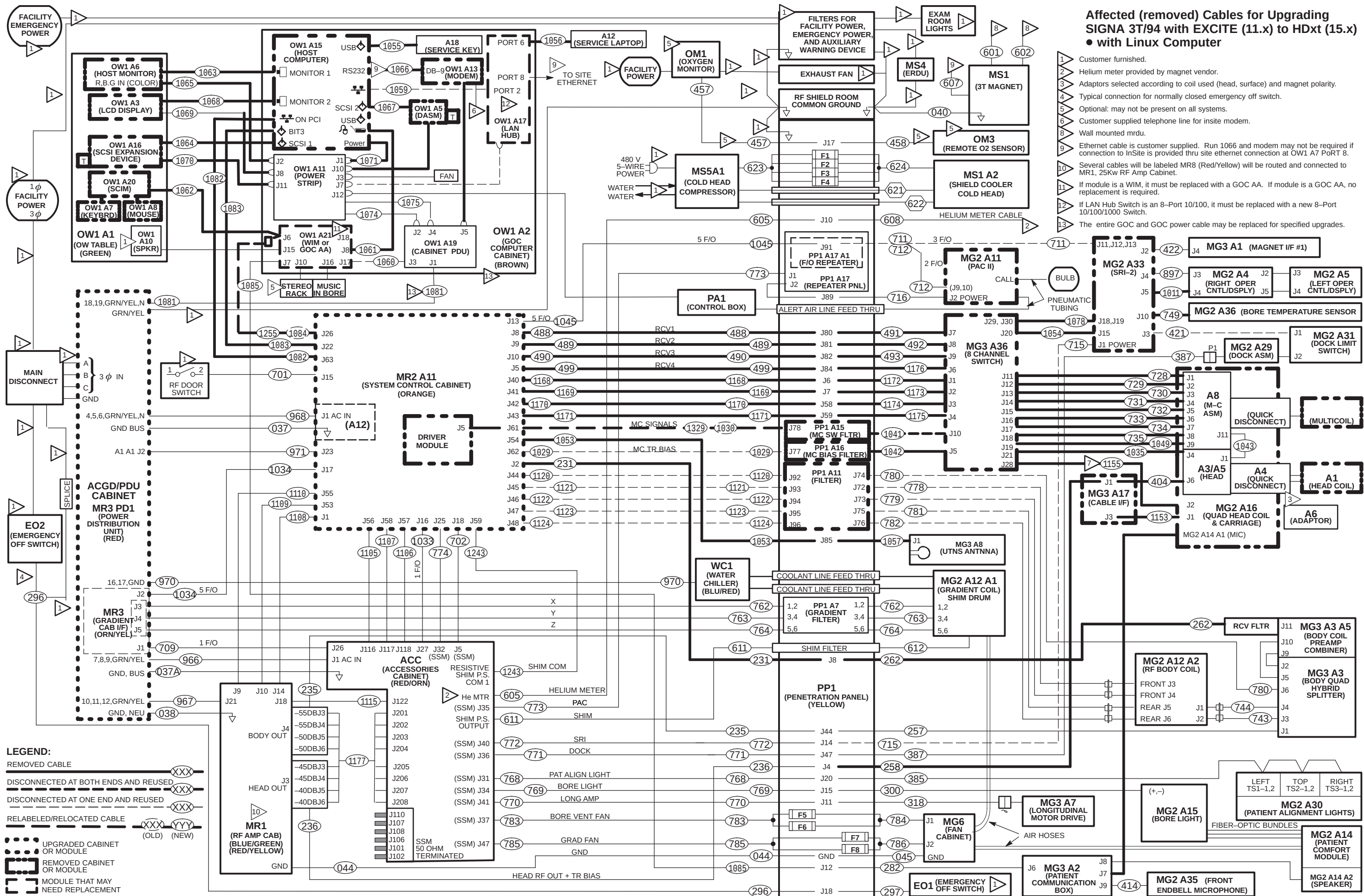
The Affected “Removed” Cable maps are located on the following pages.

- Page 4–5 Affected (removed) Cables for Upgrading to Signa 3.0T HDx (Equip. Room)
- Page 4–7 Affected (removed) Cables for Upgrading to Signa 3.0T HDx (Magnet Room)

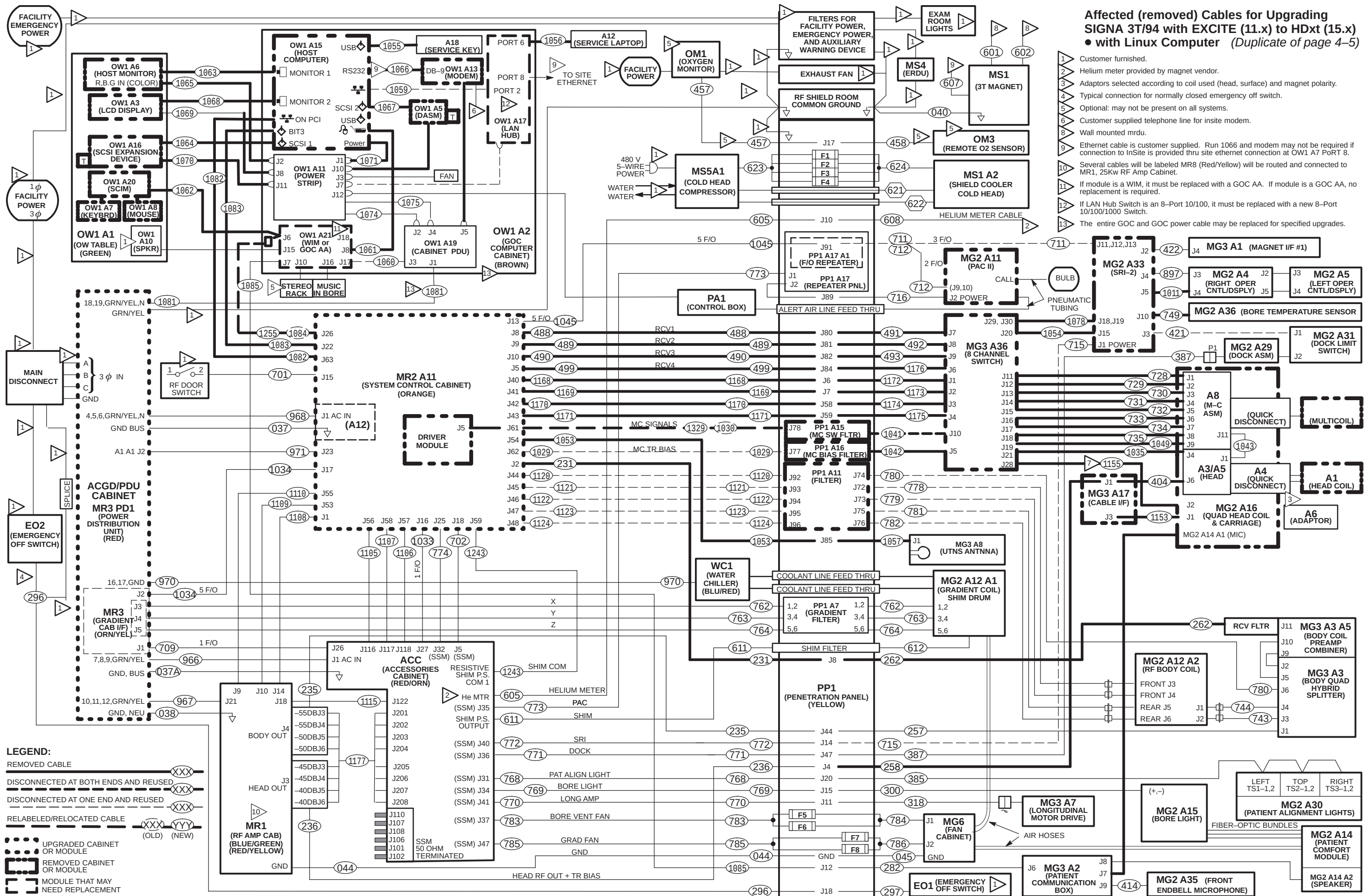
The Affected “Added” Cable maps are located on the following pages.

- Page 4–9 Affected (added) Cables for Upgrading to Signa 3.0T HDx (Equip. Room)
- Page 4–11 Affected (added) Cables for Upgrading to Signa 3.0T HDx (Magnet Room)

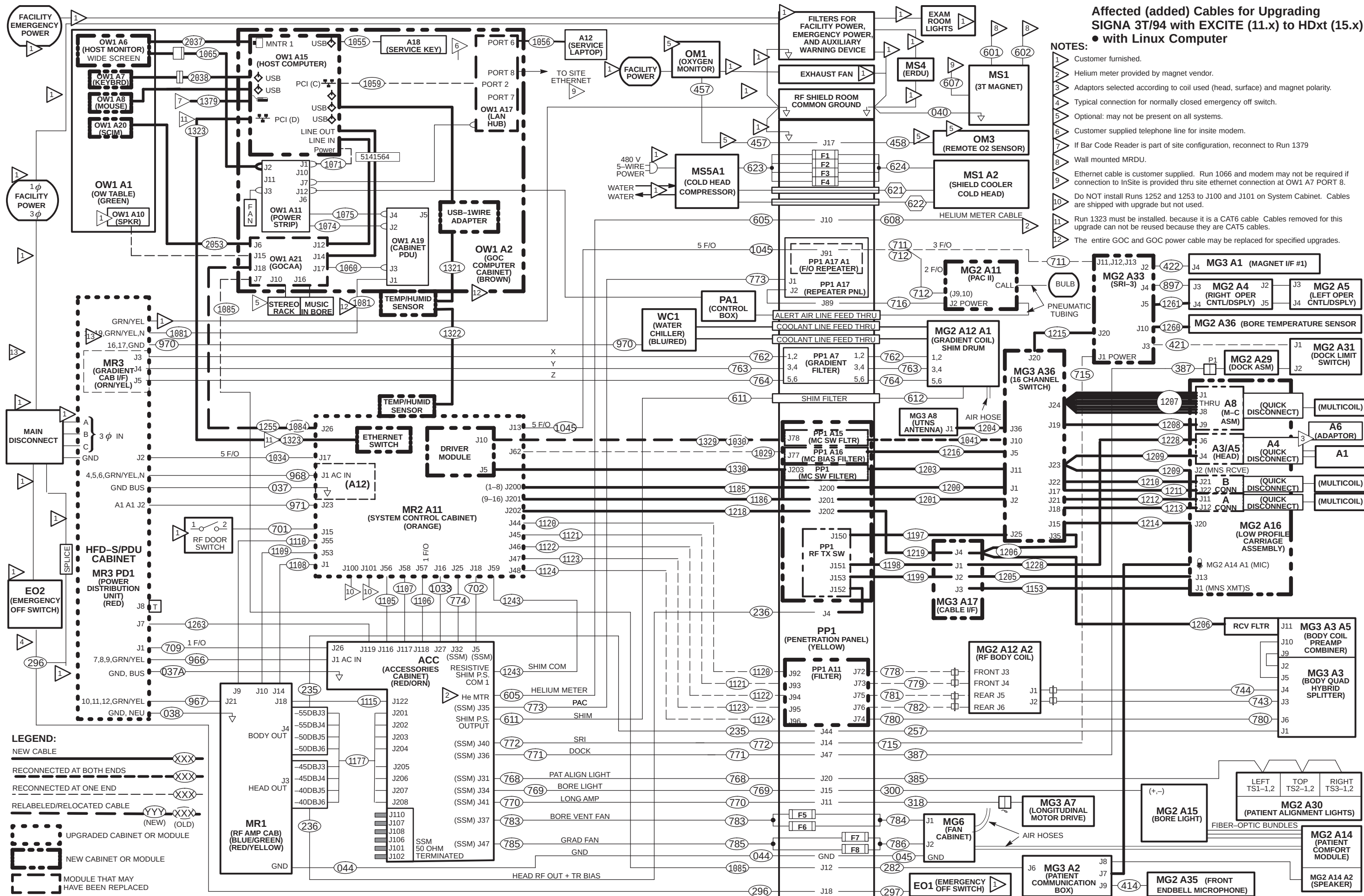
Upon completion of connecting cables, return fold-out map to the binder for future reference.



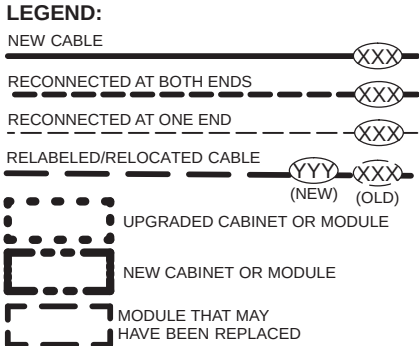
This page left blank intentionally



This page left blank intentionally



This page left blank intentionally



NOTES: *(Duplicate of page 4–9)*

- 1 Customer furnished.
- 2 Helium meter provided by magnet vendor.
- 3 Adaptors selected according to coil used (head, surface) and magnet polarity.
- 4 Typical connection for normally closed emergency off switch.
- 5 Optional: may not be present on all systems.
- 6 Customer supplied telephone line for insite modem.
- 7 If Bar Code Reader is part of site configuration, reconnect to Run 1379
- 8 Wall mounted MRDU.
- 9 Ethernet cable is customer supplied. Run 1066 and modem may not be required for connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8
- 10 Do NOT install Runs 1252 and 1253 to J100 and J101 on System Cabinet. Cables are shipped with upgrade but not used.
- 11 Run 1323 must be installed, because it is a CAT6 cable. Cables removed for this upgrade can not be reused because they are CAT5 cables.
- 12 The entire GOC and GOC power cable may be replaced for specified upgrades.

This page left blank intentionally

SECTION 5 – PENETRATION PANEL

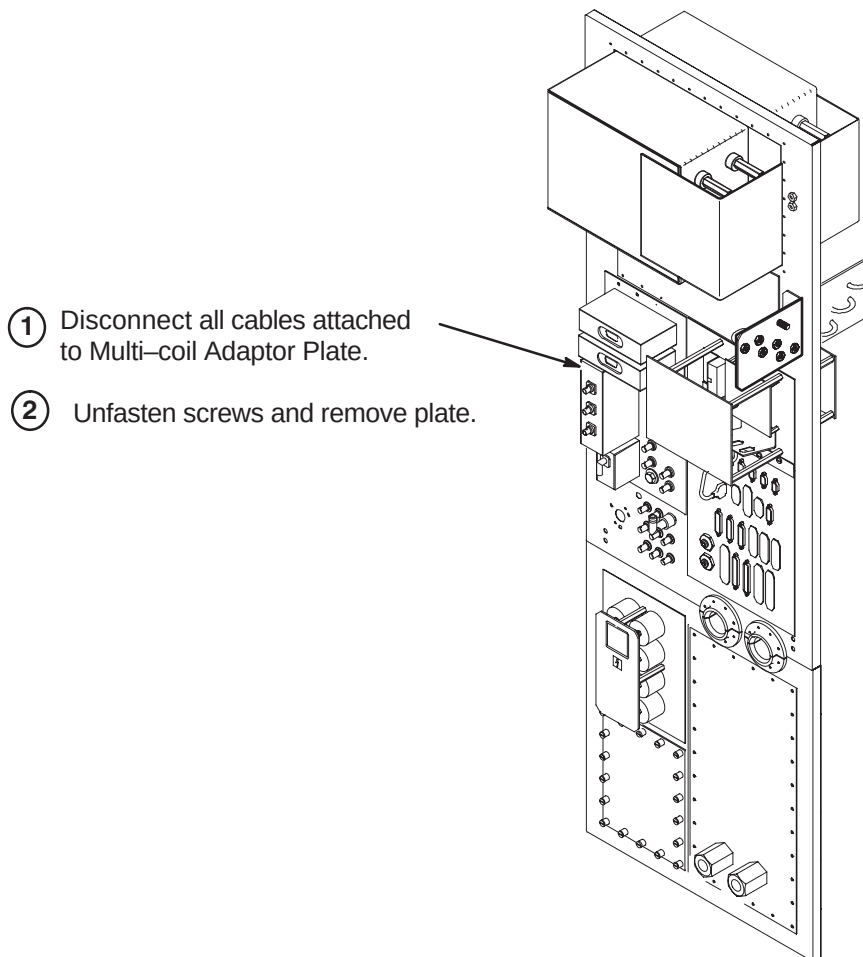
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

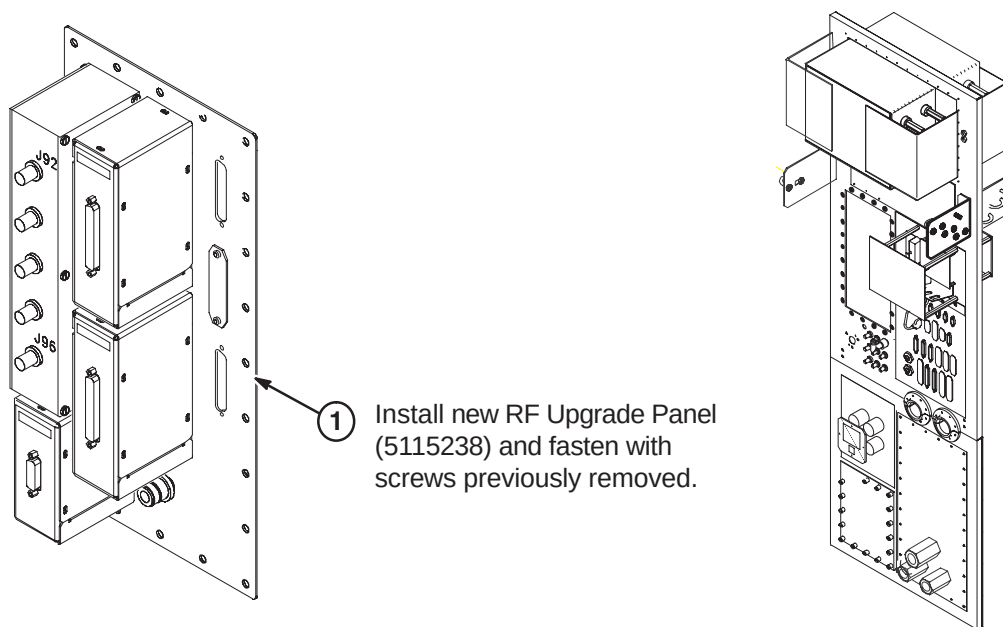
5-1 INTRODUCTION

The Multi-coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

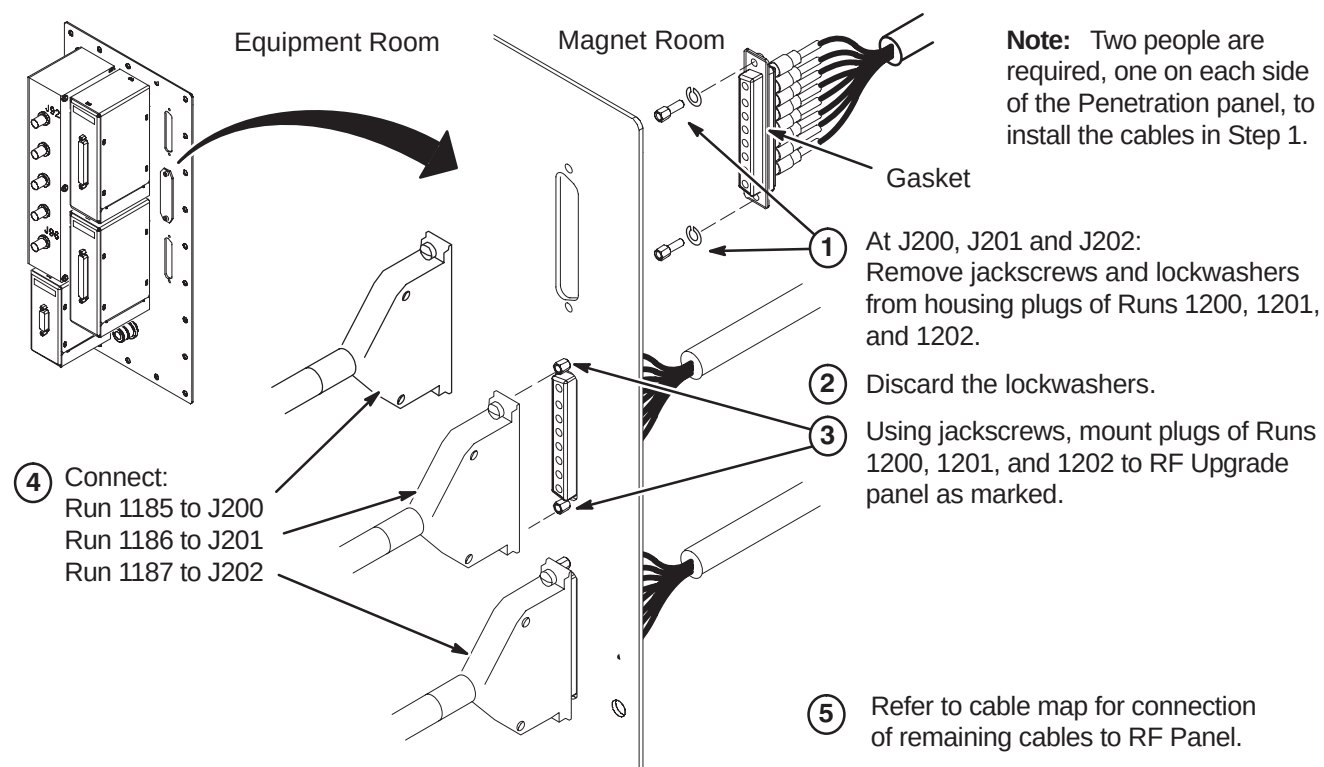
5-2 REMOVE MULTI-COIL ADAPTOR PLATE FROM PENETRATION PANEL



5-3 INSTALL NEW EXCITE RF PANEL



5-4 INSTALLATION OF CABLES AT J200 AND J201



SECTION 6 – MAGNET ROOM

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
6-1 INTRODUCTION	6-3
6-2 DE-INSTALLATION PROCEDURES	6-3
6-2-1 DISCONNECT CABLE TRACK CABLES FROM REAR PEDESTAL MODULES	6-3
6-2-2 DE-INSTALL EXISTING LOW PROFILE CARRIAGE COVER & CABLE TRACK ..	6-4
6-2-3 DE-INSTALL BRIDGE	6-5
6-2-4 DE-INSTALL SRI MODULE	6-6
6-2-5 DE-INSTALL PAC MODULE	6-7
6-3 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL	6-7
6-3-1 UNFASTEN FLAT PANEL BORE LIGHTS	6-7
6-3-2 REMOVE FRONT END BELL	6-8
6-3-3 DE-INSTALL EXISTING RUN 749	6-8
6-3-4 PREPARE END BELL FOR INSTALLATION OF DUAL SENSOR CABLE	6-9
6-3-5 ATTACH DUAL THERMAL SENSOR CABLE	6-9
6-3-6 ATTACH FRONT END BELL TO MAGNET	6-10
6-3-7 ATTACH BORE LIGHTS	6-10
6-4 INSTALL NEW COMPONENTS AND CABLES	6-11
6-4-1 INSTALL NEW DUAL TRACK BRIDGE	6-11
6-4-2 INSTALL NEW COMPONENTS ON REAR PEDESTAL	6-14
6-4-3 INSTALL LPCA AND CABLE TRACK	6-16
6-4-4 CONNECT REMAINING CABLES TO REAR PEDESTAL	6-21
6-4-5 INSTALL SRI-3 MODULE	6-22
6-4-6 INSTALL PAC MODULE	6-24
6-5 TIMES MICROWAVE RF CABLES INSTALLATION	6-25
6-5-1 CABLE INFORMATION	6-25
6-5-2 ROUTE, TERMINATE, AND CONNECT RF CABLES	6-25
6-6 INSTALL RATING PLATE ON FRONT TRIM COVER	6-26

This page left blank intentionally

CAUTION

MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6-1 INTRODUCTION

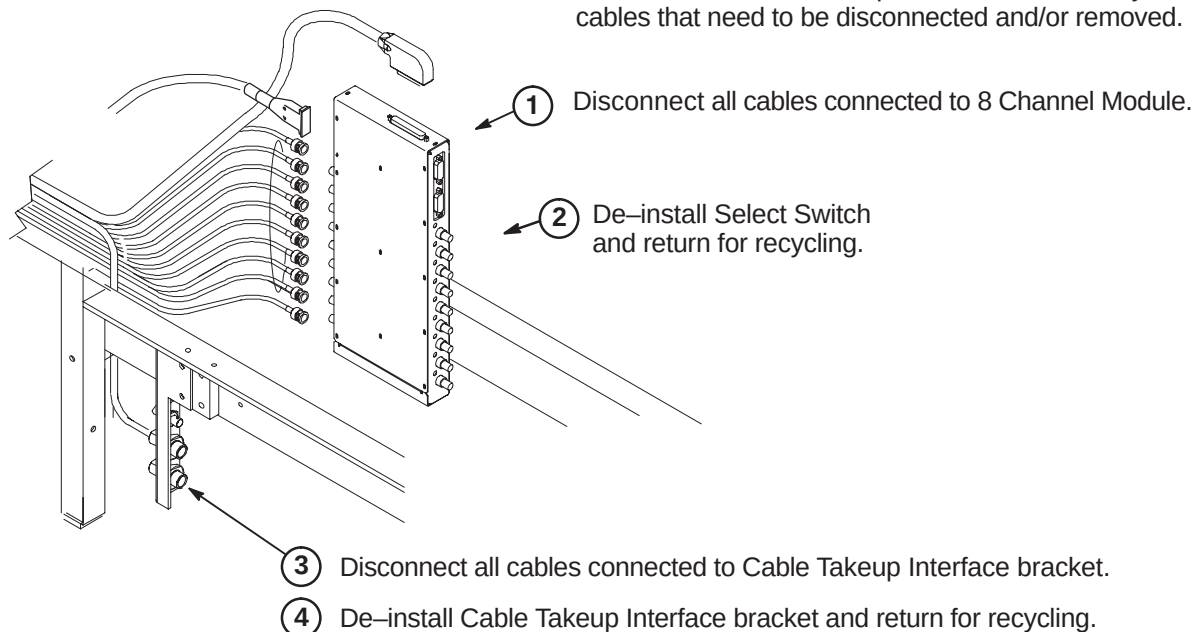
The Bridge, LPCA and Cable Track, SRI, PAC, and modules on the Rear Pedestal will be replaced.

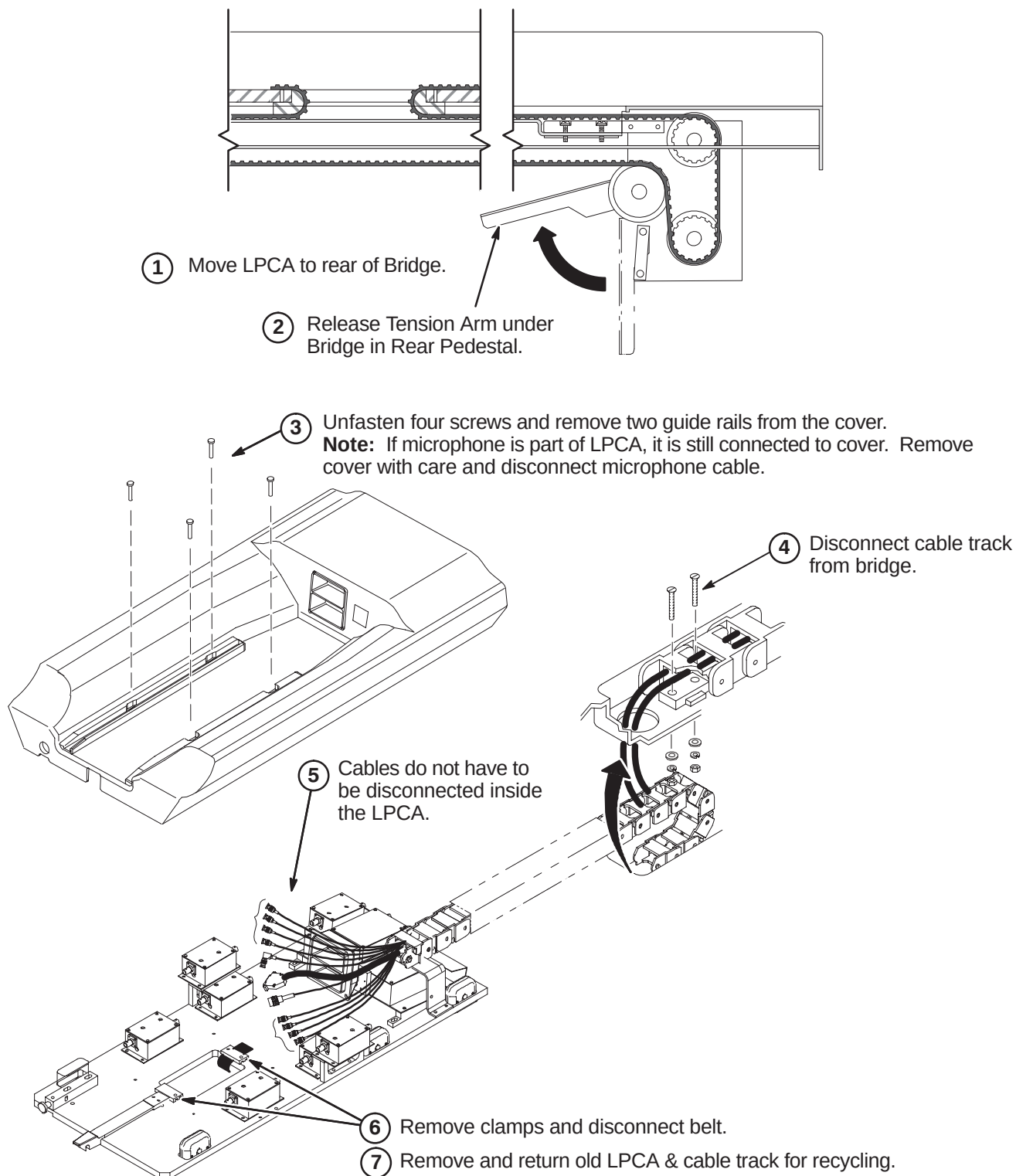
Several cables will be removed and replaced with new cables, some cables will be relabeled and reused, and new cables will be routed and connected. Refer to Section 4 cable maps for information on removing and installing cables

Also included in this section are the instructions for upgrading the Cradle Release on the Patient Table. Refer to section if the linkage mechanism does not have the same pieces as the new linkage shown in this procedure.

6-2 DE-INSTALLATION PROCEDURES**6-2-1 DISCONNECT CABLE TRACK CABLES FROM REAR PEDESTAL MODULES****Note:**

Refer to cable removal map for information on any other cables that need to be disconnected and/or removed.

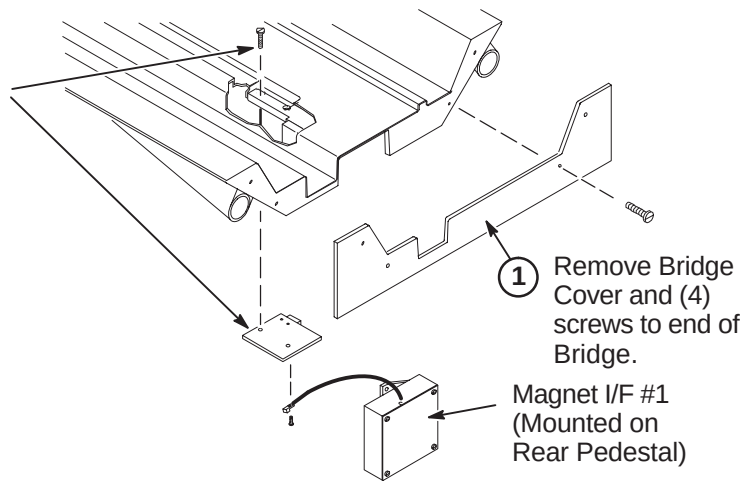
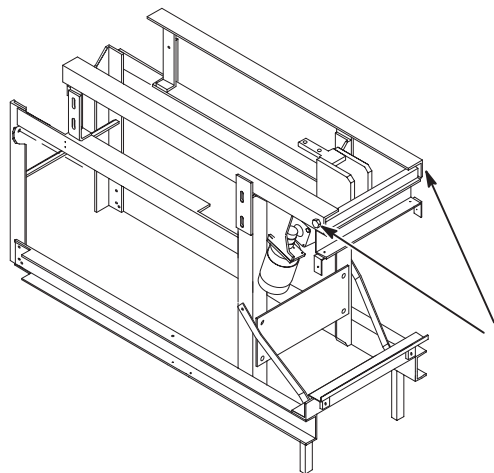


6-2-2 DEINSTALL EXISTING LPCA & CABLE TRACK

6-2-3 DEINSTALL BRIDGE

6-2-3-1 Unfasten Bridge from Rear Pedestal

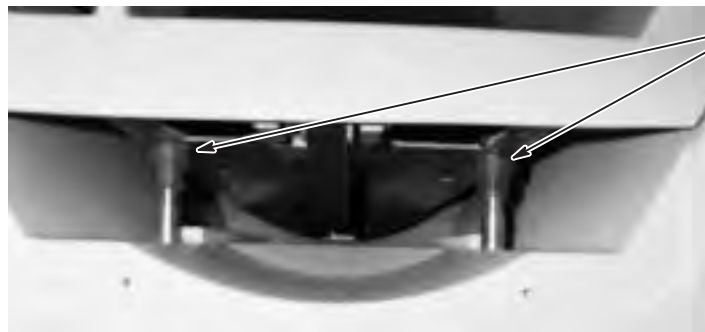
- ② Unfasten and save two 10-32 x 1-1/2 Round Head Slot Brass Screws (46-220184P47) and disconnect Bridge from existing Tongue and Rear Bridge Mount Asm.



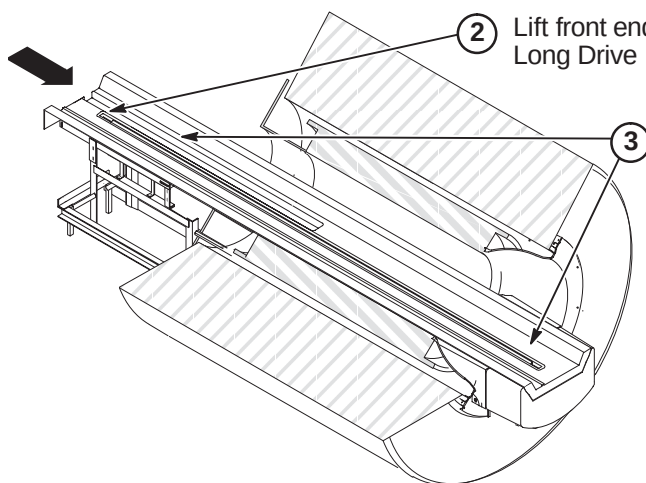
Note: If necessary, loosen, but do not remove, the screw on each side of the Longitudinal Drive Motor Asm so the Tongue will tilt up for easier removal of Bridge.

CAUTION: Longitudinal Motor contains ferrous material. Holding this assembly too close to magnet bore will forcibly attract it to magnet and possibly cause bodily injury or material damage.

6-2-3-2 Remove Bridge from Magnet



- ① Under the front of the Bridge, two Cap Support Cups (2318948) are epoxied to the bottom side of the Bridge. Remove these from the existing de-installed Bridge. If new Cups are not provided with upgrade, these will be reused on new Bridge.



- ② Lift front end of Bridge to disengage tongue on Long Drive assembly at Rear Pedestal.

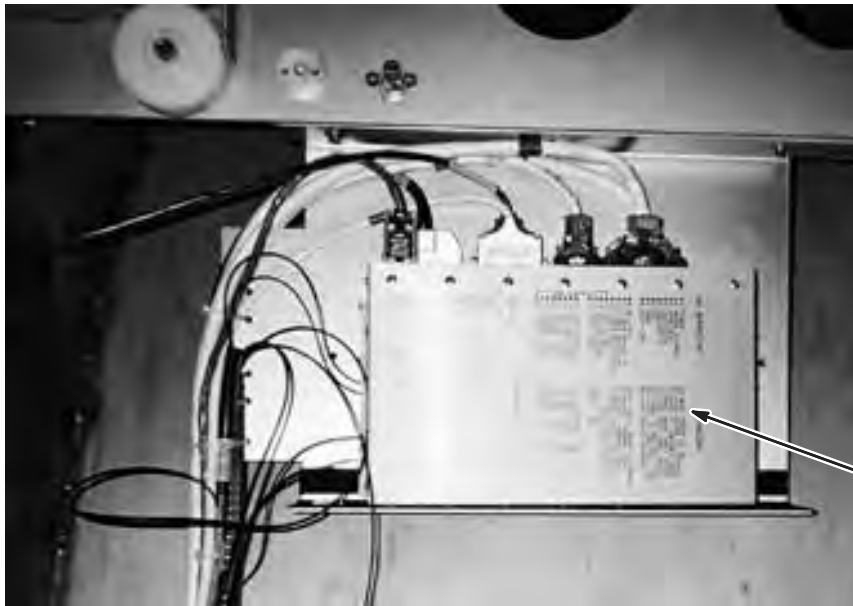
- ③ While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front as it is being removed.

- ④ Remove and save Drive Belt and Front Pulley for transfer to new Bridge.

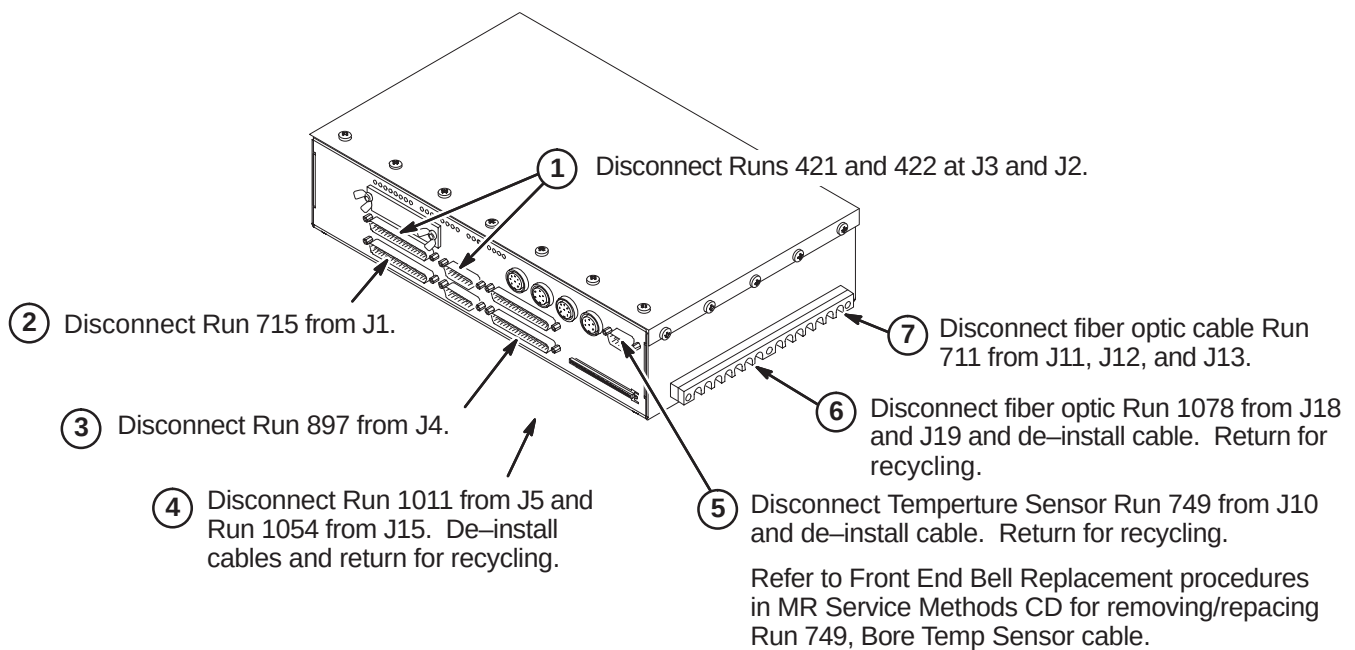
6-2-4 DEINSTALL SRI MODULE

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261.

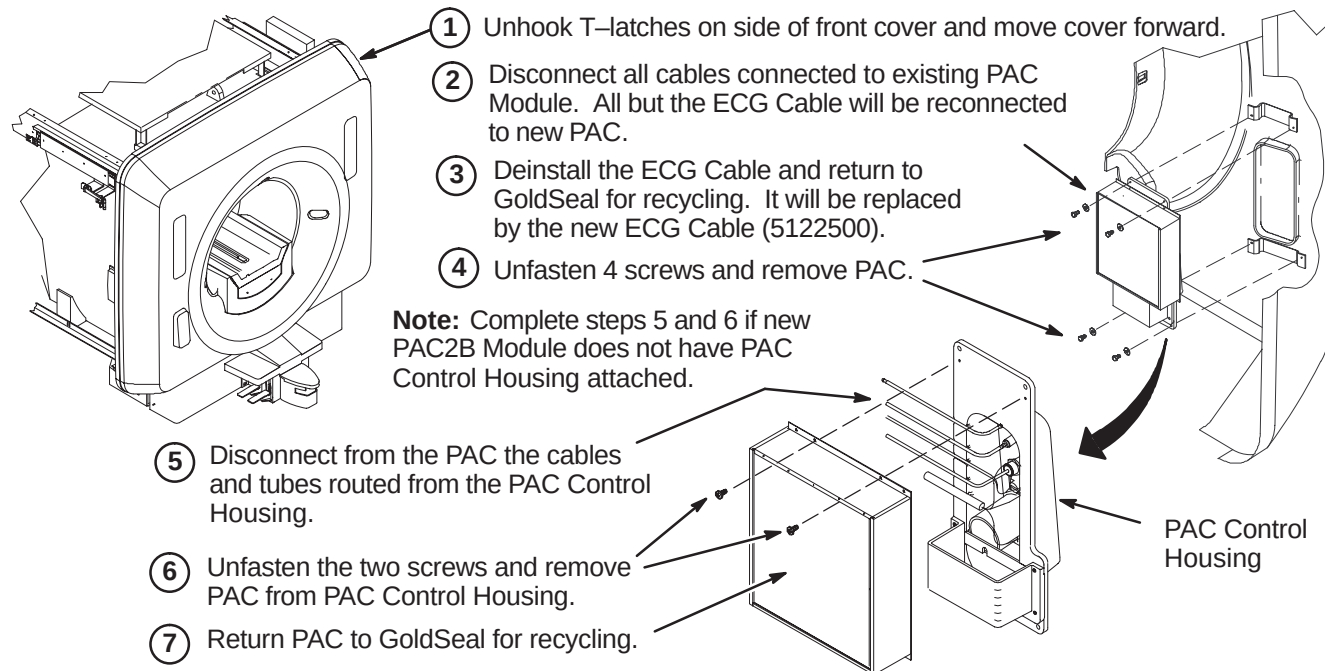


⑧ Carefully pull SRI Module off of velcro strips and return for recycling.



6-2-5 DEINSTALL PAC MODULE

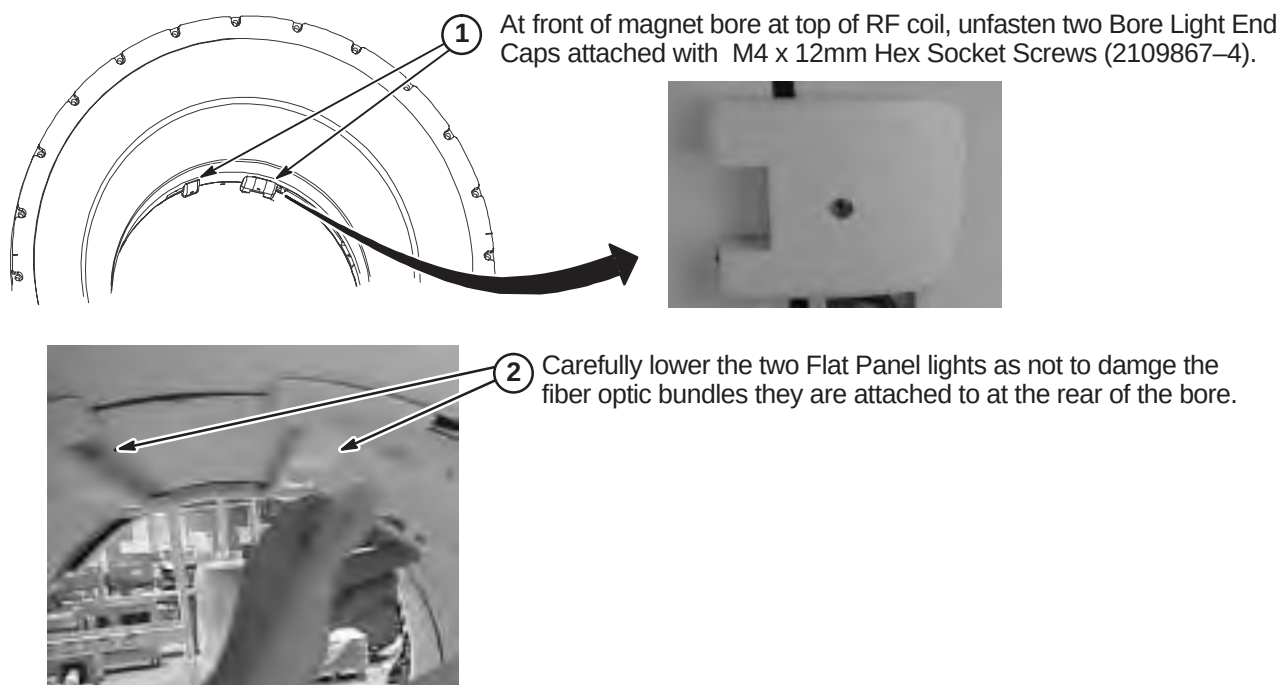
The existing PAC Module will be replaced by a new PAC 2B Module with independent vector gating.



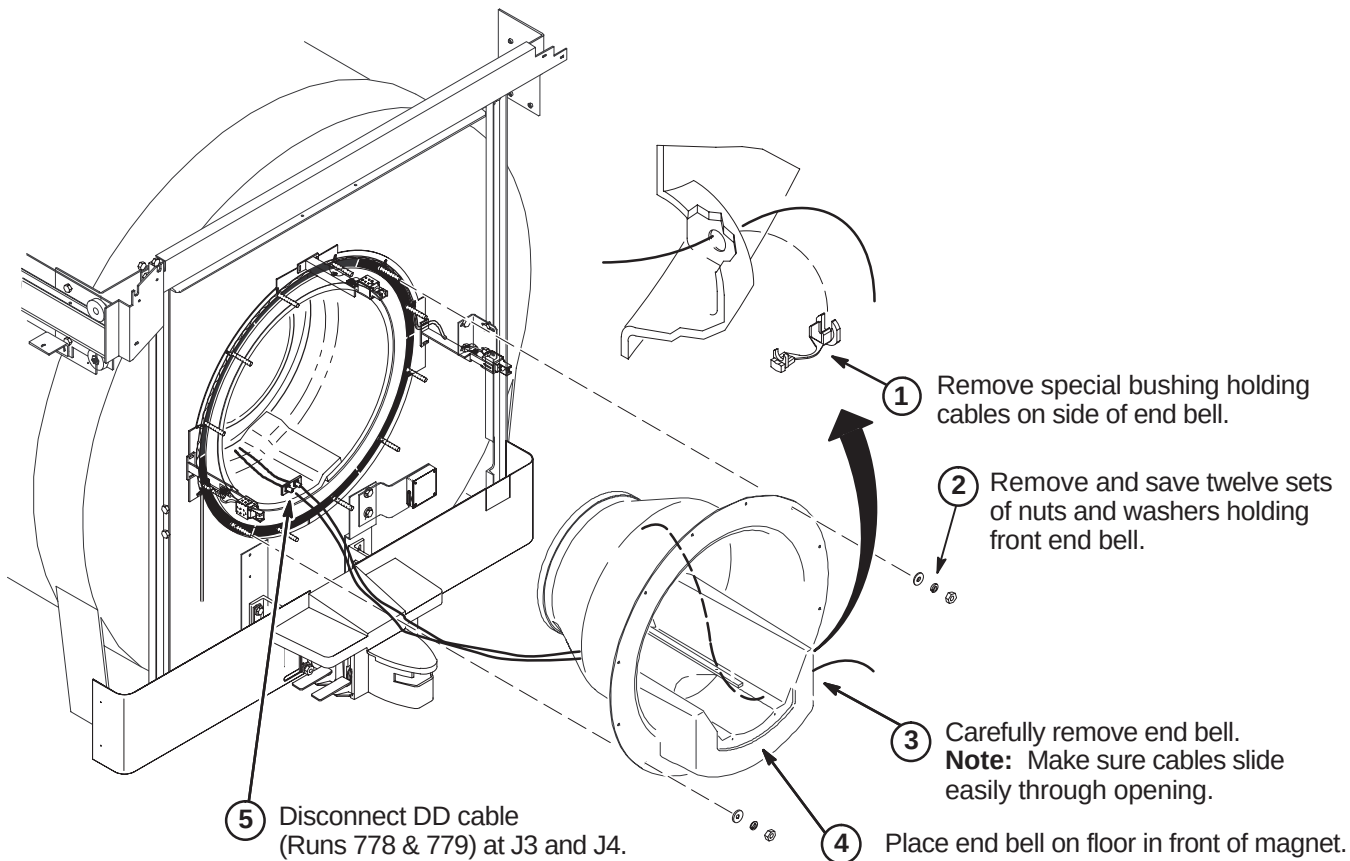
6-3 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL

The existing Temperature Sensor Cable, Run 749, needs to be removed and must be replaced with new Dual Thermal Sensor cable, Run 1260.

6-3-1 UNFASTEN FLAT PANEL BORE LIGHTS

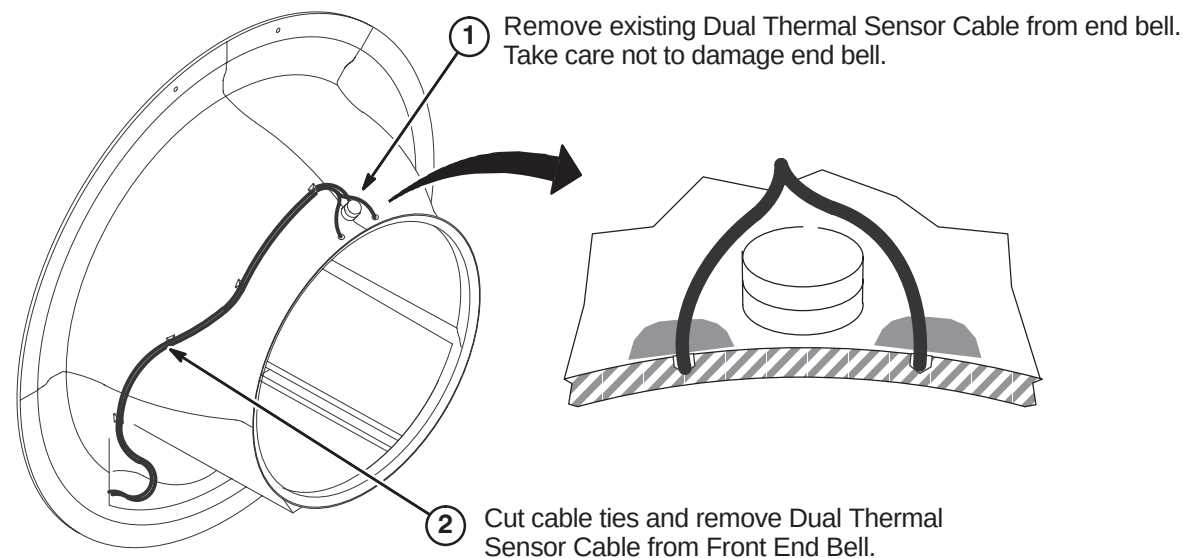


6-3-2 REMOVE FRONT END BELL



6-3-3 DE-INSTALL EXISTING RUN 749

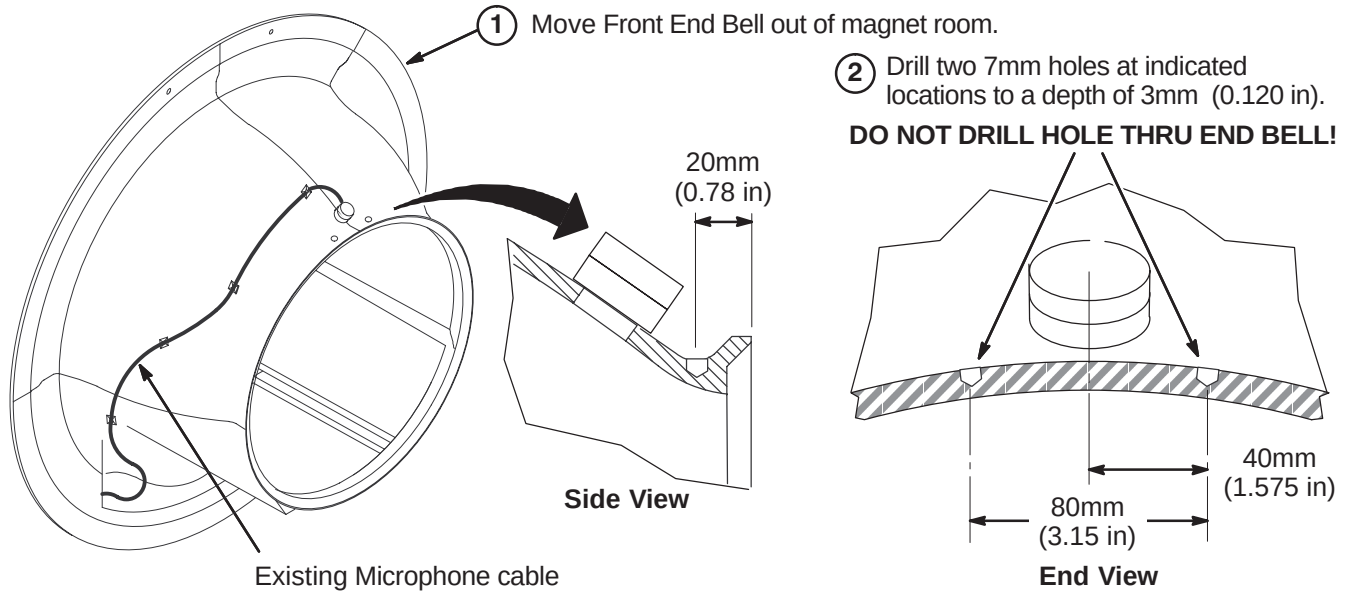
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). Signa HDx requires Run 1260 to be installed.



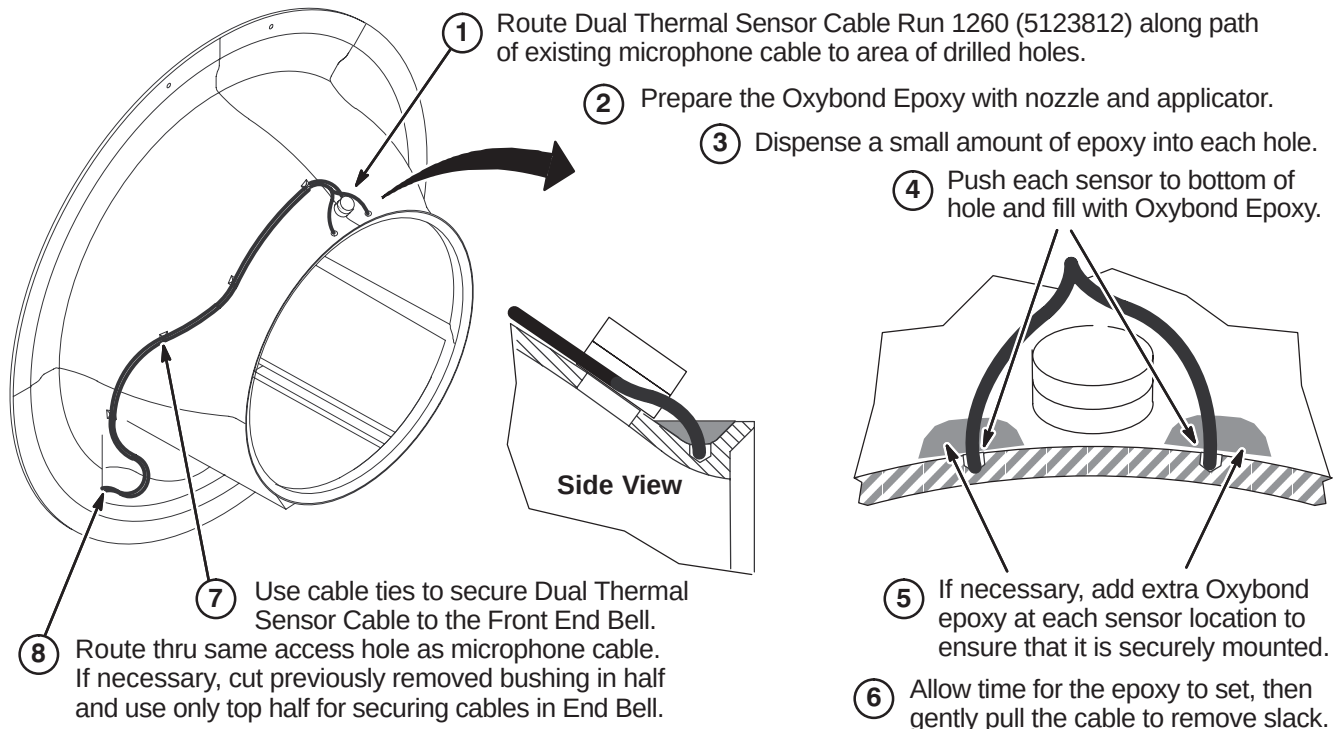
6-3-4 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

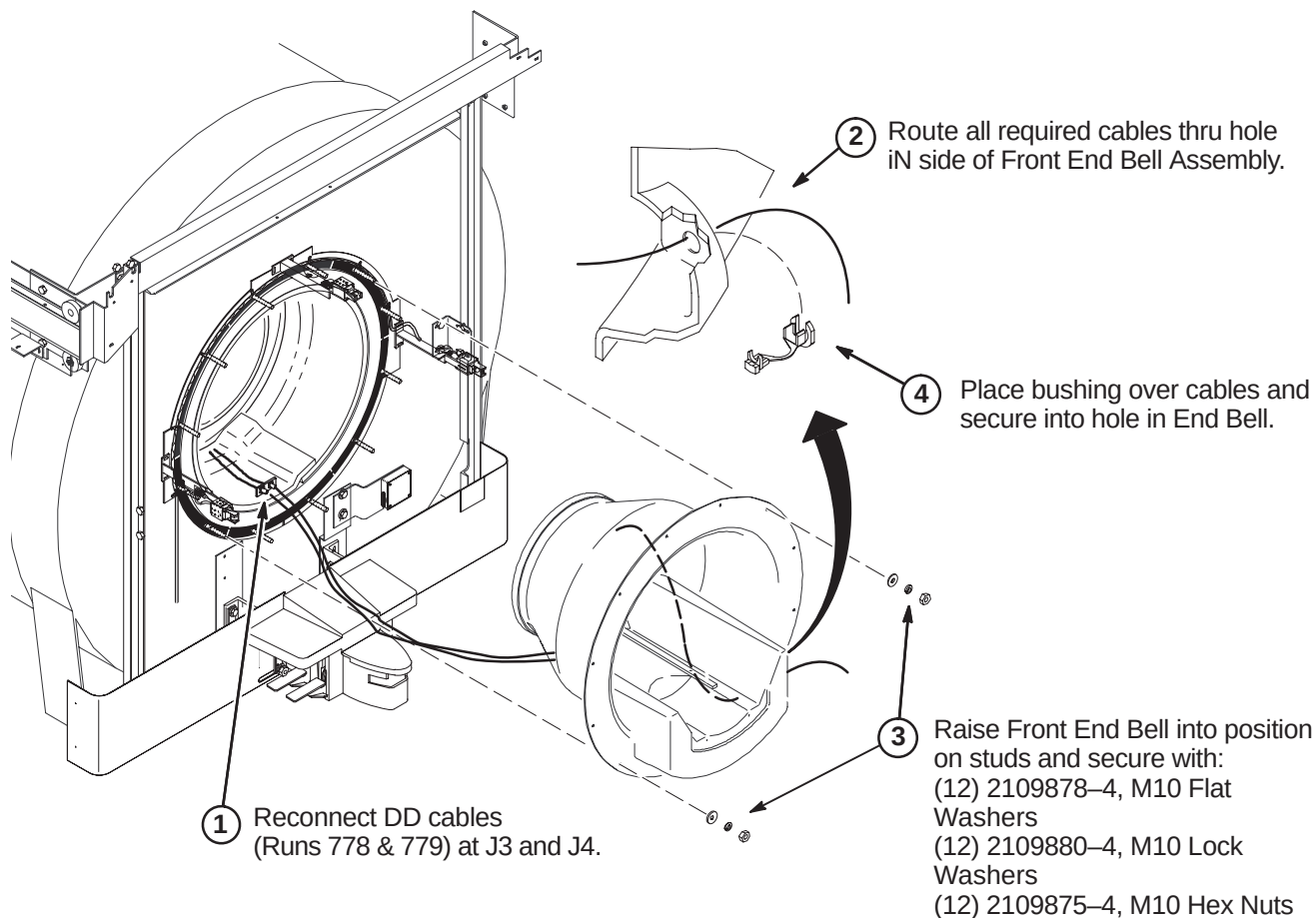


6-3-5 ATTACH DUAL THERMAL SENSOR CABLE

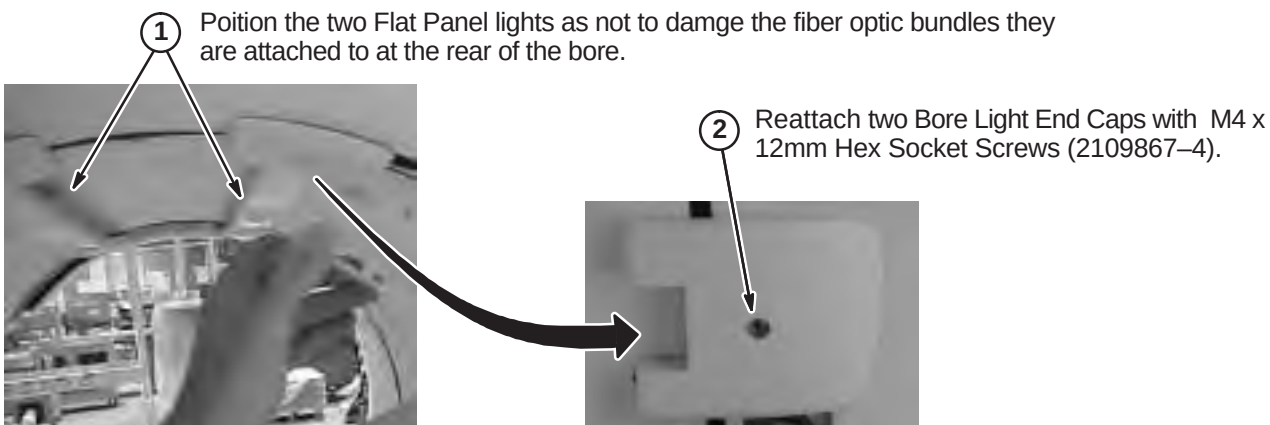


6-3-6 ATTACH FRONT END BELL TO MAGNET

Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Gradient Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6" blower hose from Blower Box to Bulkhead.



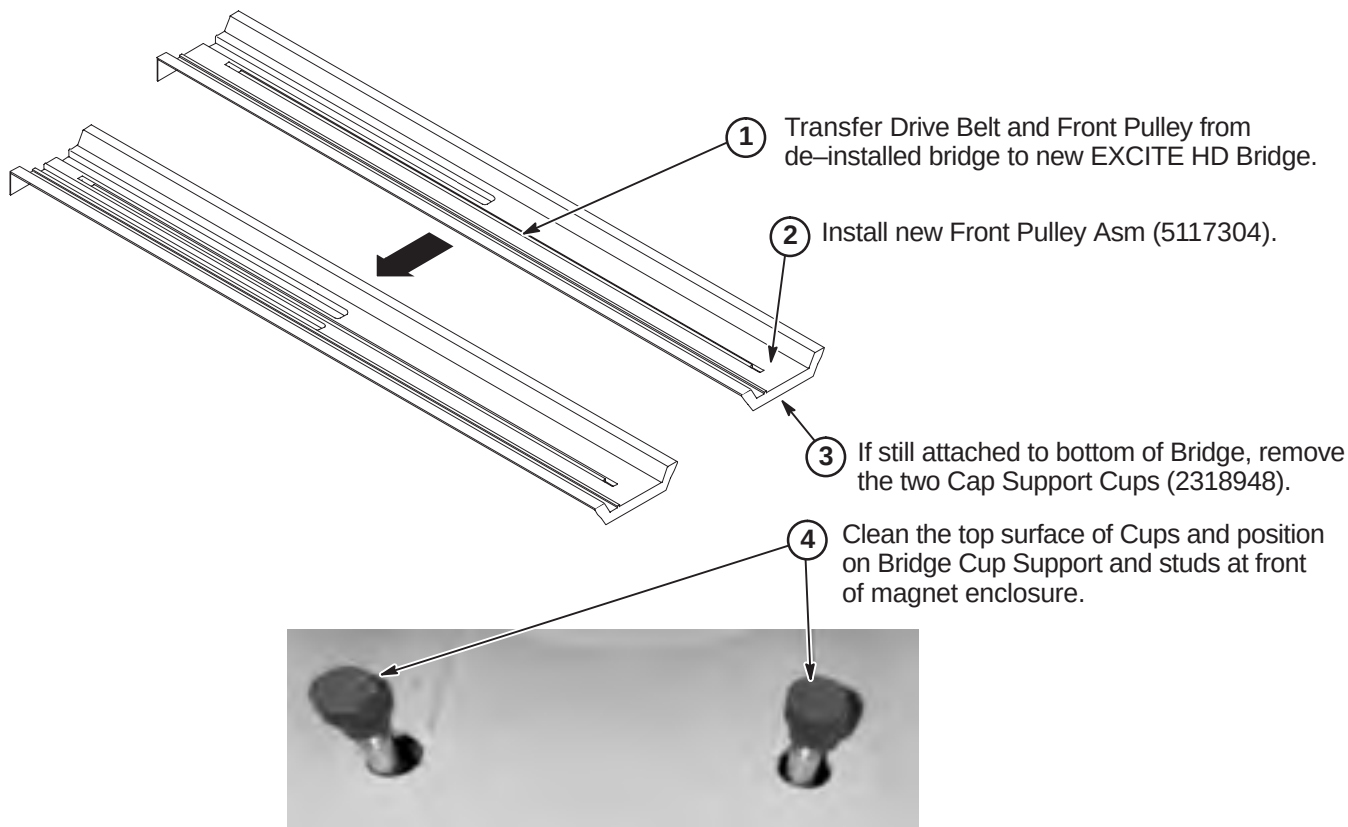
6-3-7 ATTACH BORE LIGHTS



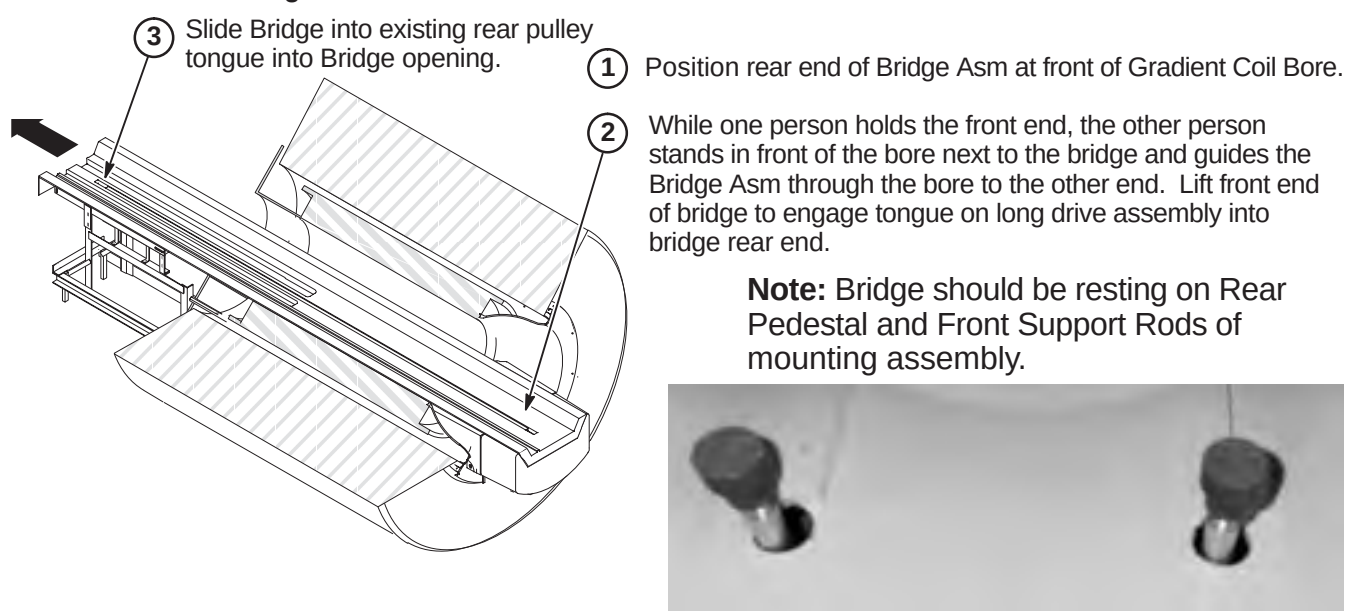
6-4 INSTALL NEW COMPONENTS AND CABLES

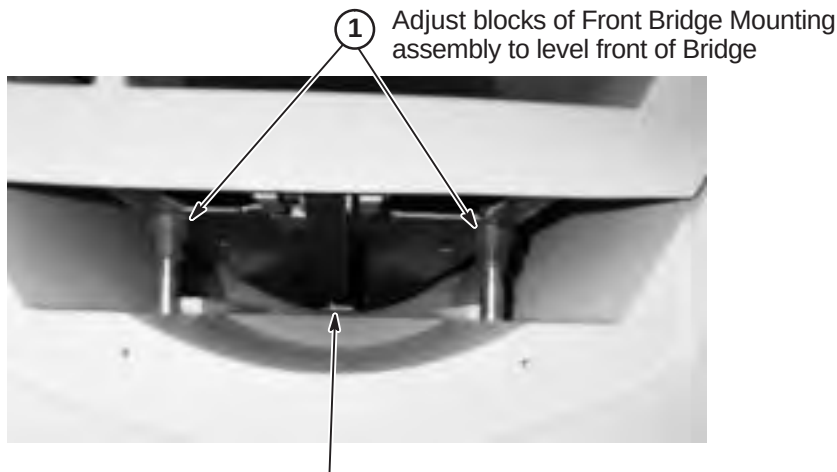
6-4-1 INSTALL NEW DUAL TRACK BRIDGE

6-4-1-1 Transfer Components to New Bridge

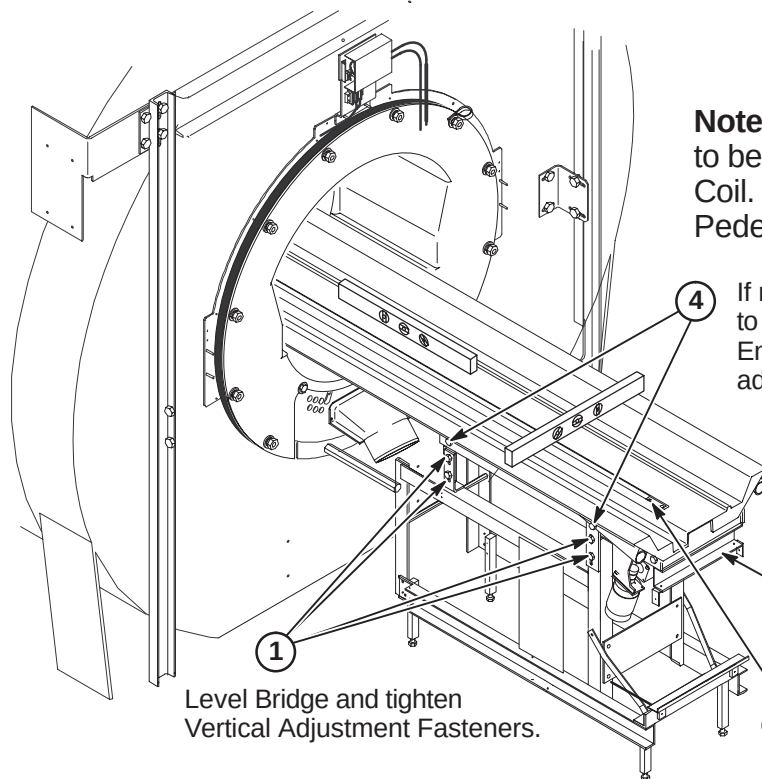


6-4-1-2 Install Bridge



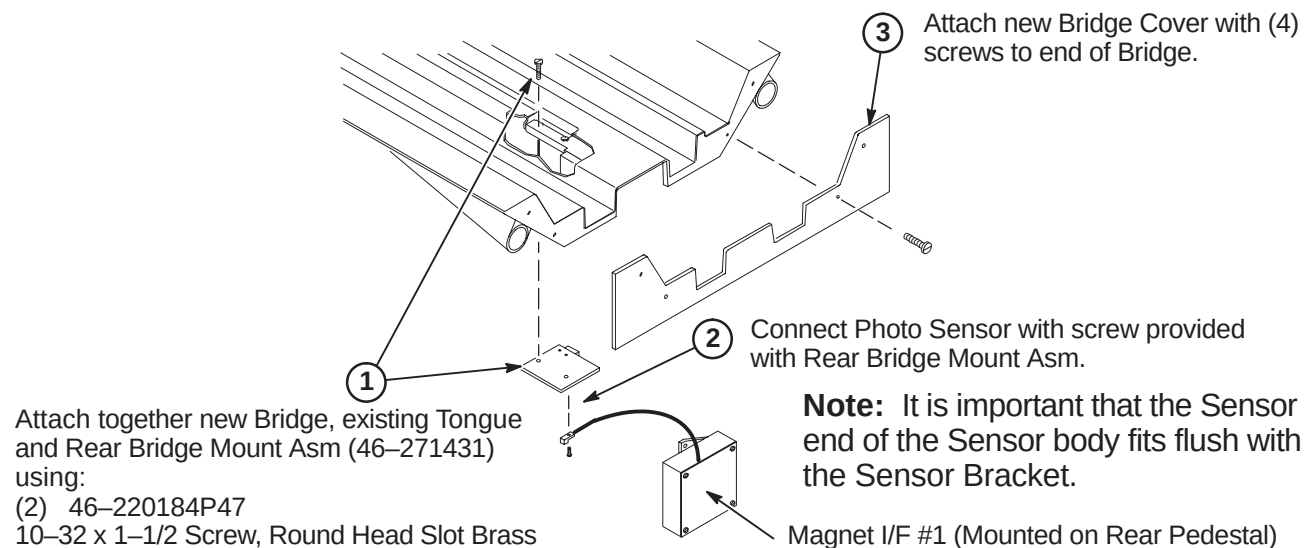
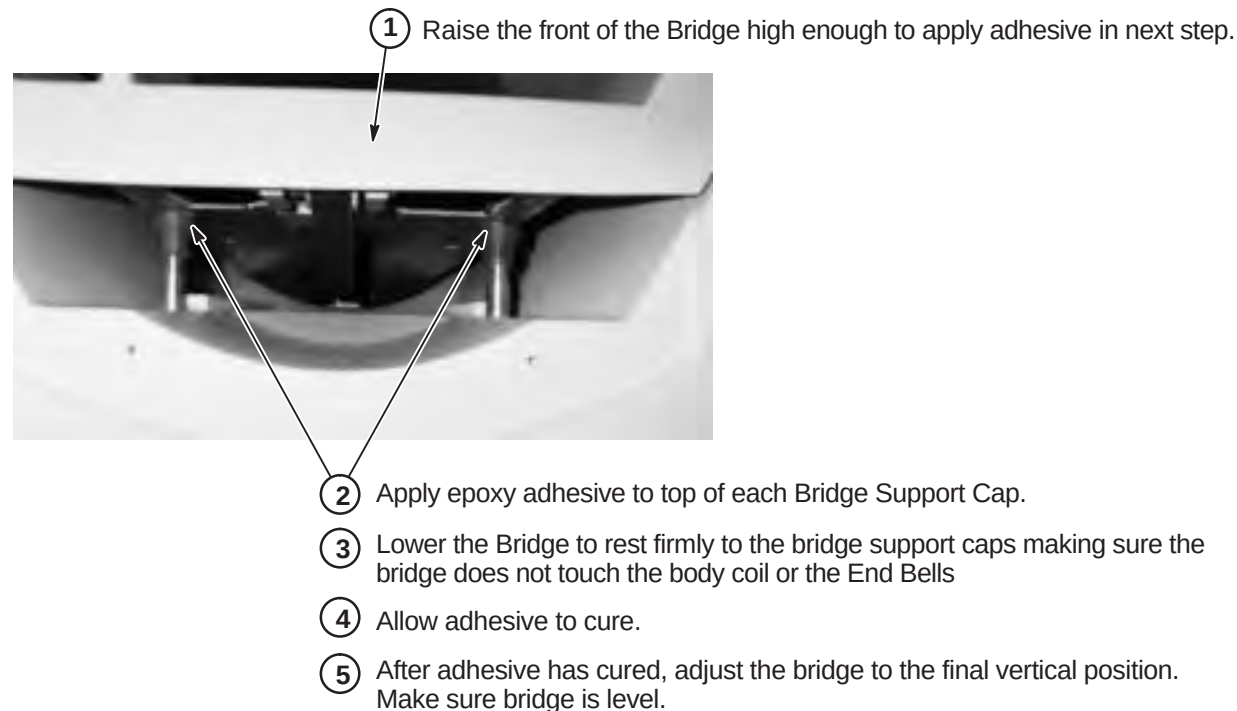
6-4-1-3 Level Front of Bridge**IMPORTANT!**

THE BRIDGE MUST NOT COME IN CONTACT WITH THE RF COIL.
There must be an air gap of 6mm to 8mm thru entire length of bore.

6-4-1-4 Level All of Bridge

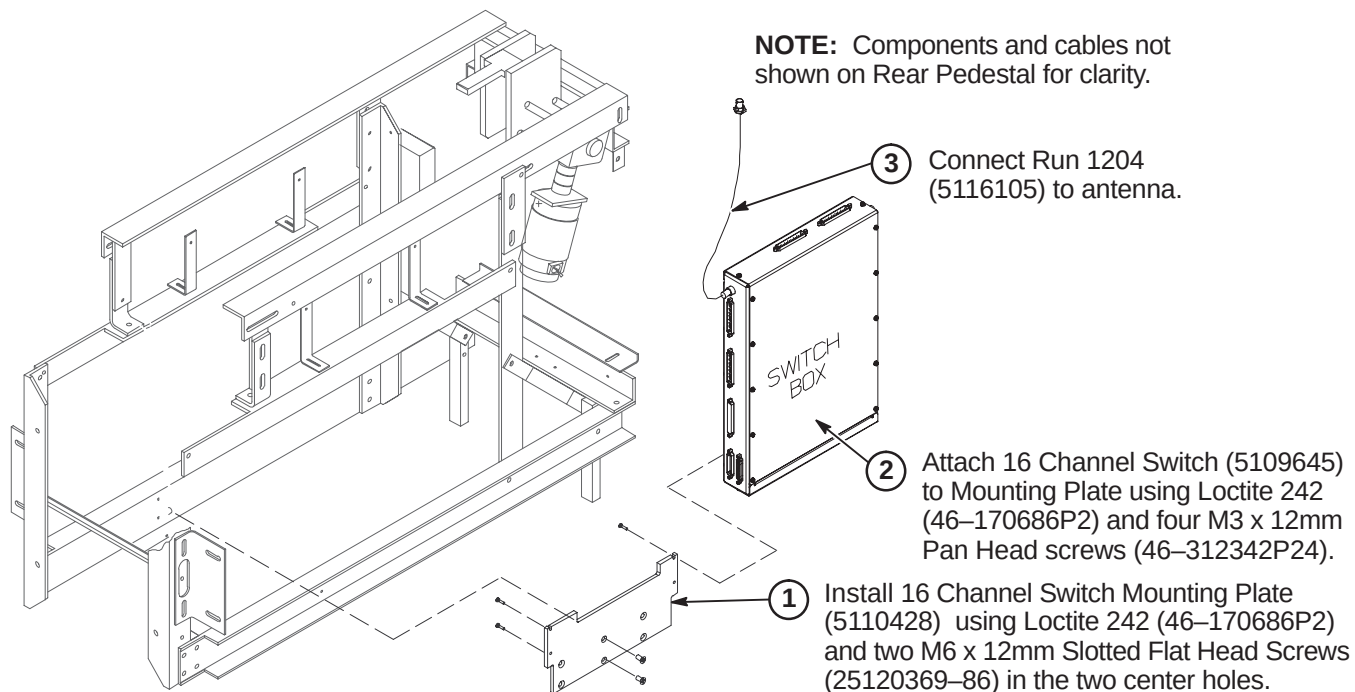
Note: The weight of Bridge should continue to be supported by the rubber strips in the Coil. The Front Bridge Mount and Rear Pedestal stabilize and level the Bridge.

Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

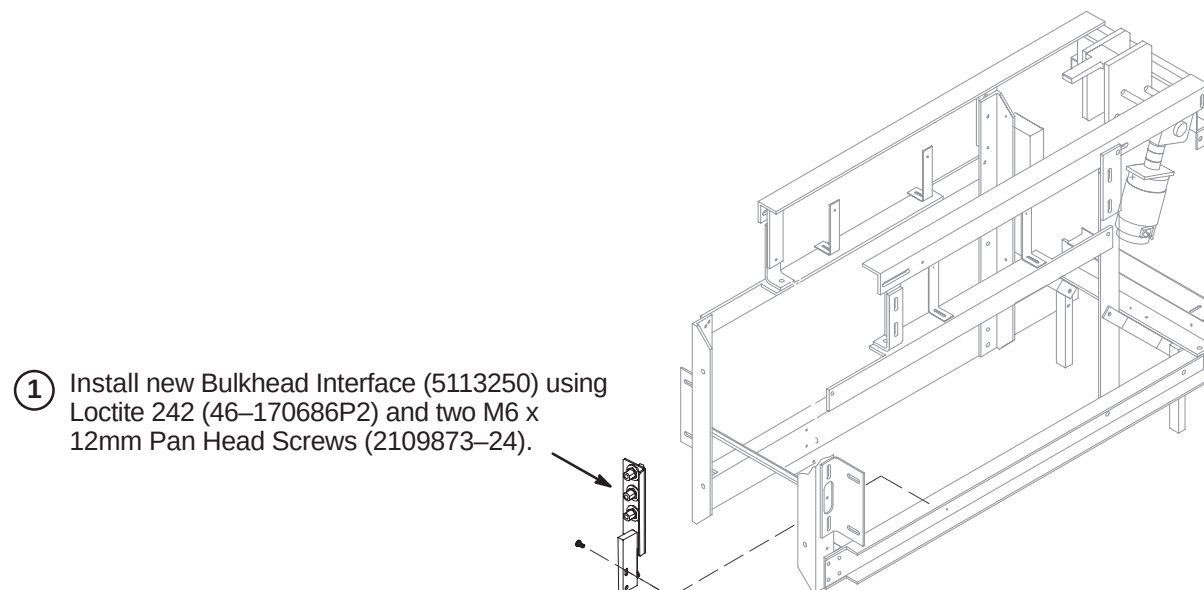
6-4-1-5 Connect Photo Sensor and Bridge End Cover**6-4-1-6 Apply Adhesive to Front Bridge Support Caps**

6-4-2 INSTALL NEW COMPONENTS ON REAR PEDESTAL

6-4-2-1 Install 16 Channel Switch

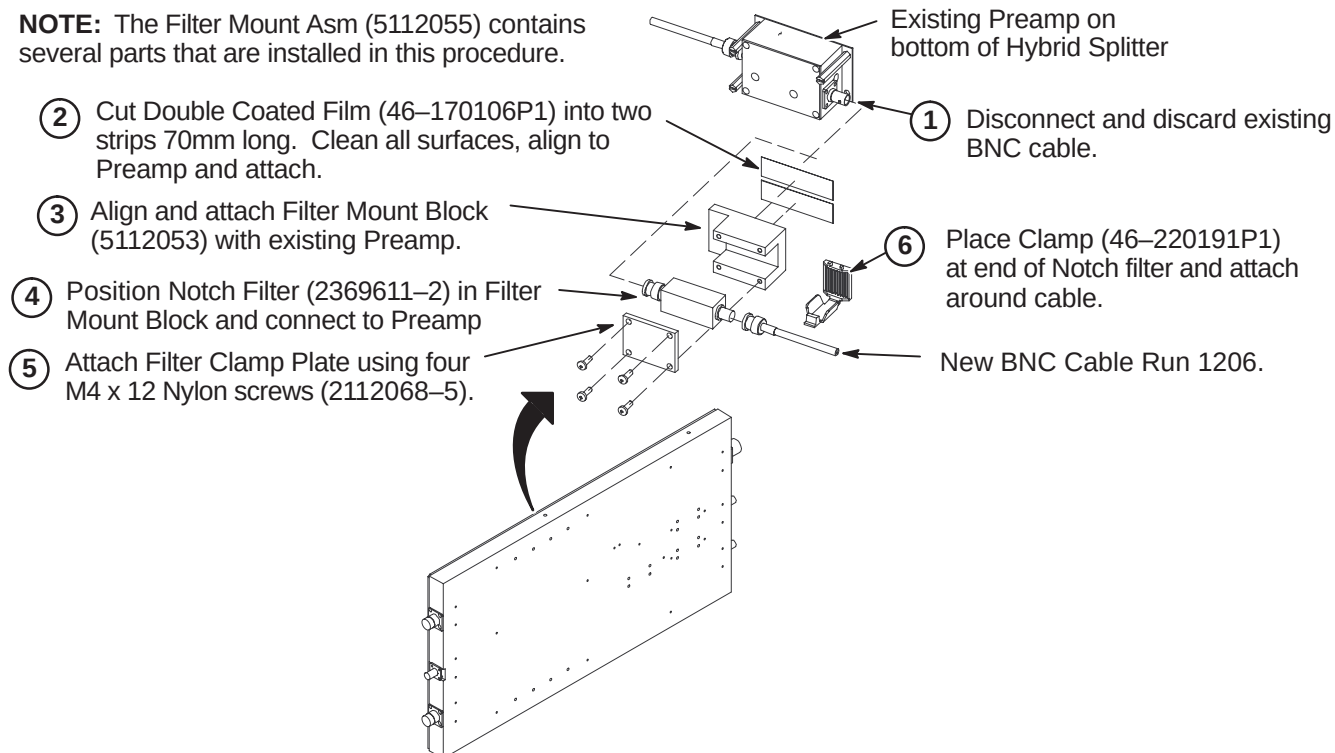


6-4-2-2 Install Bulkhead Interface



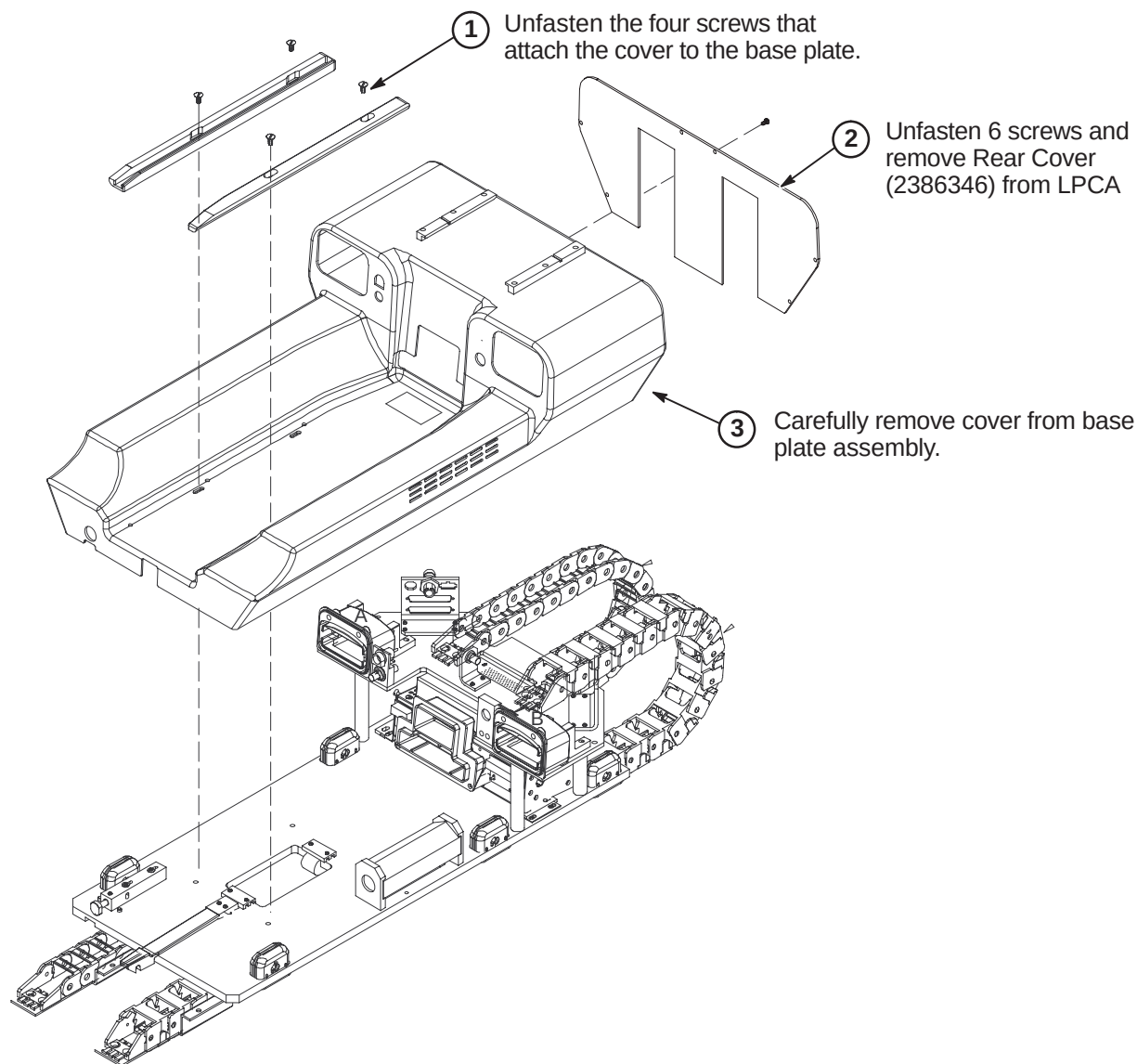
6-4-2-3 Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.



6-4-3 INSTALL LPCA AND CABLE TRACKS

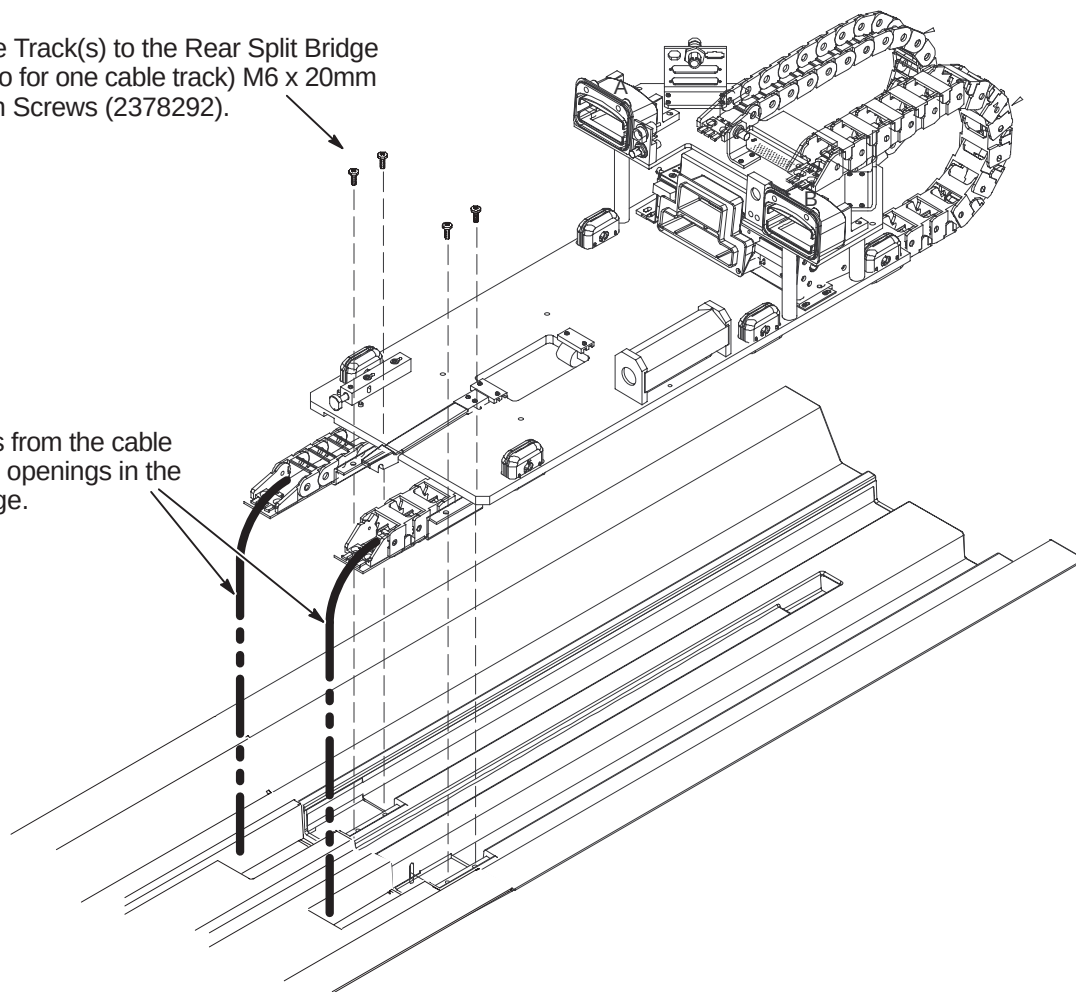
6-4-3-1 Remove LPCA Cover

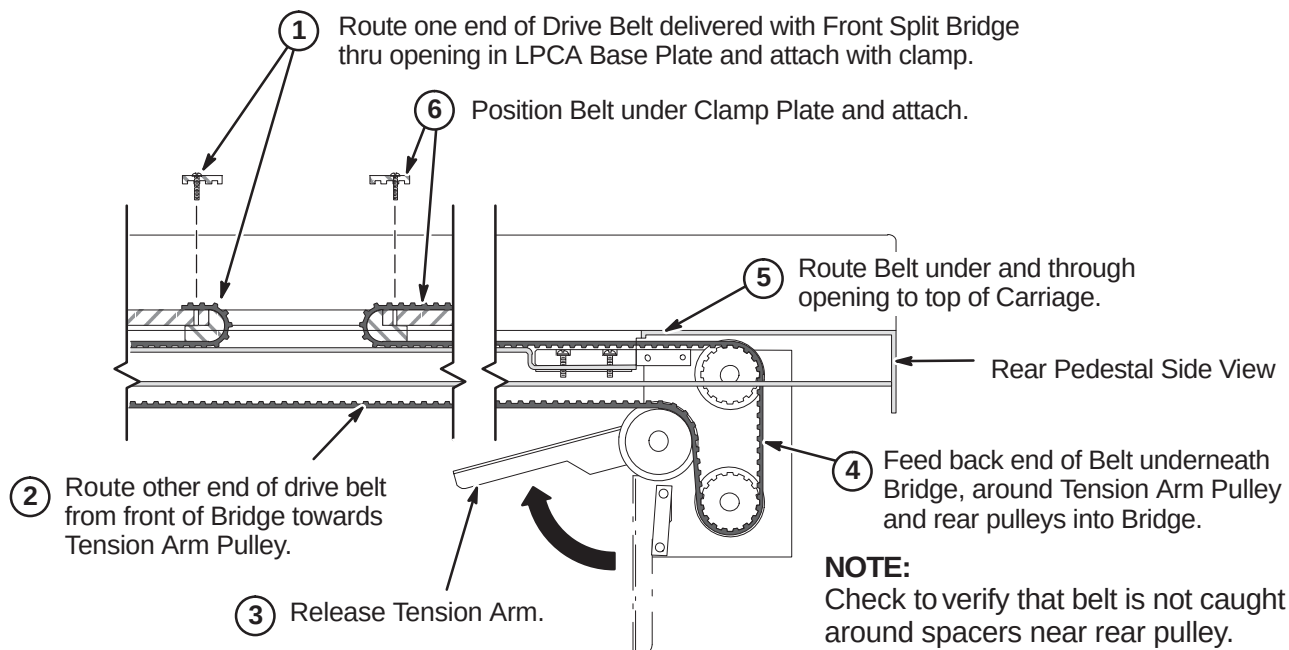
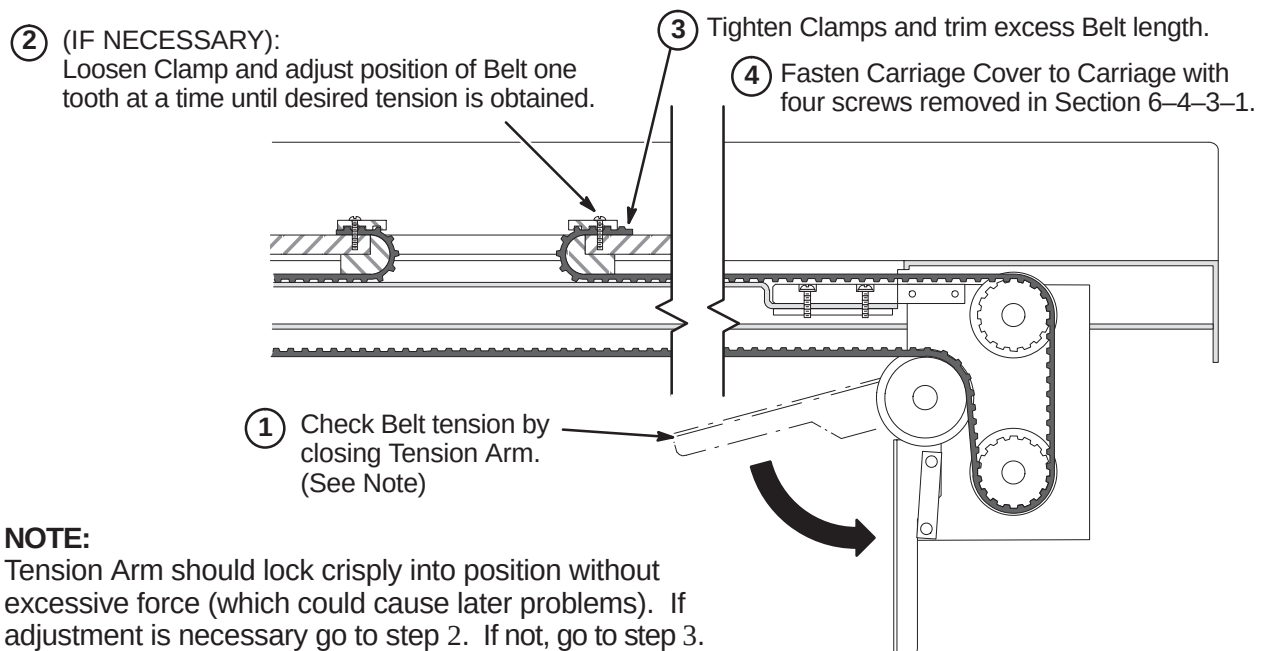


6-4-3-2 Attach Cable Tracks to Bridge

- ② Attach the Cable Track(s) to the Rear Split Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6-4-3-3 Route and Attach Drive Belt**6-4-3-4 Final Adjustment of Drive Belt**

6-4-3-5 Installation of Sensor Flag

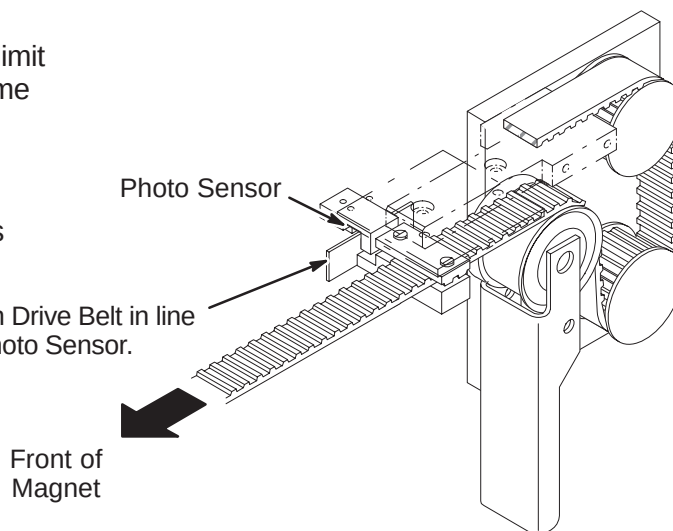
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

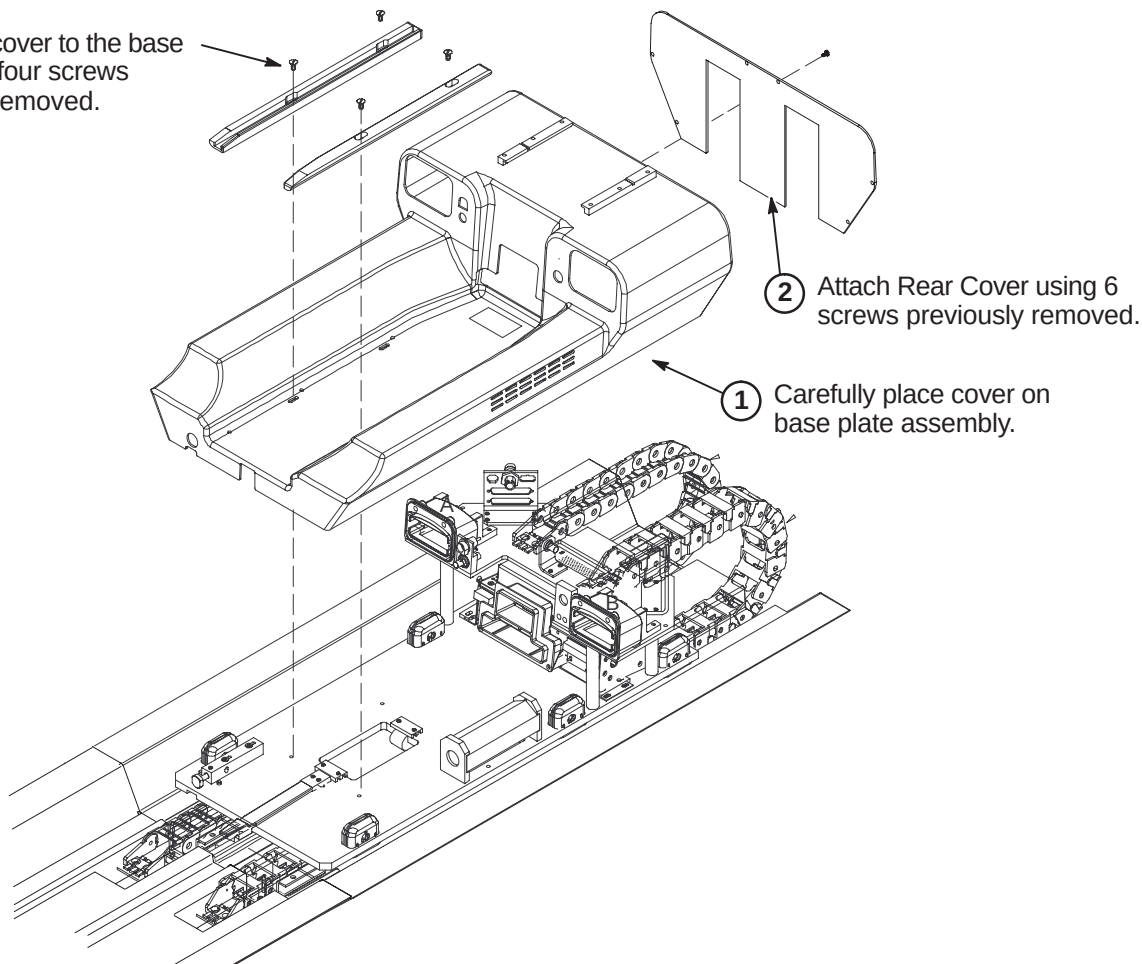
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



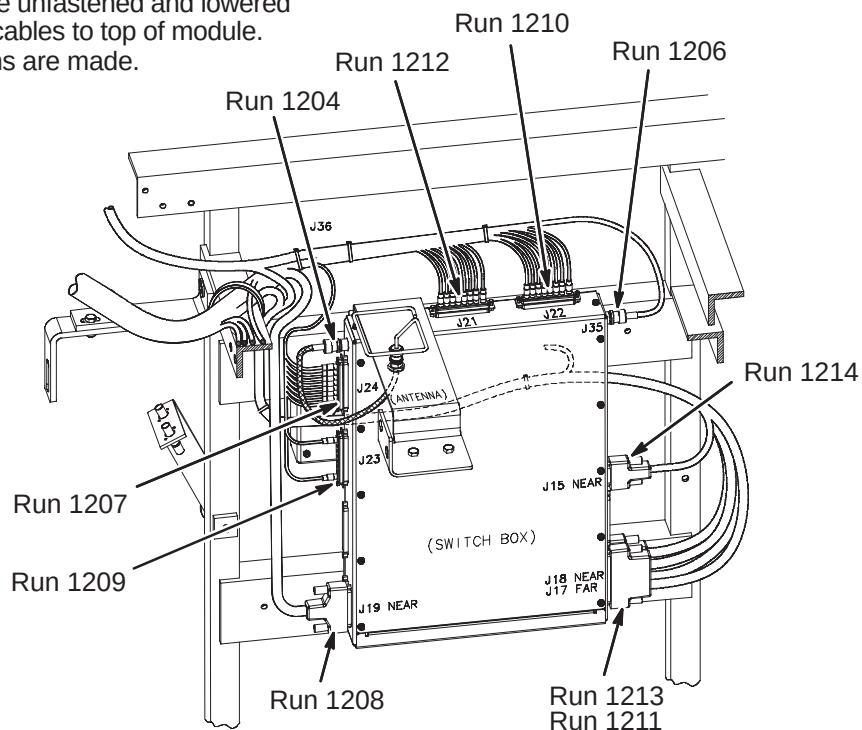
6-4-3-6 Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.



6-4-3-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**6-4-3-8 Connect Cables from Cable Track to Bulkhead Interface**

- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

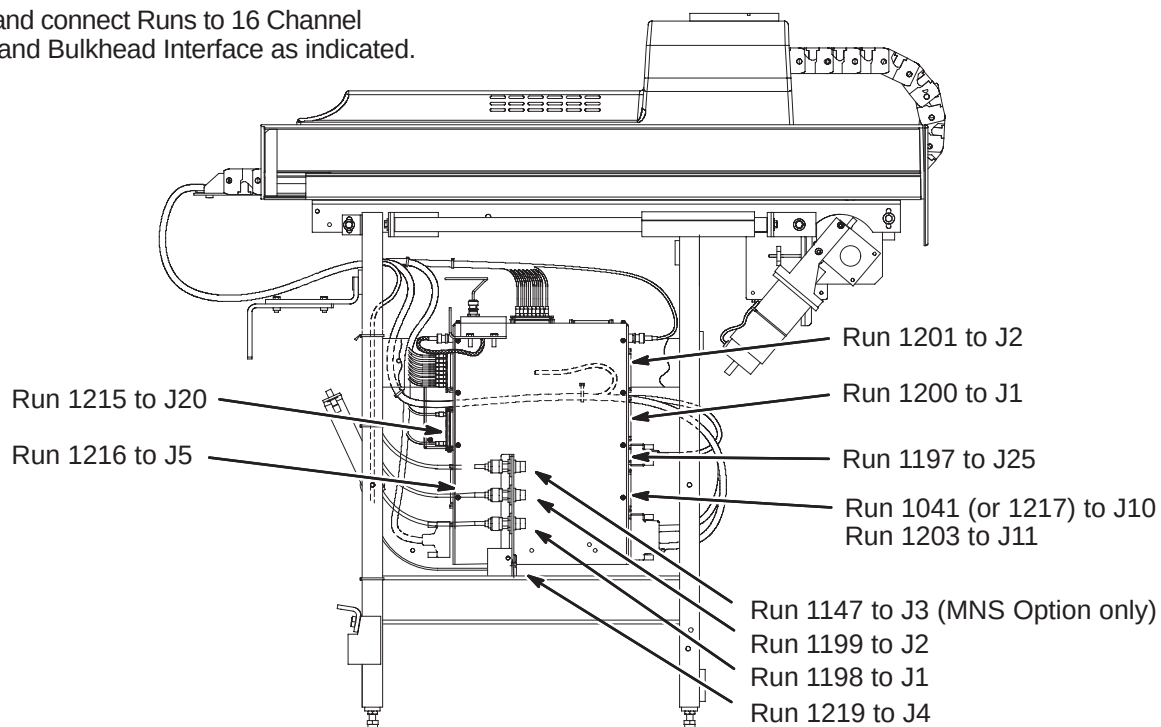
- ② Connect Runs to Bulkhead Interface as indicated.

Run 1153 to J3
 Run 1205 to J2
 Run 1228 to J1
 Run 1206 to J4

6-4-4 CONNECT REMAINING CABLES TO REAR PEDESTAL

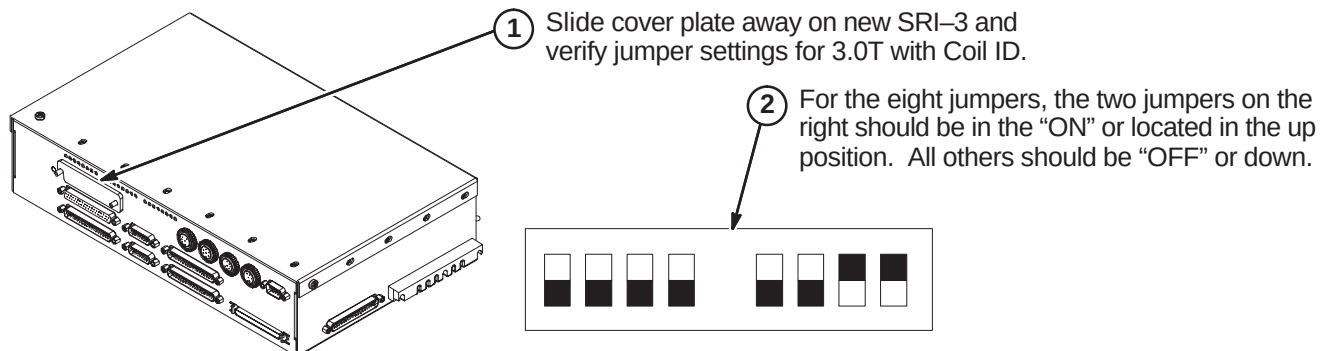
6-4-4-1 Cable Connections to 16 Channel Switch and Bulkhead Interface

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.



6-4-5 INSTALL SRI-3 MODULE

6-4-5-1 Verify Jumper Positions for SRI-3



6-4-5-2 Position SRI-3 in L Bracket

Note:

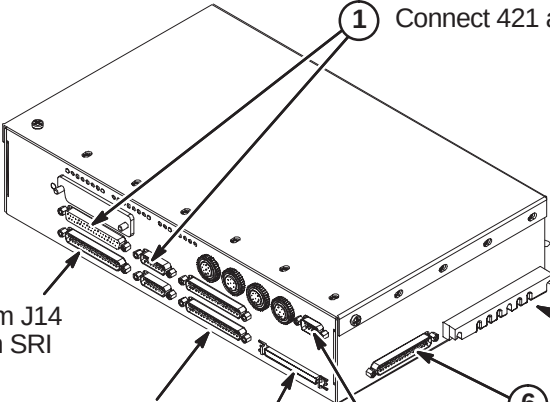
SRI-3 Module to be installed so that it can be easily pulled off for the installation and removal of cables.

- 1 Install Loop Fastener strips (5110749) in four places as shown. Remove protective covers.

- 3 Press the SRI-3 Module on bracket and press into place against the adhesive of the velcro.

- 2 On back of SRI-3 Module, remove protective cover from Velcro at the four locations.

6-4-5-3 Route and Connect Cables to SRI-3

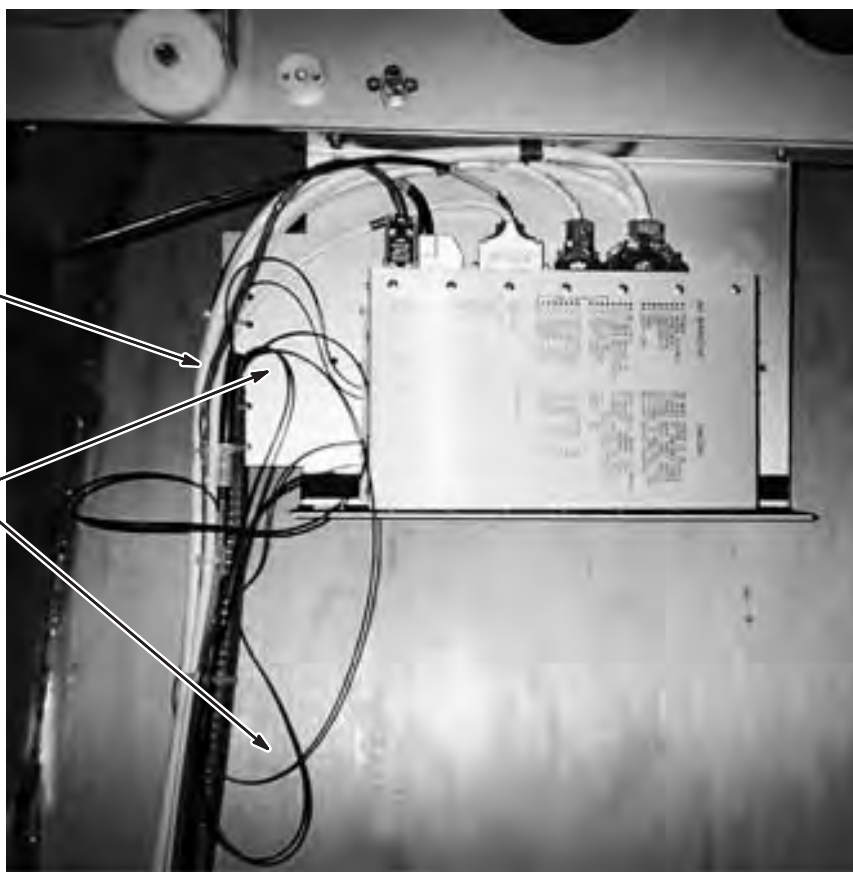
- 
- 1 Connect 421 and 422 to J3 and J2.
- 2 Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
- 3 Connect Run 897 from Operator Display to J4.
- 4 Route Run 1261 from Operator Display and connect to J5.
- 5 Route Run 1260, Dual Temperature Sensor, from Front End Bell and connect to J10.
- 6 Route Run 1215 from J20 on 16 Channel Switch and connect to J20.
- 7 Connect fiber optic cable Run 711 to J11, J12, and J13.

- 8 Route all cables and fiber optic cables as shown.

Note:

Fiber optic cables loops must not be smaller than two inch (50mm) radius.

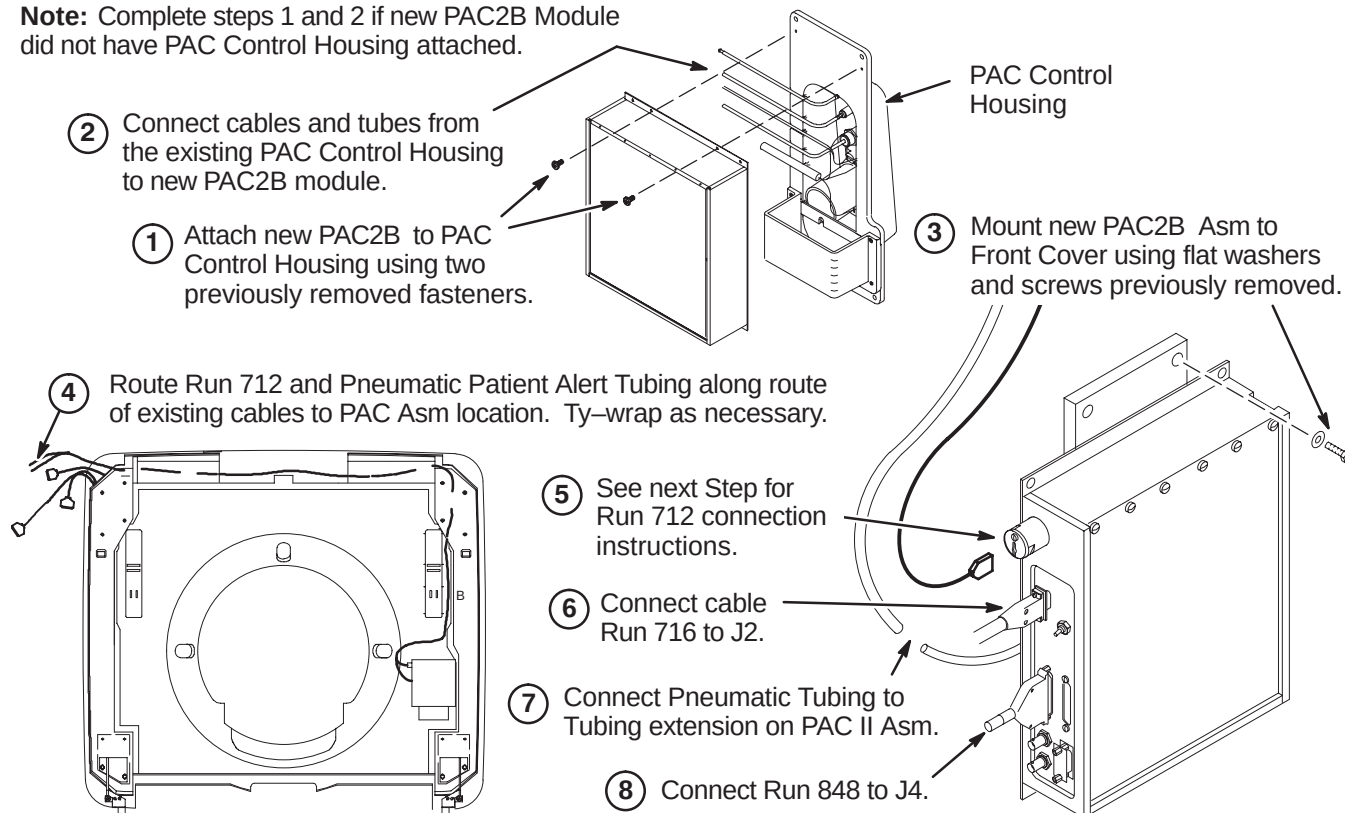
- 9 Reapply spiral wrap around all cables and fiber optics.



6-4-6 INSTALL PAC MODULE

6-4-6-1 Install PAC2B and Connect Run 712, 716, & Pneumatic Patient Alert Tubing

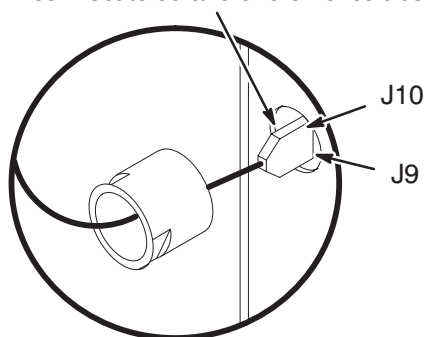
Note: Complete steps 1 and 2 if new PAC2B Module did not have PAC Control Housing attached.



6-4-6-2 Connect Run 712

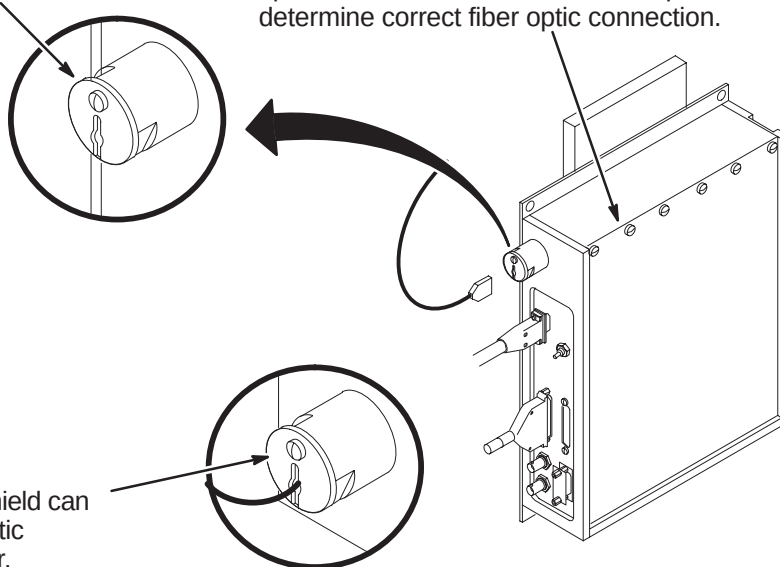
1 Remove shield cover by removing one screw and unscrew shield from module.

2 Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.



3 Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.



6-5 TIMES MICROWAVE RF CABLES INSTALLATION

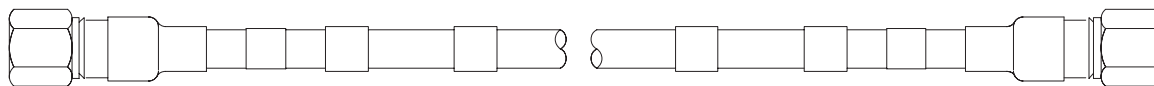
6-5-1 CABLE INFORMATION

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).



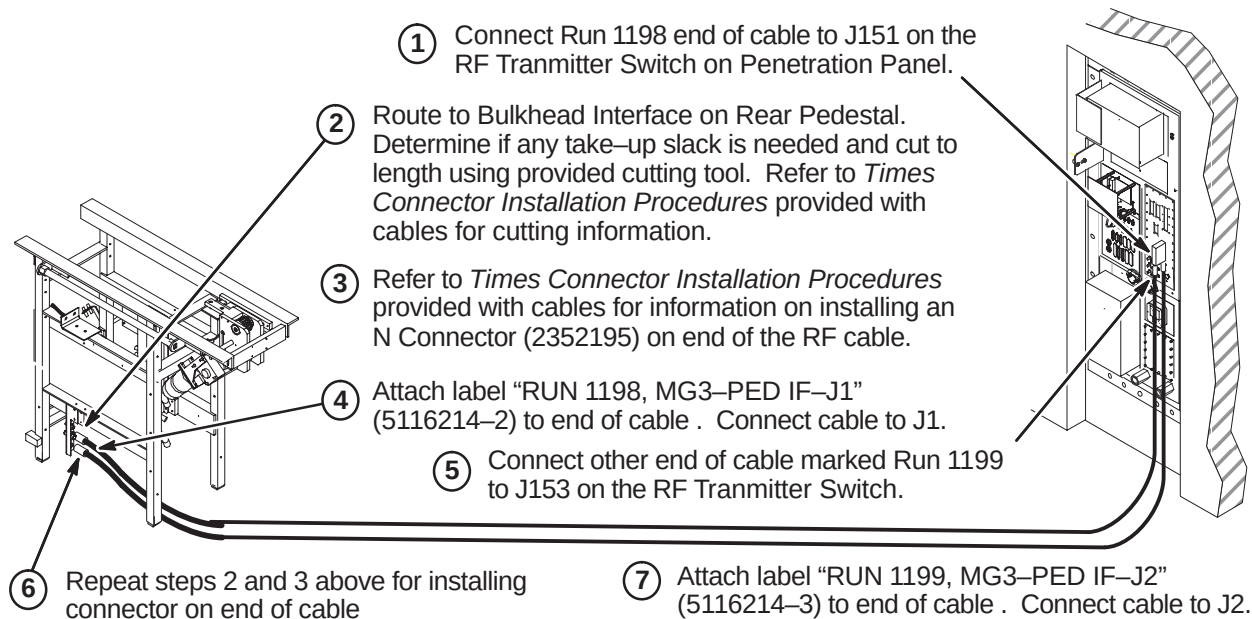
Do not subject LMR-600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

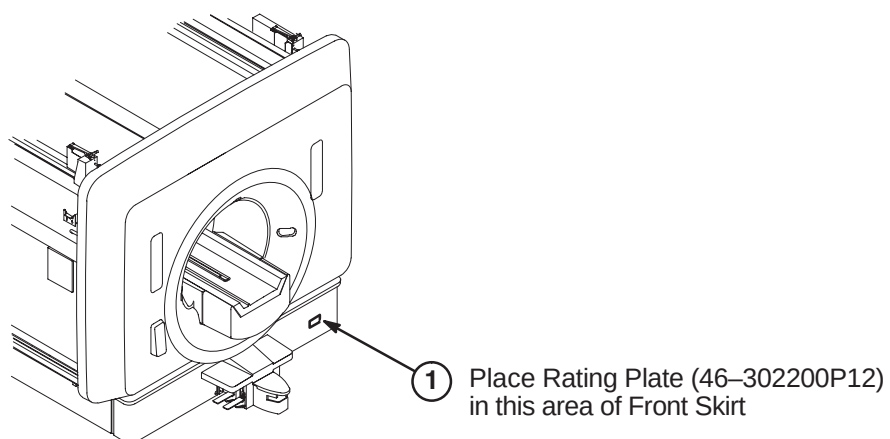
Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

6-5-2 ROUTE, TERMINATE, AND CONNECT RF CABLES



6-6 INSTALL RATING PLATE ON FRONT TRIM COVER

SECTION 7 – CRADLE RELEASE UPGRADE

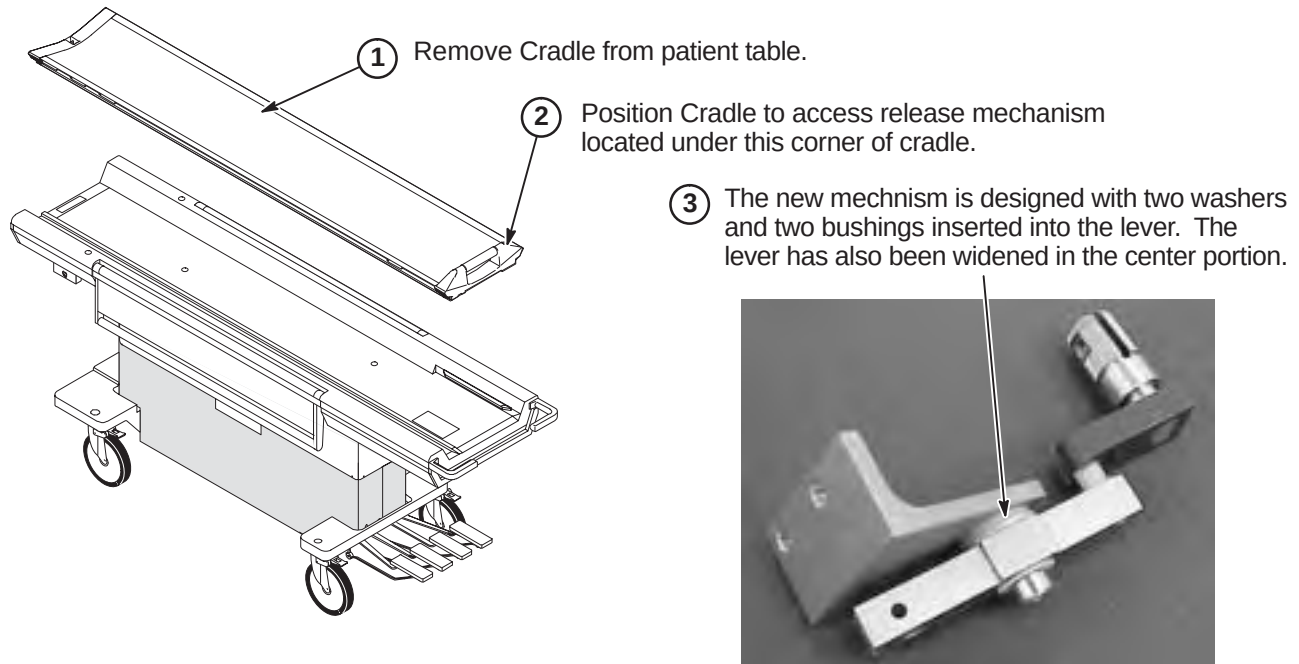
TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
7-1 PATIENT TABLE CRADLE RELEASE UPGRADE	7-2
7-1-1 PREPARE CRADLE FOR UPGRADE	7-2
7-1-2 UPGRADE PROCEDURE	7-2

7-1 PATIENT TABLE CRADLE RELEASE UPGRADE

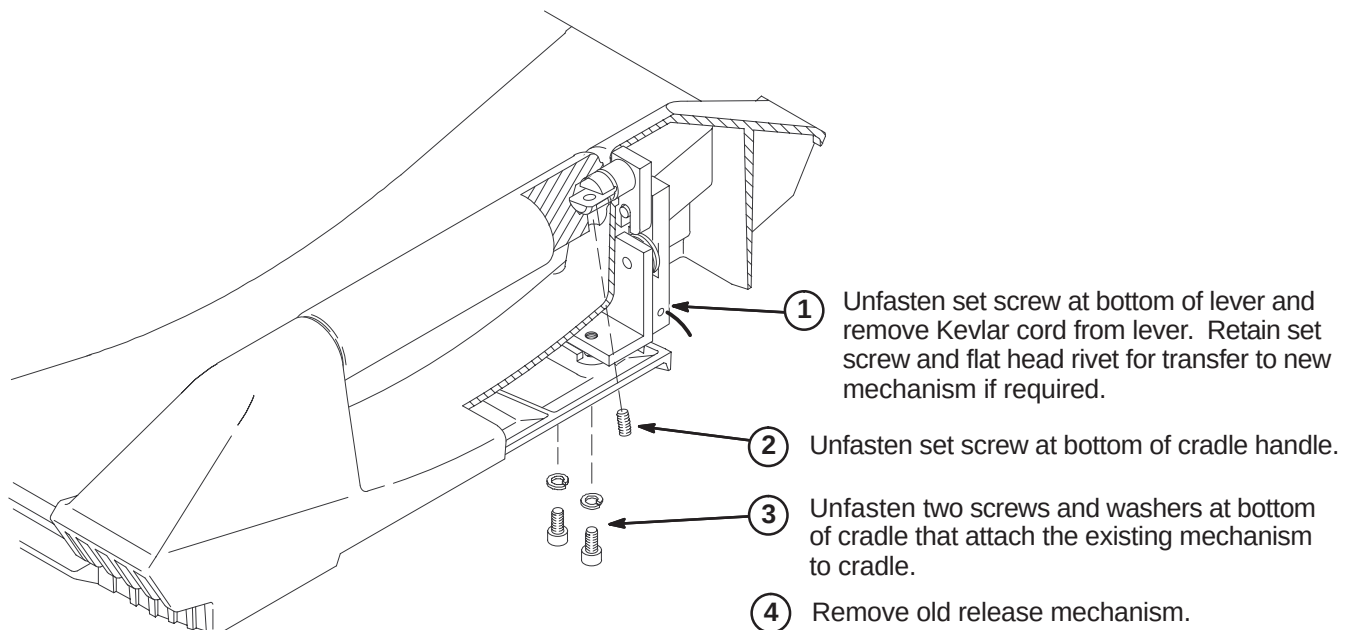
This upgrade is performed to improve the release mechanism and improve image quality.

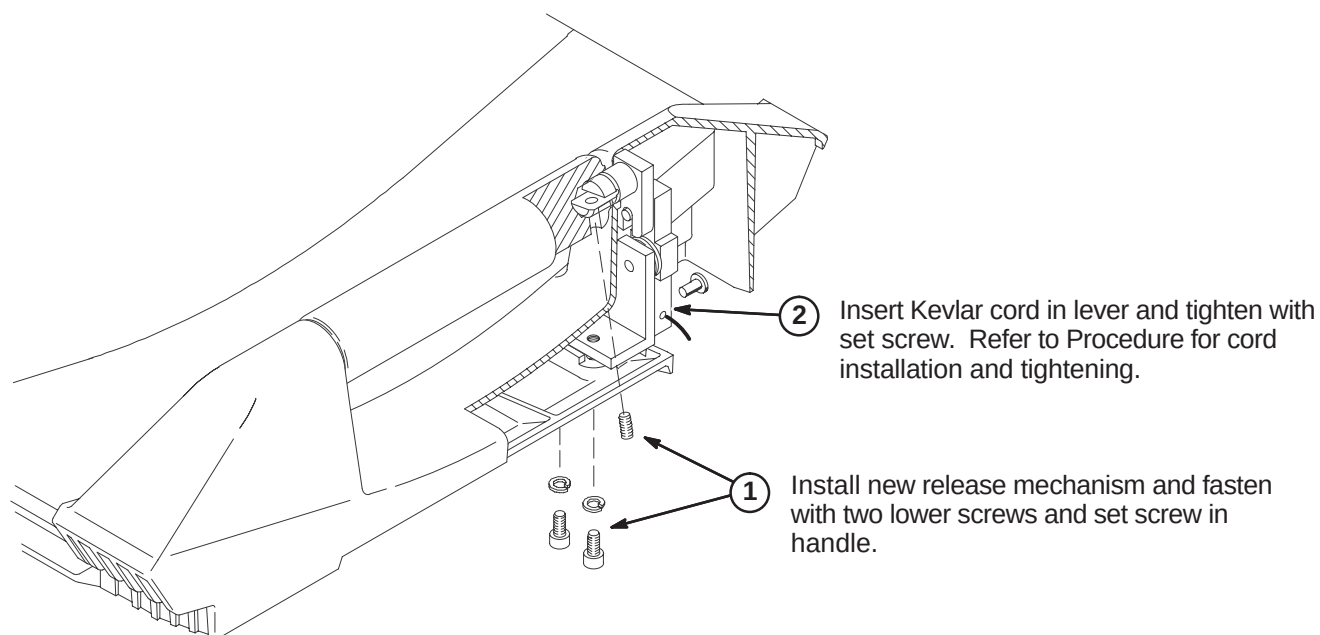
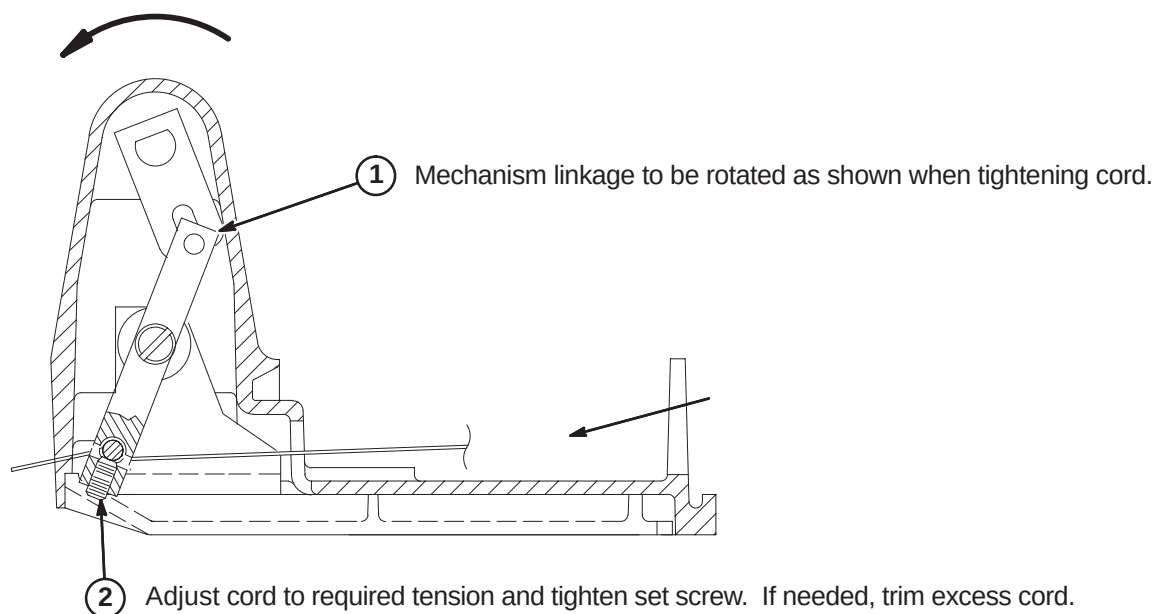
7-1-1 PREPARE CRADLE FOR UPGRADE



7-1-2 UPGRADE PROCEDURE

7-1-2-A Remove Existing Release Mechanism



7-1-2-B Install New Release Mechanism**7-1-2-C Attach Cord**

SECTION 8 – OPERATOR WORKSPACE

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
8-1 INTRODUCTION	8-2
8-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER	8-3
8-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS	8-3
8-2-2 POSITION COMPUTER CABINET	8-4
8-2-3 SEISMIC ANCHORING (IF REQUIRED)	8-4
8-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS	8-5
8-2-5 GOC CABINET COVER REMOVAL	8-6
8-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)	8-7
8-2-7 GOC CABLE CONNECTIONS	8-8
8-2-8 POWER CABLE CONNECTIONS	8-9
8-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC	8-10
8-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS	8-10
8-3-2 GOC CABINET COVER REMOVAL	8-11
8-3-3 GOC COMPONENT REPLACEMENT	8-12
8-3-4 INSTALLATION OF NEW GOC COMPONENTS	8-12
8-3-5 GOC COMPUTER REPLACEMENT	8-14
8-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION	8-16
8-5 ROUTE RUN 1323	8-18
8-6 INSTALL NEW OW TABLE TOP COMPONENTS	8-19
8-7 CONNECT BAR CODE READER (If part of site configuration)	8-20
8-8 INSTALL SCIM/KEYBOARD	8-20
8-9 CONNECT CABLES TO COMPUTER	8-21
8-10 FINAL POSITIONING OF GOC CABINET	8-22
8-11 POWER-ON INFORMATION	8-23

8-1 INTRODUCTION

A new Monitor, SCIM, and Keyboard will replace existing components on the Operator Workspace Table. Also, the new Temperature and Humidity kit will be installed along with the new Z400 computer.

A new Z400 computer is also installed for this upgrade. Depending on site requirements, there are two options for this computer upgrade. They are:

- A replacement of the existing GOC with a new GOC containing the Z400 computer. Refer to Section 8-2.
- A replacement of only the existing computer with the new Z400 computer. Refer to Section 8-3.

After completing one of the above options, continue the Operator Workspace upgrade with Section 7-4.

8-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER

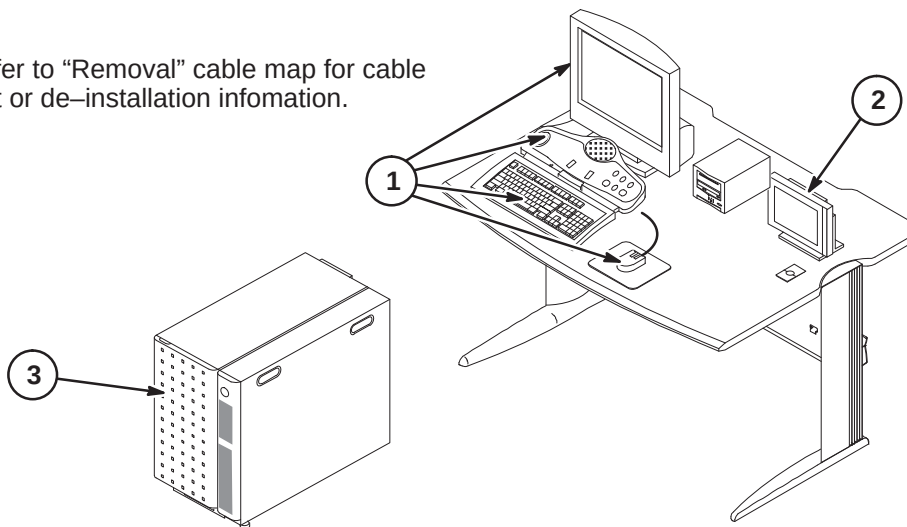
8-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

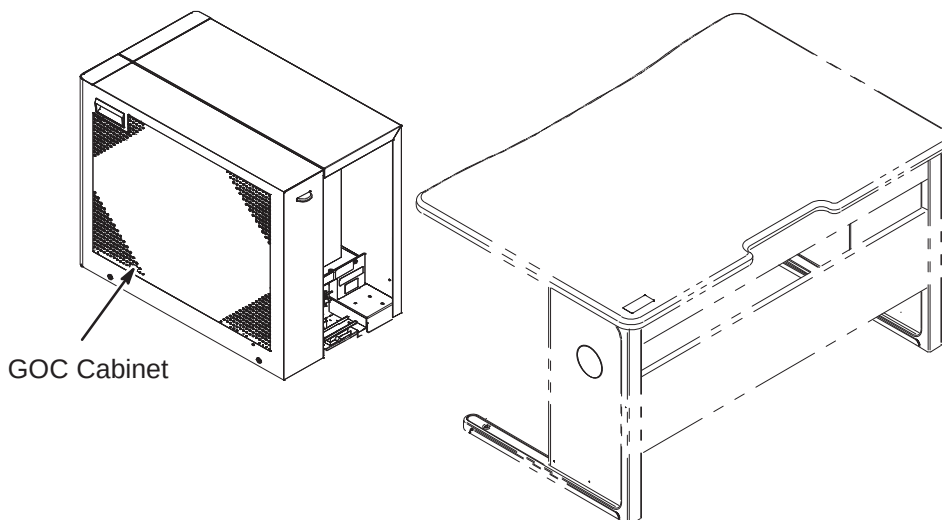
- 1 Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.
Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.
- 2 De-install LCD Display, Runs 1068 and 1069, and return for recycling.
- 3 Existing GOC will be replaced by a new GOC containing the Z400 computer:

Note: Refer to "Removal" cable map for cable retainment or de-installation information.



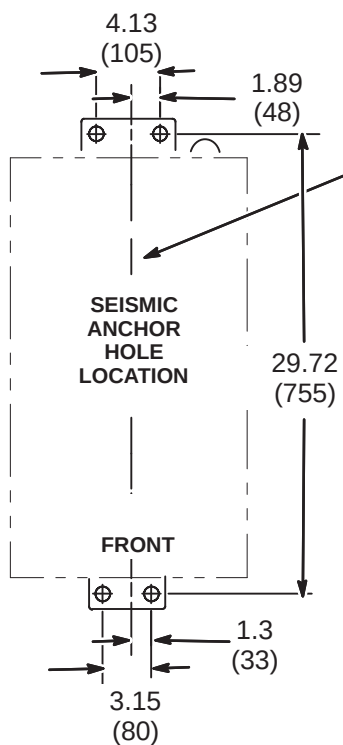
8-2-2 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



8-2-3 SEISMIC ANCHORING (IF REQUIRED)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

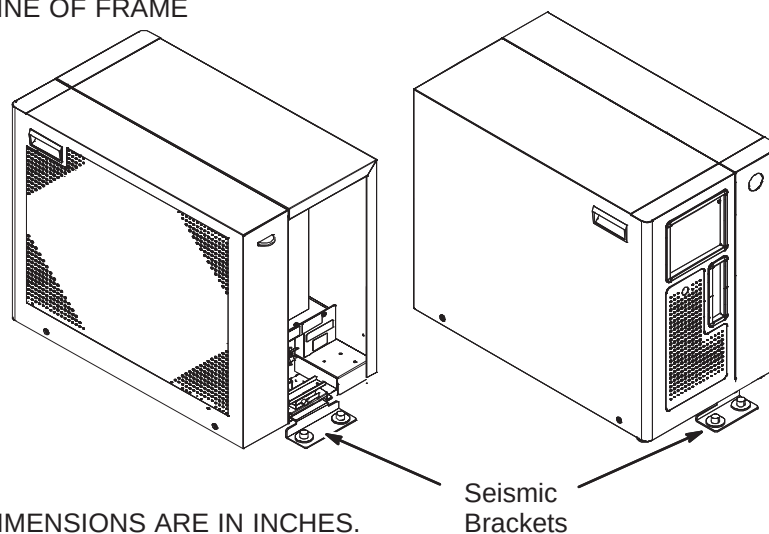


If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets

CENTERLINE OF FRAME

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



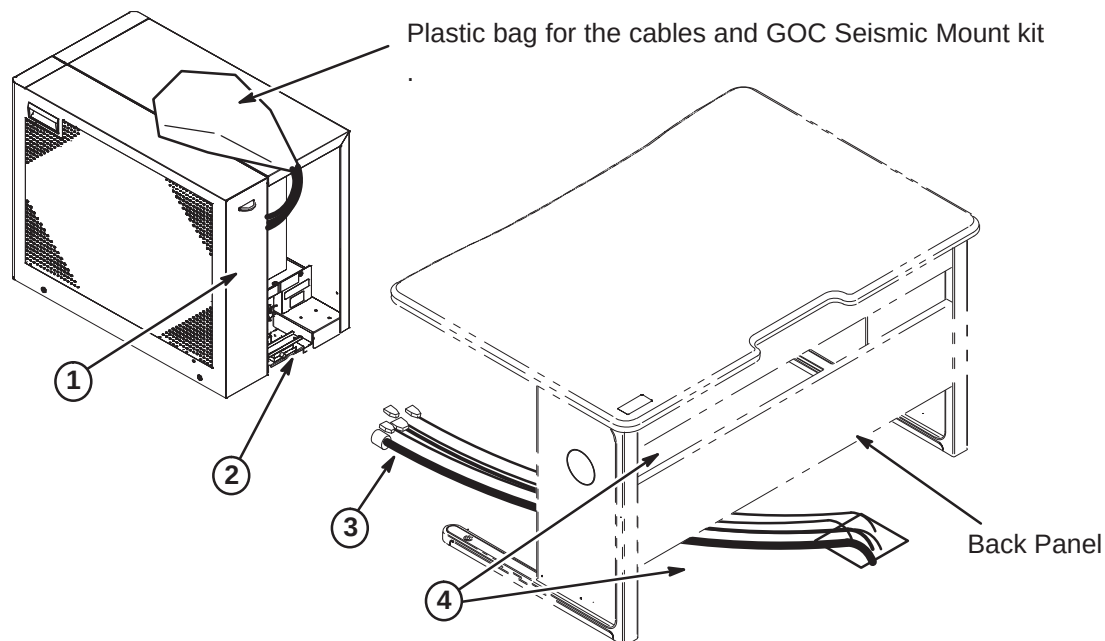
8-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- SCSI output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

- 1 Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
- 2 Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
- 3 Route all cables before connecting any cables. Refer to cable map for connection information.
- 4 If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



8-2-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

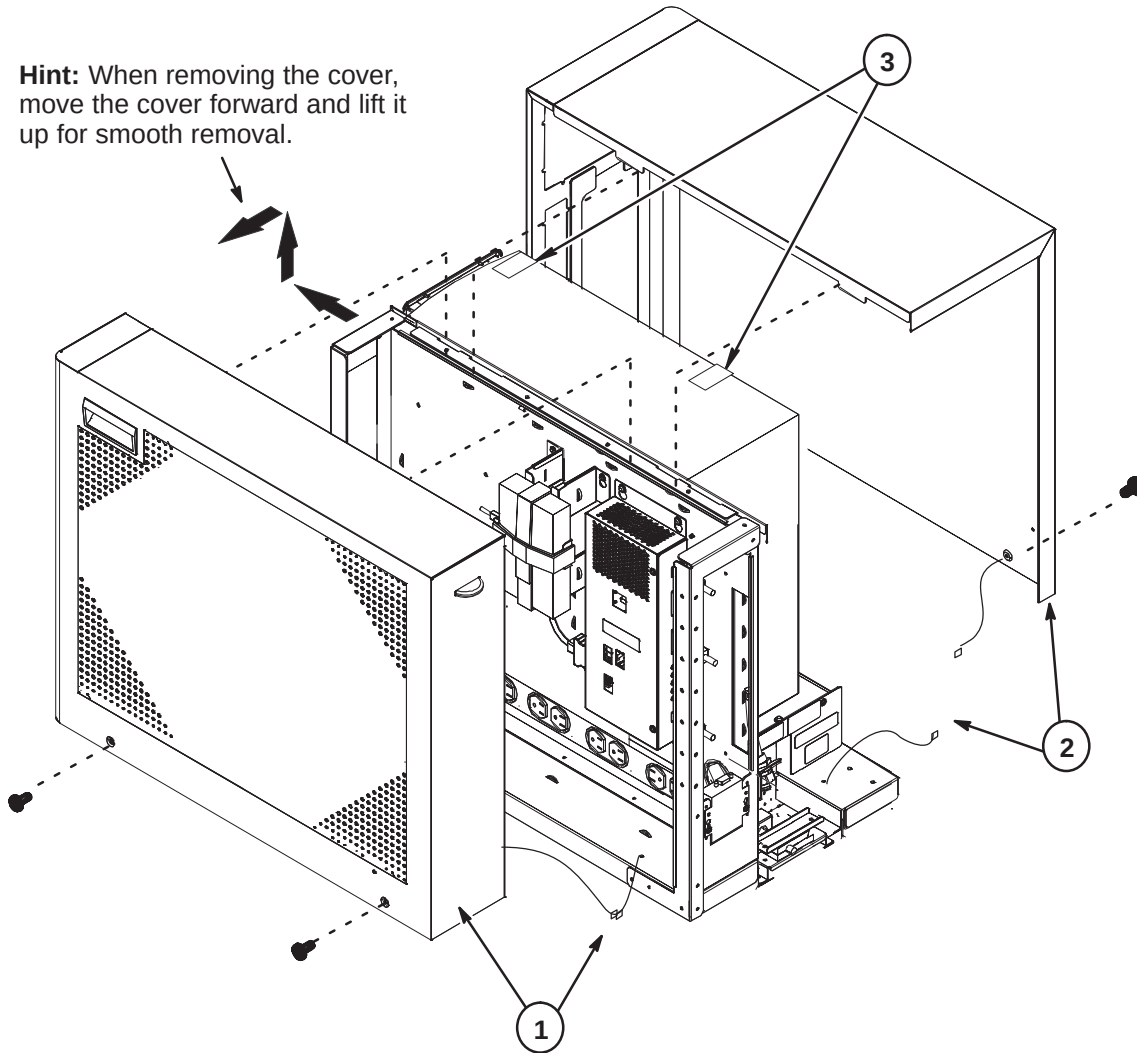
- 1 For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
- 2 For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

- 3 Peel the two shipping tapes off from PC.

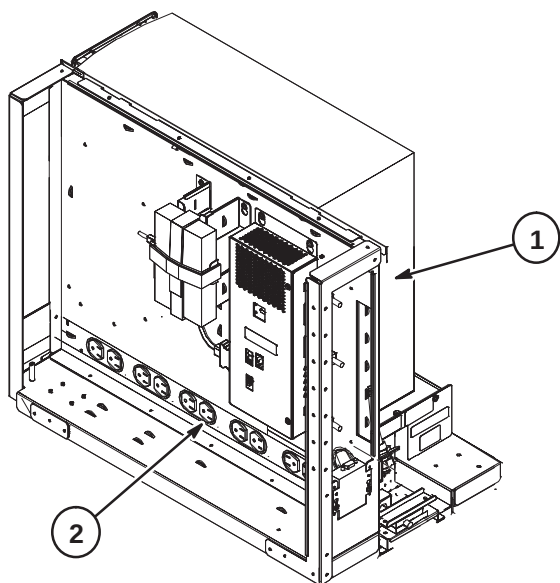
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



8-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

- 1 Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
- 2 Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



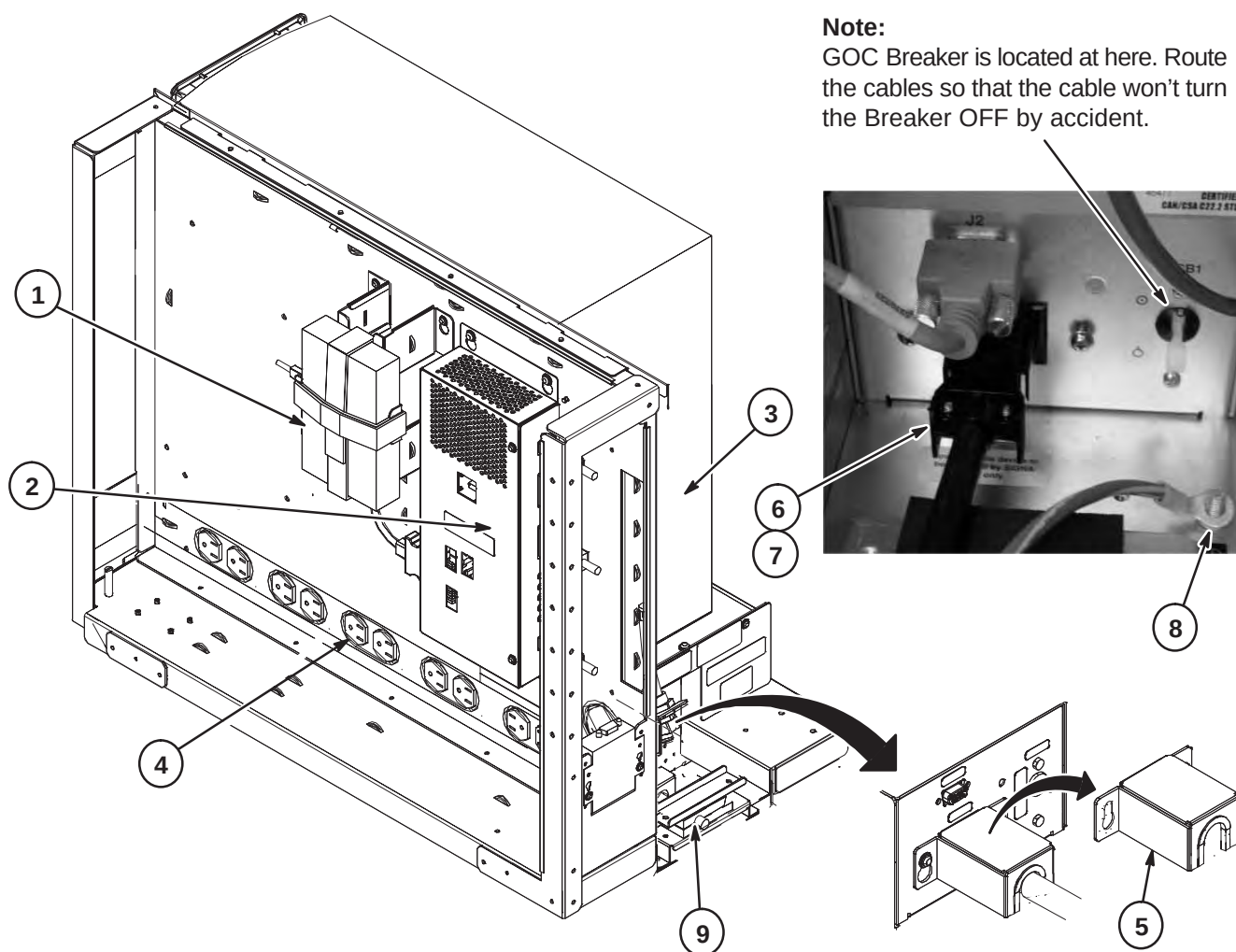
8-2-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

- 1 Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

- 2 Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
- 3 Refer to Section 8-2-9 for all cables connected to Linux PC computer.
- 4 Refer to Section 8-2-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
- 5 Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
- 6 Route and connect GOC power cable Run 1381 to J1 on PDU.
- 7 Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5 after connecting cable and tighten screws.
- 8 Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
- 9 Tighten clamp to secure cables in place.

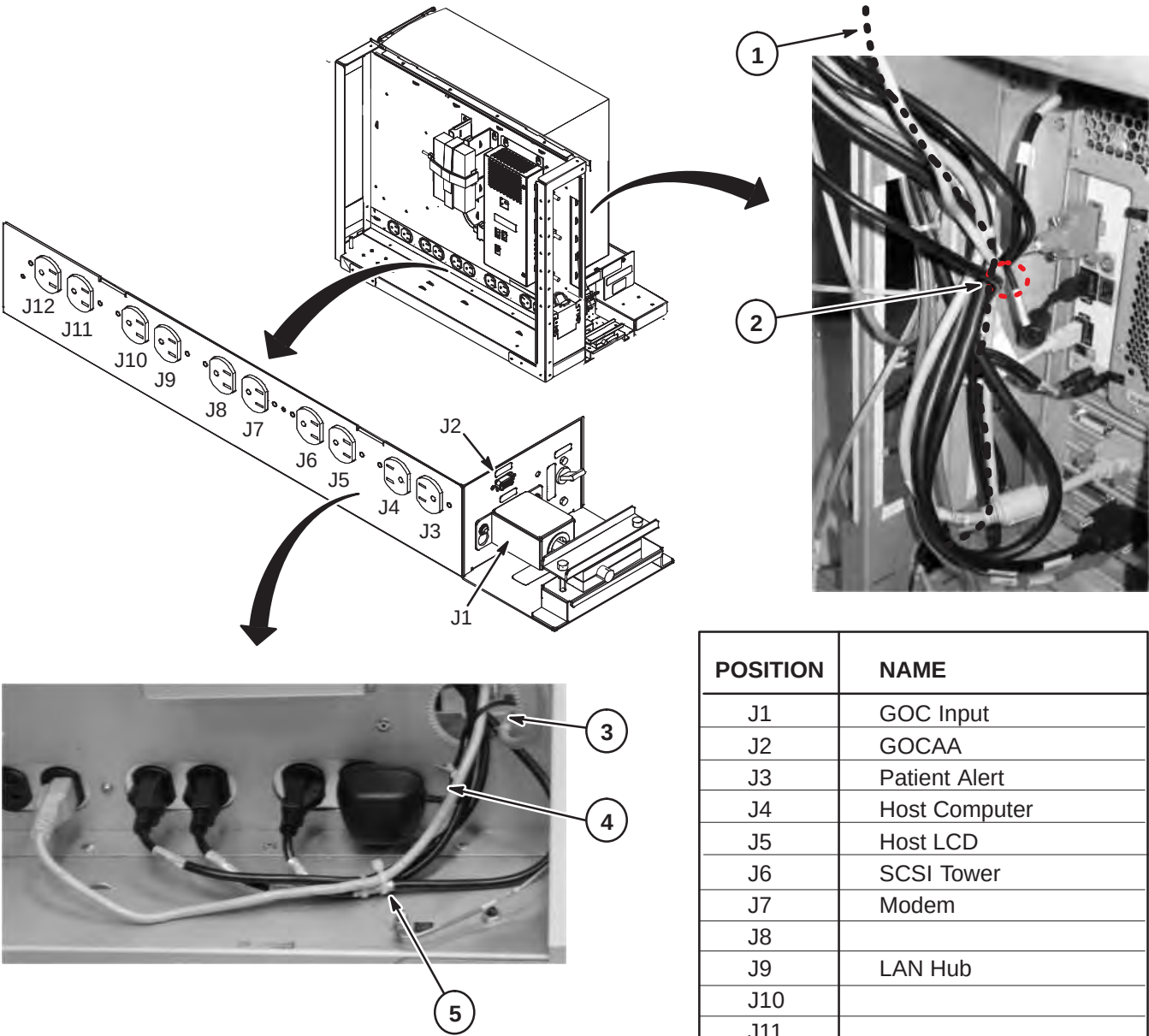


8-2-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1 If Modem is installed, route power cable and connect to J7.
- 2 Tie-wrap point for Modem power cable.
- 3 Route power cables connected to J3 thru J7 through access hole.
- 4 route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5 Tie-wrap power cables J4 to J7 at this point.



8-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC

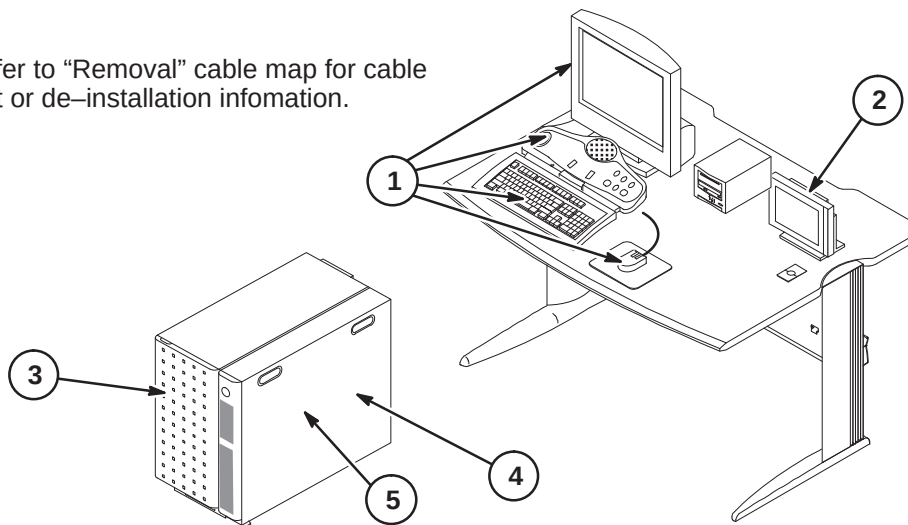
8-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

- 1 Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.
Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.
- 2 De-install LCD Display, Runs 1068 and 1069, and return for recycling.
- 3 Existing PC computer in the GOC will be replaced by a new computer:
- 4 The GOC AA module has been delivered as part of the upgrade kit.
 - If the existing GOC already has a GOC AA module installed, the new GOC AA is not required.
 - If the existing GOC has a WIM installed, it must be replaced by the new GOC AA module.
- 5 An 8-Port 10/100/1000 Mbps Gigabit Ethernet Switch has been delivered as part of the upgrade kit.
 - If the existing GOC already has this Switch installed, the new Switch is not required.
 - If the existing LAN Hub Switch is an 8-Port 10/100, then it must be replaced by the new 10/100/1000 Switch.

Note: Refer to "Removal" cable map for cable retainment or de-installation information.

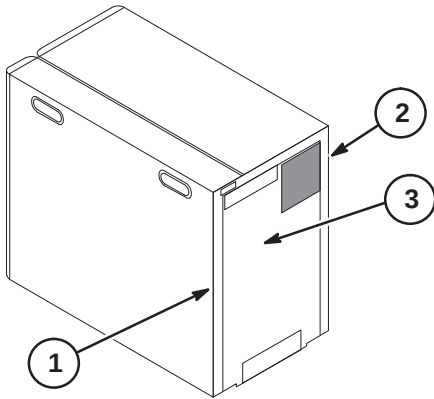


8-3-2 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

8-3-2-A OLDER STYLE GOC COVERS

- 1 For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
- 2 For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
- 3 Loosen four screws holding the back cover. Lift and remove cover.

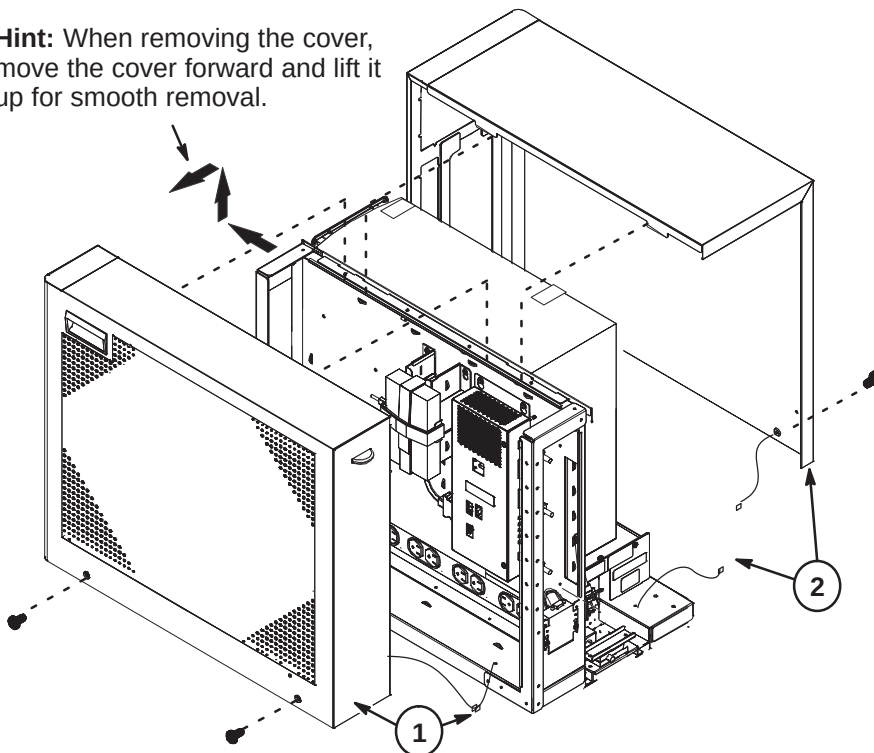


8-3-2-B NEWER STYLE GOC COVERS

Note: A ground wire is connected to each side cover. Use care when removing side covers.

- 1 For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
- 2 For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

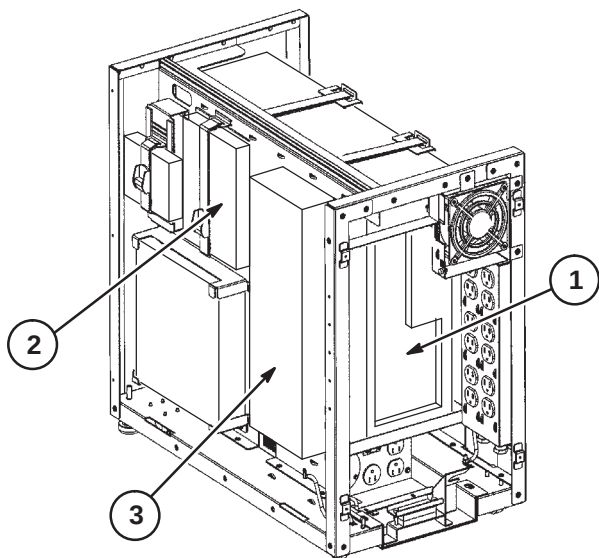
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



8-3-3 GOC COMPONENT REPLACEMENT

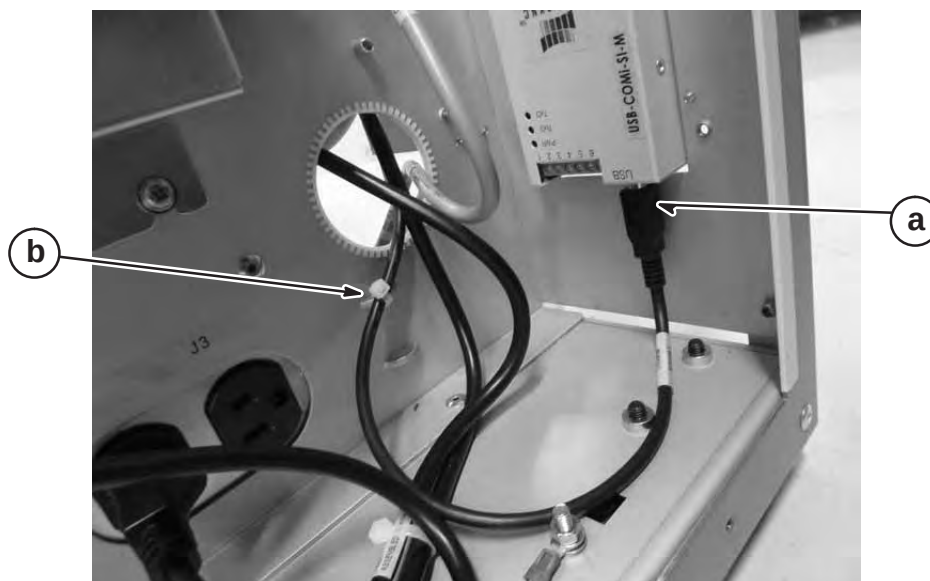
Refer to applicable cable removal map for de-installation of existing cables that will be replaced by new cables. The cable maps also show which cables will be disconnected from existing components and reconnected to new components.

- 1 Disconnect all cables from existing computer. Make sure they are properly labeled.
- 2 If required, replace existing LAN Hub Switch with new 10/100/1000 Mbps Gigabit Ethernet Switch (5139480-2).
- 3 If required, replace existing WIM with new GOC AA module (2395043).



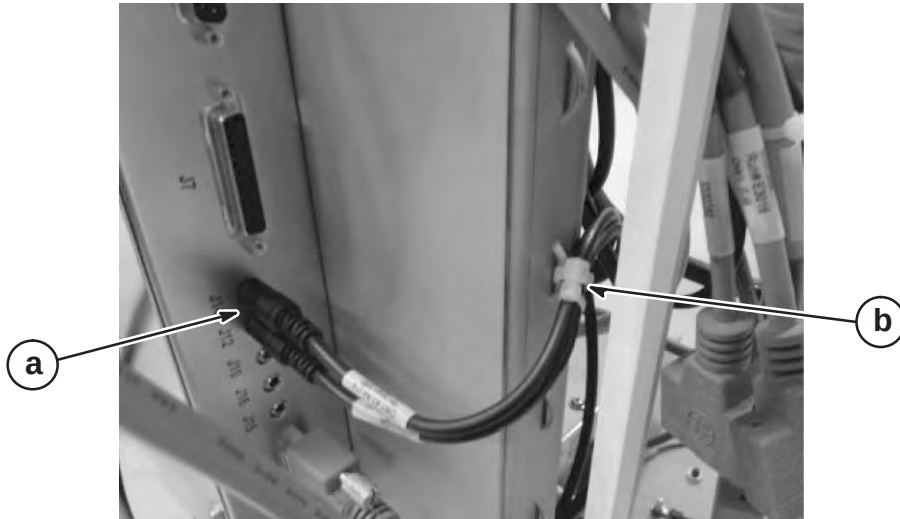
8-3-4 INSTALLATION OF NEW GOC COMPONENTS

- 1 Replace the Host to USB-Serial Converter USB Cable.
 - a. Reconnect the Host to USB-Serial Converter USB Cable.
 - b. Cut and restore tie-wrap(s) as required during cable replacement.

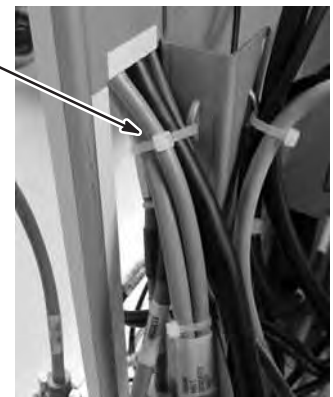
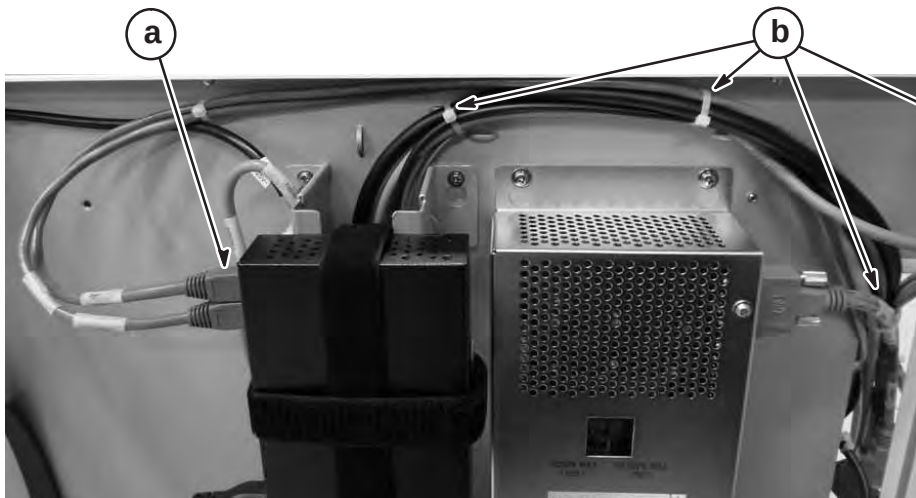


8-3-4 INSTALLATION OF NEW GOC COMPONENTS (Continued)**2 Replace the Audio Cable Assembly.**

- a. Reconnect the Audio Cable Assembly, connected to J12 and J14 on GOCAA .
- b. Cut and restore tie-wrap(s) as required during cable replacement.

**3 Replace the Host-Hub Ethernet Cable.**

- a. Replace the Ethernet Cable between the Host and Port 1 of Ethernet Switch Hub.
- b. Cut and restore tie-wrap(s) as required during cable replacement.



View from GOC rear side.

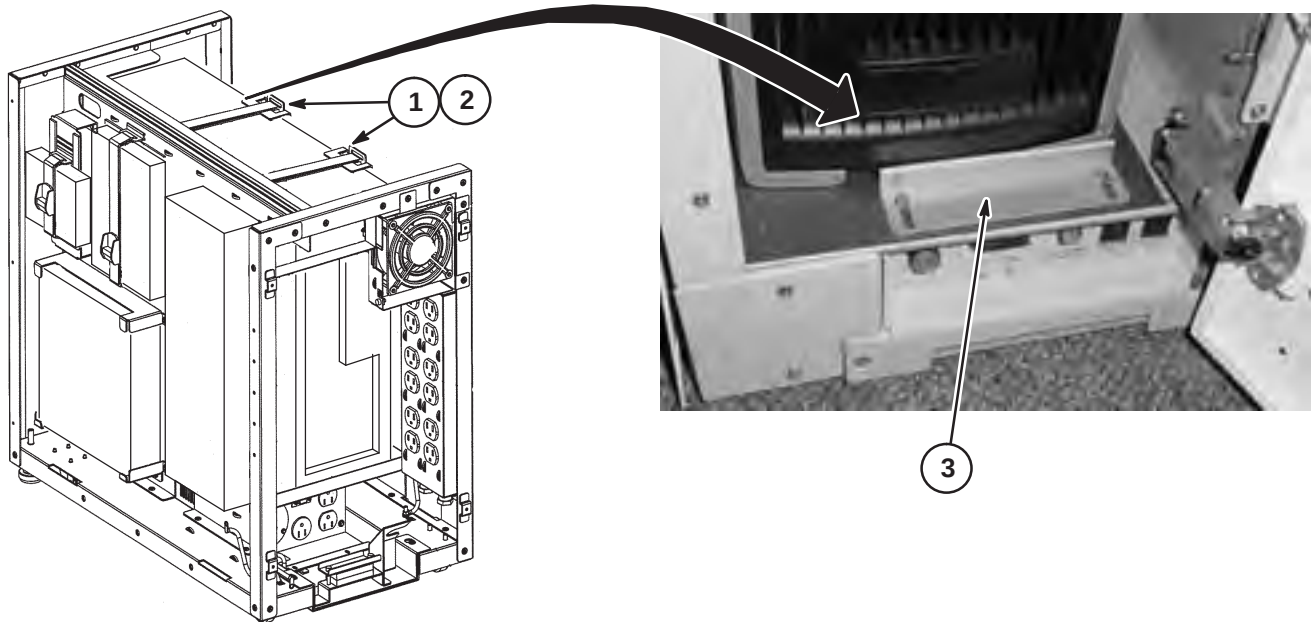
8-3-5 GOC COMPUTER REPLACEMENT**IMPORTANT!**

XW8000, 8200, 8400, or 9300 PC internal hardware, such as any hard drive, is not compatible with the Z400 computer. Do not attempt to transfer any internal hardware between any of these and the Z400 computer.

CAUTION

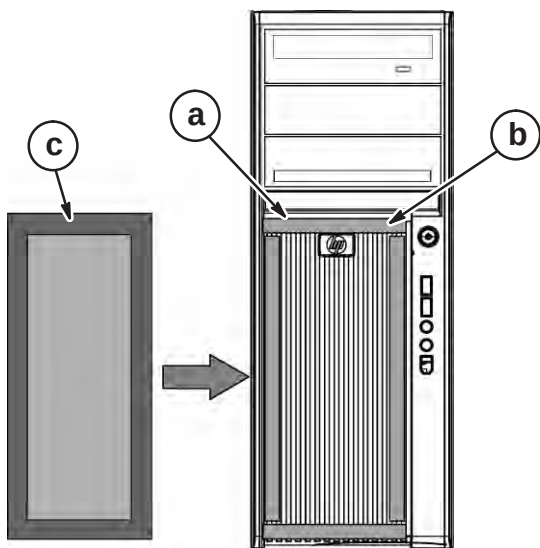
The computer weighs about 50 pounds (23Kg). This requires two people to handle the computer.

- 1 Unfasten straps and carefully remove computer.
- 2 Because the new Z400 computer is shorter than the original PC being replaced, install the new computer to the rear-most position and secure in place with straps.
- 3 If required, install PC Stopper Bracket (5127954) at front of computer using existing screws.



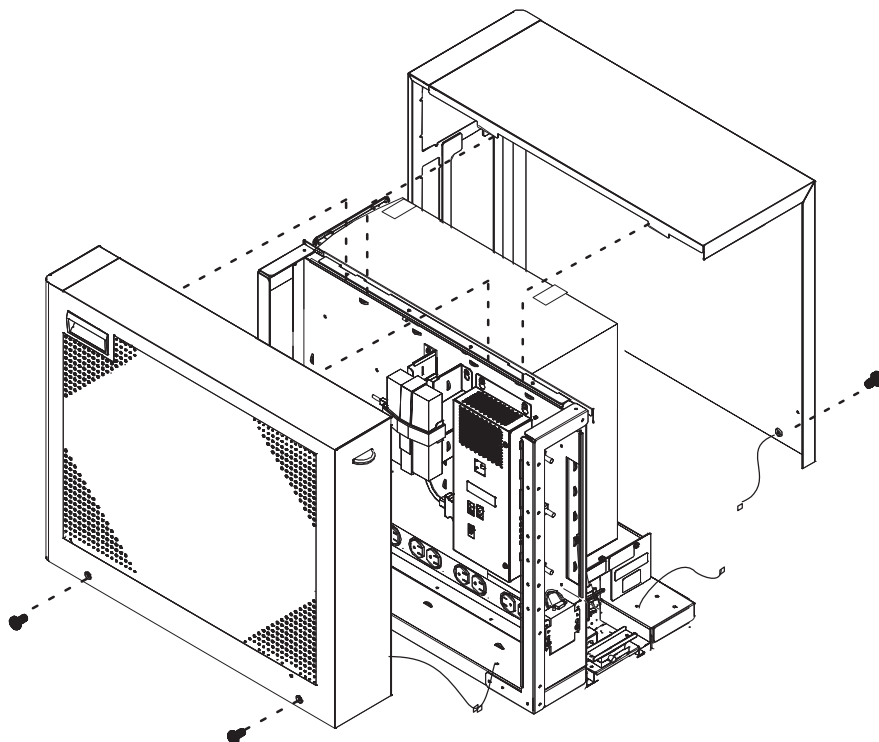
8-3-5 GOC COMPUTER REPLACEMENT (Continued)

- 4 If applicable, attach the Dust Filter to the front panel after following the cleaning instructions below.
- Clean the velcro area of front panel with alcohol and cloth. Peel the velcro from backside of filter.
 - Attach the velcro around front panel.
 - Attach the filter to front panel.



IMPORTANT! Make sure the Z400 computer is installed at the rear-most position before installing covers. This position will aid in connecting existing cables to rear of Z400 computer.

- 5 Reinstall the GOC covers,.



8-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION

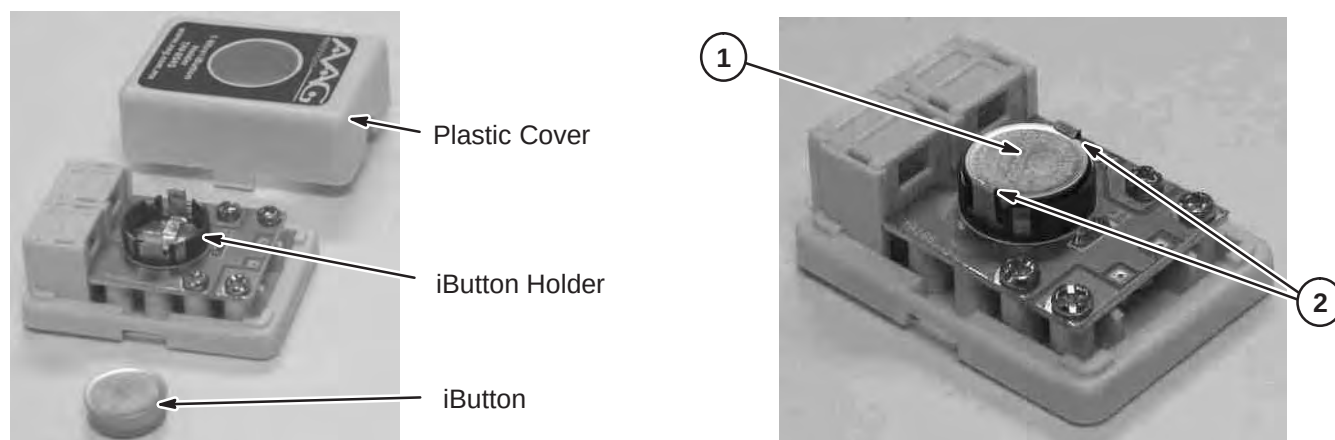
The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

8-4-1 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

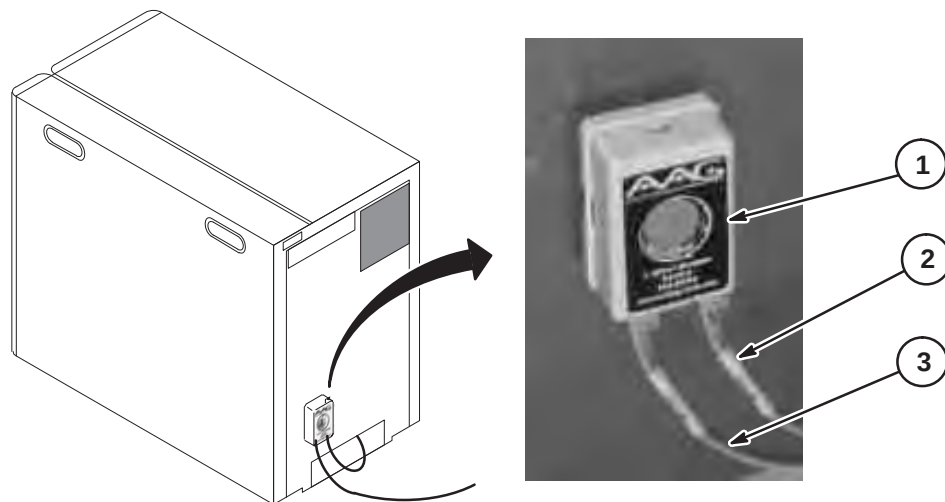
Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

- 1 Install the iButton in the holder so that the solid part of the case is exposed.
- 2 Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
- 3 If removed, reinstall the Plastic Cover.



8-4-2 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

- 1 Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
- 2 Connect cable Run 1321 (5143794-2) to Sensor and route other end thru bottom opening of GOC and connect to USB to 1-Wire Adapter (5307784) installed in GOC USB port. Refer to step 1 in Section 8-2-9.
- 3 Route and connect Run 1322 (5143794-2) from System cabinet to Sensor at back of GOC.



8-4-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

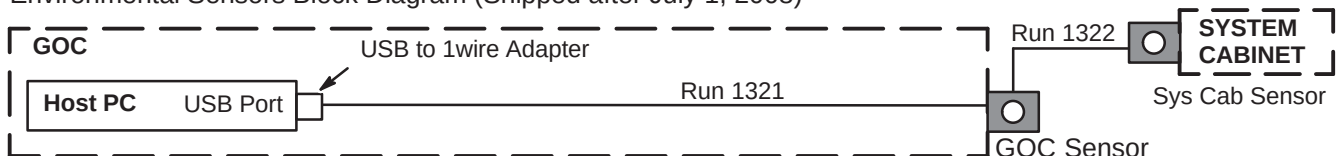
Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

- 1 Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
- 2 Route and connect cable Run 1322 (5143794) from GOC rear cover.

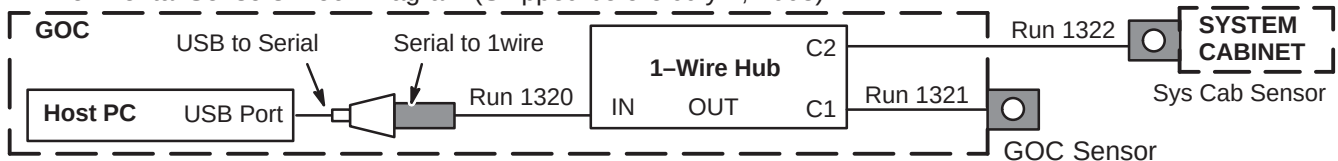


8-4-4 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after July 1, 2008)



Environmental Sensors Block Diagram (Shipped before July 1, 2008)

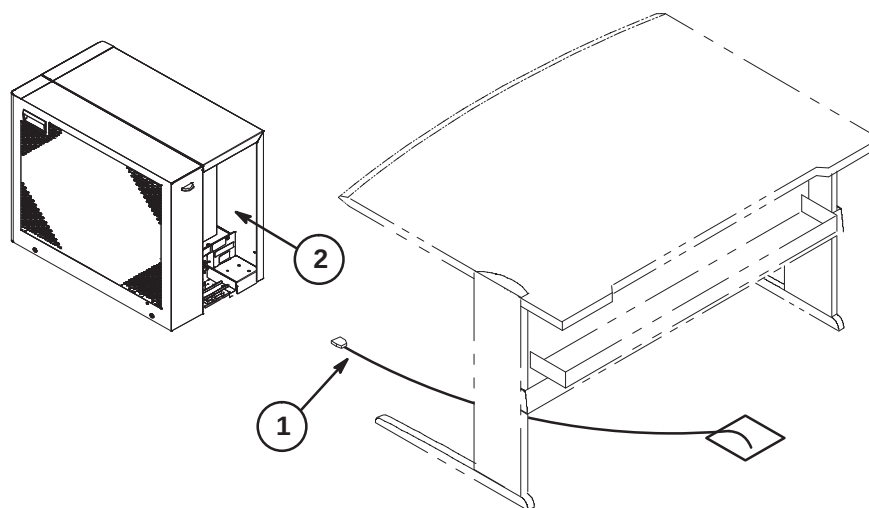


8-5 ROUTE RUN 1323

- 1 Route Run 1323 from RFS or System Cabinet to location of GOC. This is a new CAT6 cable that must be installed. Existing CAT5 cables cannot be reused.

NOTE: For optimal efficiency, route Run 1322, Section 8-4-3, at the same time as Rune 1323. It is also routed between RFS/System Cabinet and GOC.

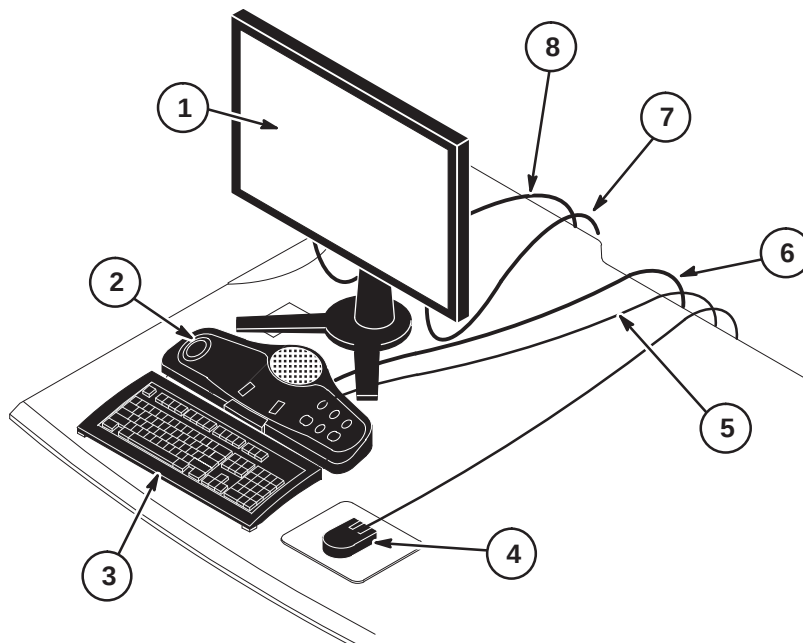
- 2 Cable will be routed thru the lower rear opening.



8-6 INSTALL NEW OW TABLE TOP COMPONENTS

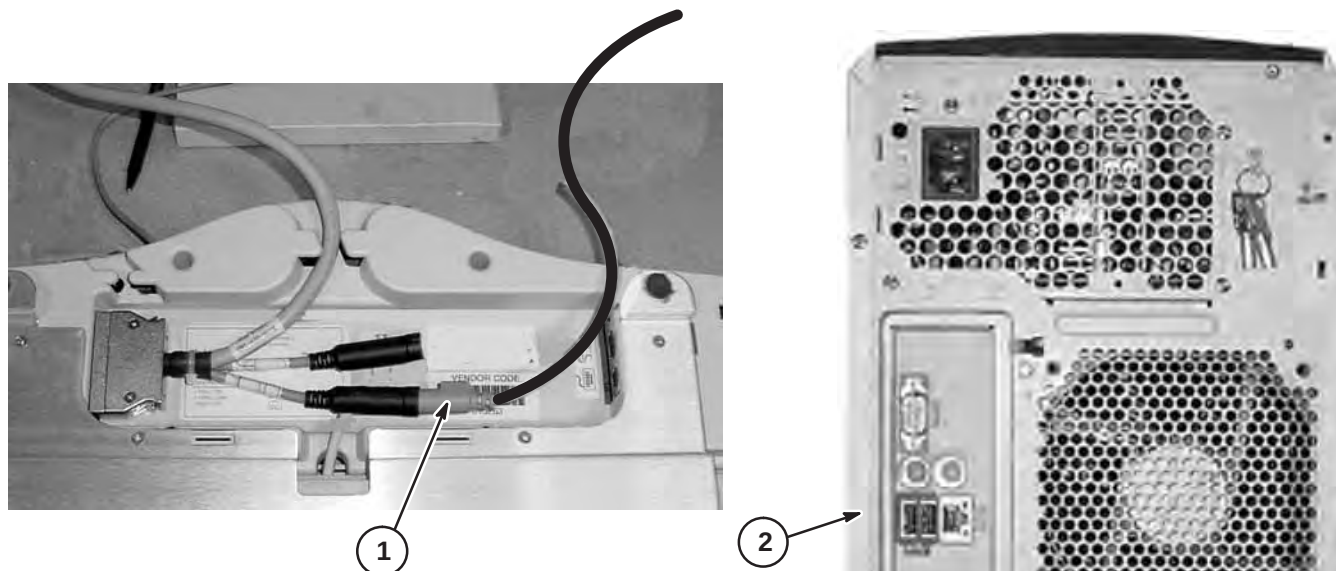
- 1 Install new 23 Inch Monitor on table.
- 2 Place new SCIM (5148388-x) on table.
- 3 Place new Keyboard (5143798-x) on table.
- 4 Install new Mouse (5143798-6) on table and route USB cable to computer in GOC. Attach provided extension USB cable if necessary.
- 5 Connect extension cable Run 2038 to USB Keyboard cable and route to computer in GOC.
- 6 Connect Run 2053 (or 1256) to SCIM and route to rear of GOC.
- 7 Connect provided monitor power cable to monitor. Connect other end of cable to Run 1065, which is connected to OW power strip.
- 8 Connect DVI cable, Run 2037, to monitor and route to rear of computer in GOC.

Note: Do not use or connect provided VGA cable.

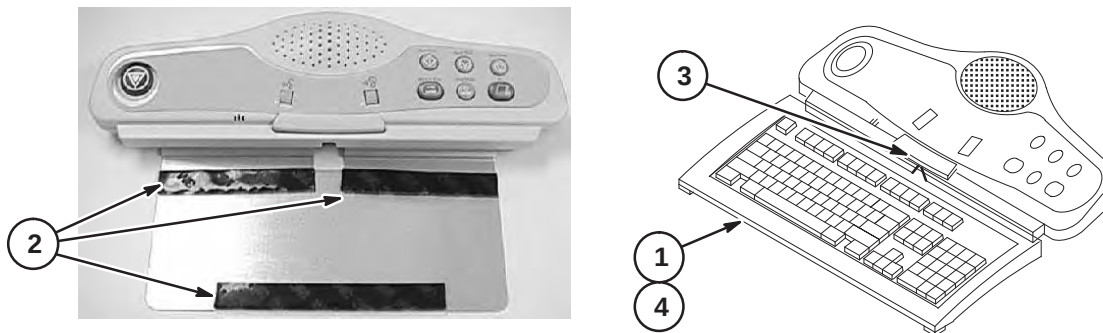


8-7 CONNECT BAR CODE READER (If part of site configuration)

- 1 If Run1256 is present – Disconnect the Bar Code Reader cable from Run 1256 Y–connector under SCIM.
- 2 Connect cable Run 1379 to rear of computer and Bar Code Reader.

**8-8 INSTALL SCIM/KEYBOARD**

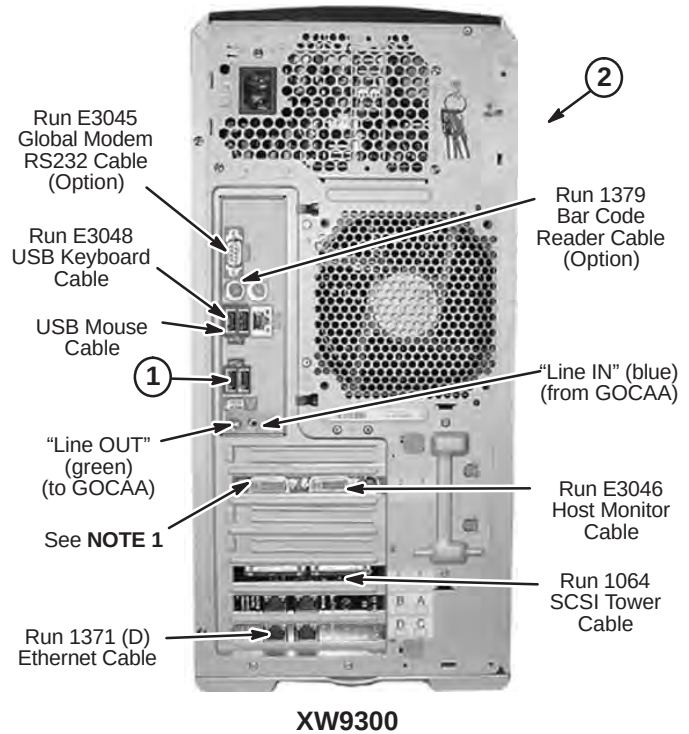
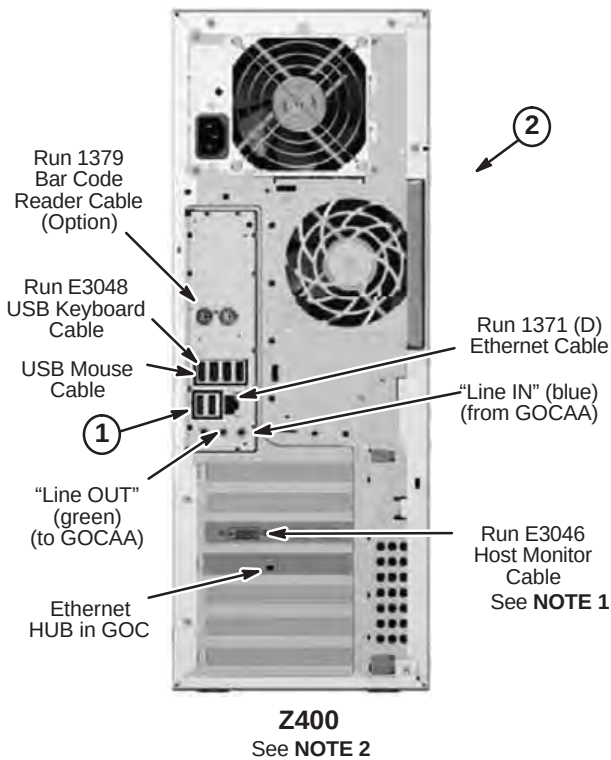
- 1 Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
- 2 Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
- 3 Guide the cable back into the center notched opening until all the cable is under the SCIM base.
- 4 Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

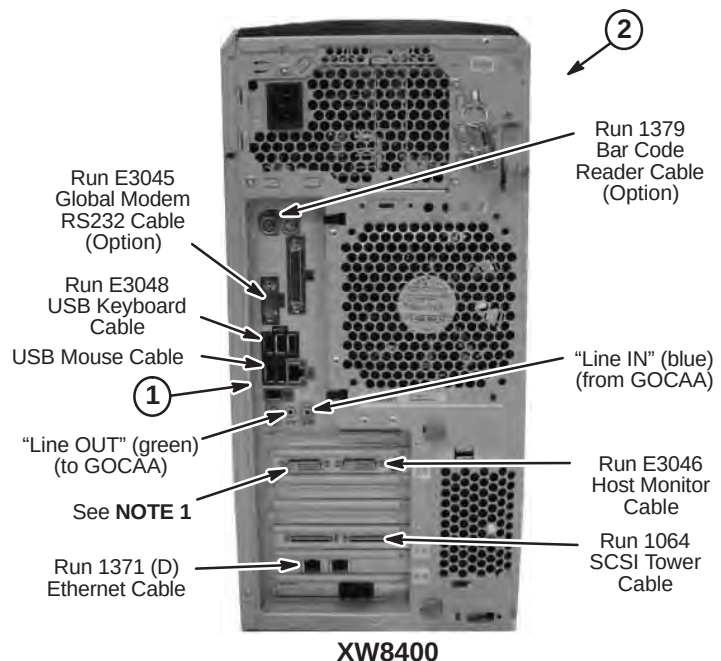
8-9 CONNECT CABLES TO COMPUTER

- 1 Refer to Section 7-4 for installing the USB to 1-Wire Adapter (5307784) to USB port.
- 2 Connect remaining cables as marked at rear of computer.



NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:
SCSI Tower and DASM are not supported for Z400PC.



8-10 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

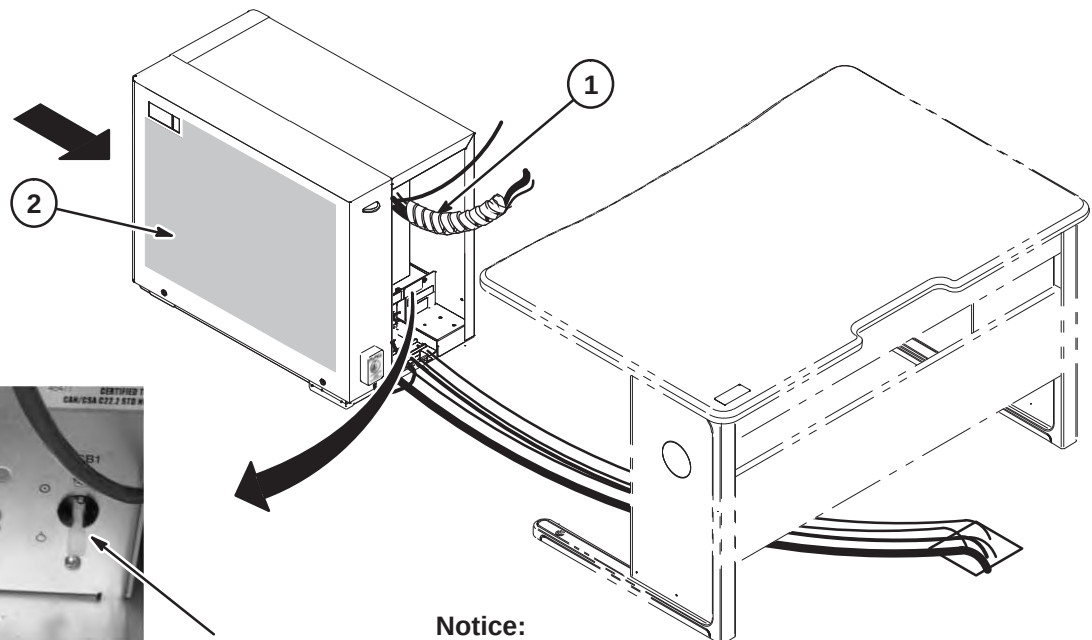
- 1 Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m).

Notice

Do Not include Modem power cable in Spiral Tube.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (if applicable)

- 2 Place cabinet in final position as determined by customer.



Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.



8-11 POWER-ON INFORMATION

- 1 Remove LOTO from the PDU.

IMPORTANT

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the GOC AA and Modem power.

- 2 Restore power to the GOC, boot software, and ensure proper system operation.

NOTE: If necessary, reconnect the optional UPS accessory and ensure power is restored.

- 3 Perform the applicable Loading Host System Software procedure to load software and applications onto the operating system.

NOTE: The Load from Cold must be performed to properly transfer customer options

- 4 Update the elicense with the new Host ID.
- 5 Repackage the replaced computer for return
- 6 Perform a scan to ensure the system and applications are working properly.

SECTION 9 – SYSTEM CABINET

TABLE OF CONTENTS

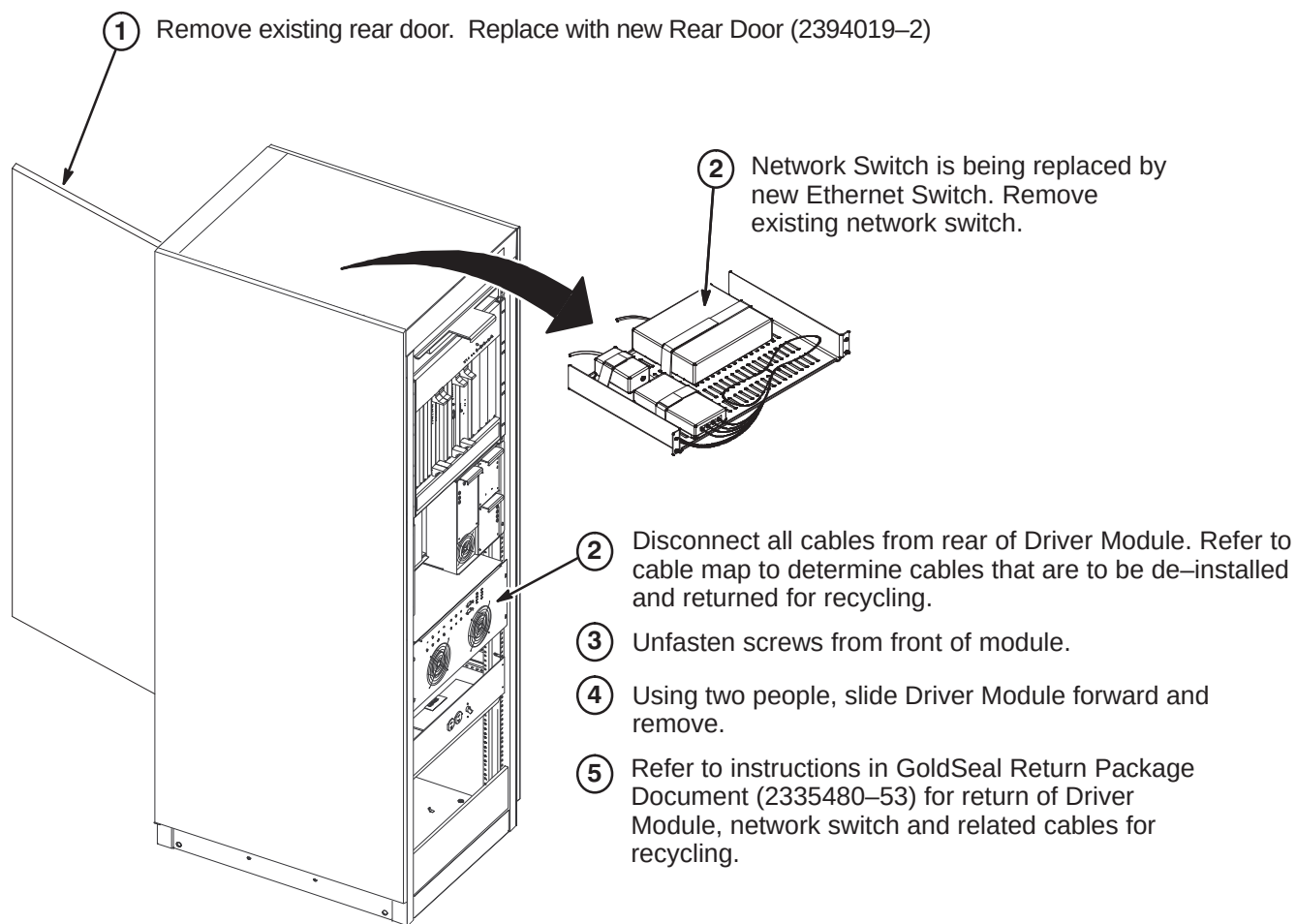
<u>SECTION</u>	<u>PAGE</u>
9-1 INTRODUCTION	9-2
9-2 PREPARE SYSTEM CABINET FOR UPGRADE	9-2
9-3 INSTALL NEW DRIVER MODULE	9-3
9-4 INSTALL COAX POWER SPLITTER	9-3
9-6 INSTALL NEW INTERFACE PANEL AND CONNECT CABLES	9-5
9-7 UPGRADE BOARDS IN MGD CHASSIS	9-6
9-8 INSTALL VRE COMPONENTS	9-7
9-8-1 INSTALL VRE MODULE RAILS	9-7
9-8-2 INSTALL VRE MODULES	9-8
9-8-3 INSTALL VRE CABLES	9-9
9-9 PROCEDURES FOR UPGRADING TO 16 CHANNEL (M3000PE)	9-10
9-9-1 REMOVE EXISTING COMPONENTS FROM RRF CHASSIS	9-10
9-9-2 INSTALL NEW MODULES IN RRF CHASSIS	9-10
9-9-3 INSTALL CABLES AT REAR OF RRF CHASSIS	9-11
9-9-4 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS	9-11
9-10 INSTALL NEW REAR DOOR AND RATING PLATE	9-12

CAUTION

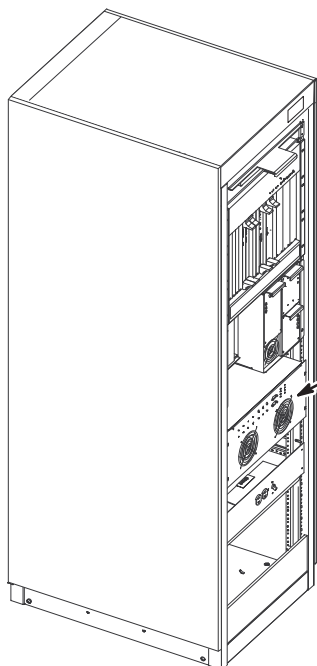
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

9-1 INTRODUCTION

The System cabinet will be modified for the upgrade to HDx Release 14.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

9-2 PREPARE SYSTEM CABINET FOR UPGRADE

9-3 INSTALL NEW DRIVER MODULE

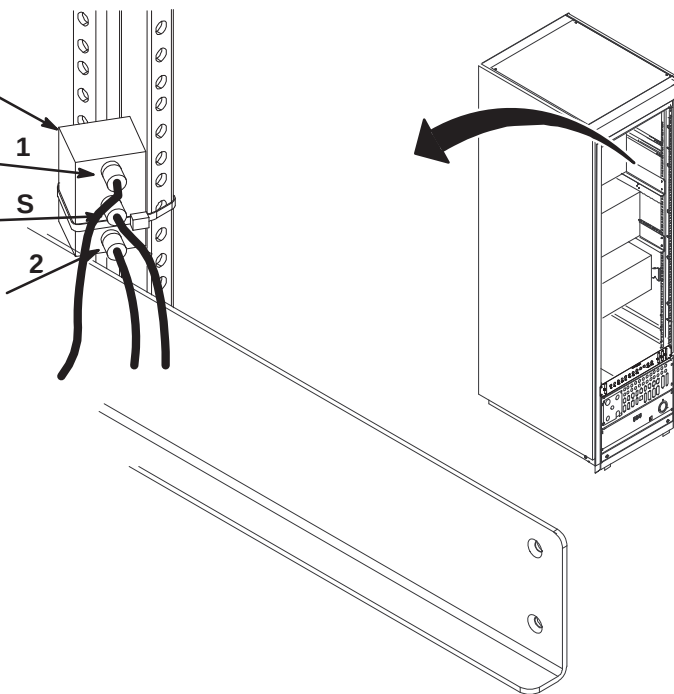


- ① Using two people, slide new Driver Module into position.
- ② Secure in place with previously removed screws.
- ③ Refer to cable maps in Section 4 and table in Section 9-6 for information on connection of cables to Driver Module.

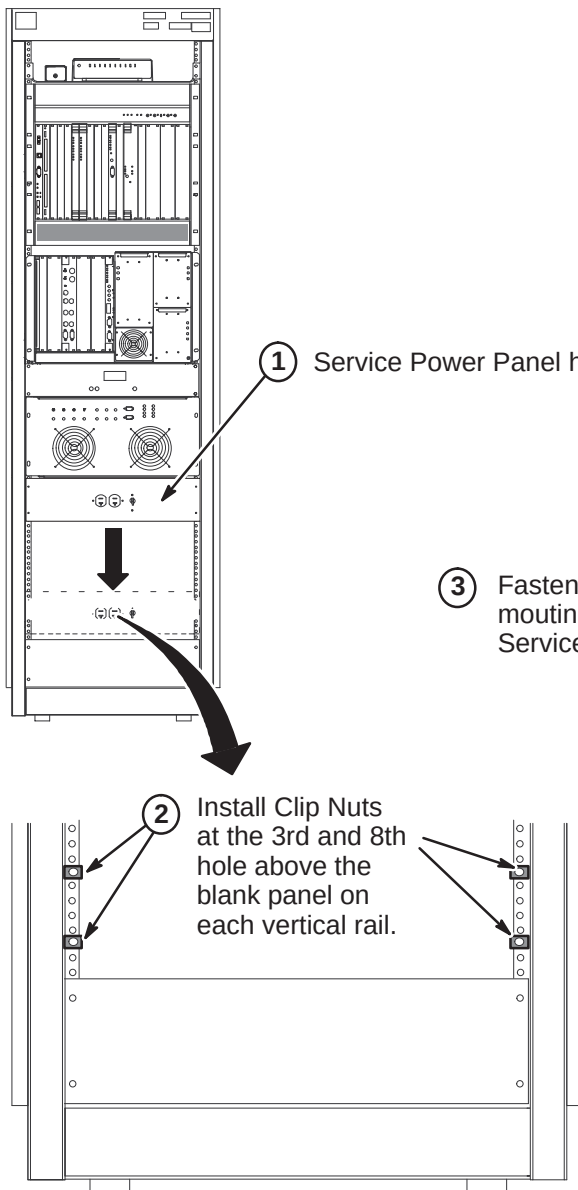
9-4 INSTALL COAX POWER SPLITTER

- ① Install BNC Coax Power Splitter (5125176) on vertical rail.
- ② Connect cable from slot 12R J31 at rear of RRF Chassis to "1" on Splitter.
- ③ Connect cable from Interface Panel J202 to "S" on Splitter.
- ④ Install 50 Ohm BNC Terminator (46-296817P1) on "2" of Splitter if site is NOT being upgraded to a 16 channel system.

If 16 Channel is installed, connect cable from slot 3R J31 at rear of RRF Chassis to "2" on Splitter.



9-5 RELOCATE SERVICE POWER PANEL

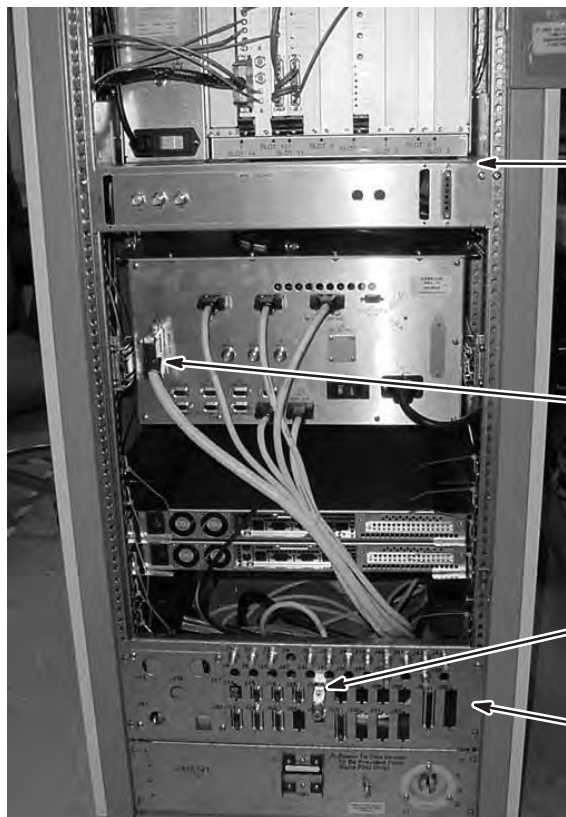


- ④ Cut tie-wraps holding Service Power Panel wires to vertical rail, re-position cables and tie-wrap as shown.

- ③ Fasten the Panel in place using the same mounting hardware. There will be a gap between Service Power Panel and the blank panel below.



9-6 INSTALL NEW INTERFACE PANEL AND CONNECT CABLES



- ① At rear of cabinet, install 3.0T upgrade Interface (I/O) Panel Assembly (5151361-2). Align top of Panel to with bottom of RRF Chassis.
 - ② Fasten with four (4) M5 x 16mm Pan head screw (2279742-2).
 - ③ Reconnect door ground wire.
 - ④ Make sure cable routed from Driver Module J5 to Interface Panel J61 has been disconnected and removed. New cable Run 1330 will connect directly to J5.
 - ⑤ Use table below for connecting cables in the System Cabinet. All the cables connected to the new I/O Panel are pre-installed on the I/O Panel.
- Note:** If ACGD Cabinet is upgraded to HFD-S, and Run 1105 is installed and connected to J56, move CAN Terminator to J8 on HFD-S Cabinet.
- ⑥ Refer to "added" cable map for connection of cables to System Cabinet Interface Panel.

The following table provides information for connecting existing and new cables within the System Cabinet.

CABLE PART #	ONE END	OTHER END
2363363	Driver Module J1	I/F Panel J56
2323878-3	Driver Module J2	RF Slot 10F J23
5117295	Driver Module J3	MGD Slot 11R J21
2374675-3	Driver Module J6	I/O Panel J55
2374675-2	Driver Module J7	I/O Panel J53
2394870-3	RF Slot 14R J5	I/O Panel J200
2394870-4	RF Slot 5R J5	I/O Panel J201
5119896	Splitter S RF Slot 14R J13	I/O Panel J202

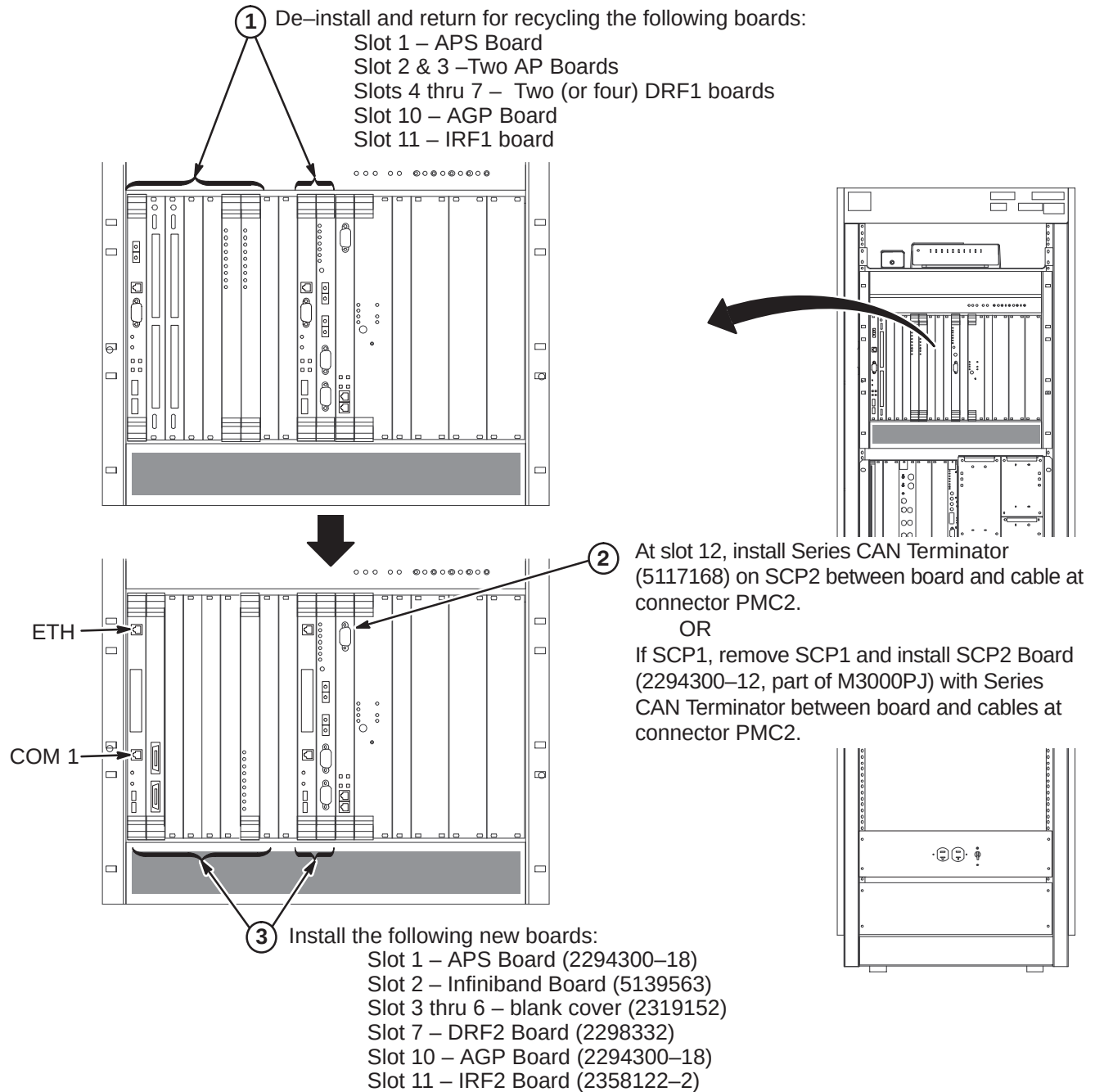
9-7 UPGRADE BOARDS IN MGD CHASSIS

After upgrade, all systems will have one DRF2 board and IRF2 board.



Note: Modules are static sensitive components and need to be handled as such (wrist strap).

Note: to remove modules – loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.



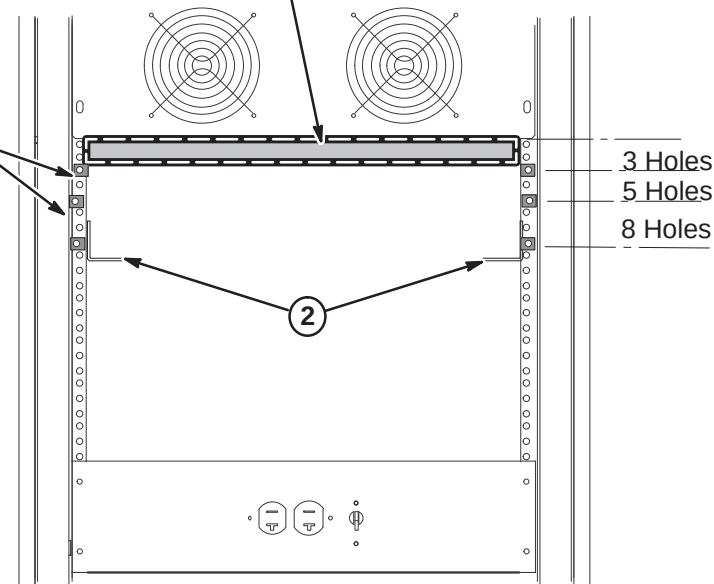
9-8 INSTALL VRE COMPONENTS

9-8-1 INSTALL VRE MODULE RAILS

- ① Make sure that the clip nuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.

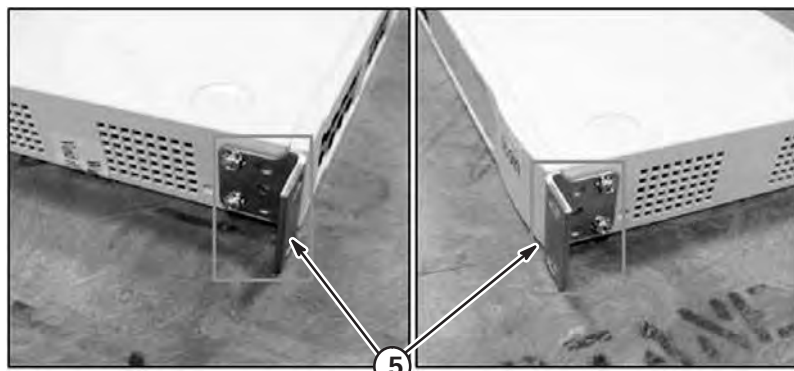
- ④ Install 3 Clip Nuts (2180698-2) on each side at locations indicated.

- ② Location of rails (5158758), one on each side at the position shown for the ICN modules.



- ③ Position Rails behind vertical rails and attach with M5 Screws (2279742-2). Rails are attached at hole location 8 on the vertical rail.

NOTE: The lower rail on each side is installed now for future installation of addition VRE option.



Install brackets on front corners of Ethernet switch as shown. Hardware kit comes with unit.

9-8-2 INSTALL VRE MODULES

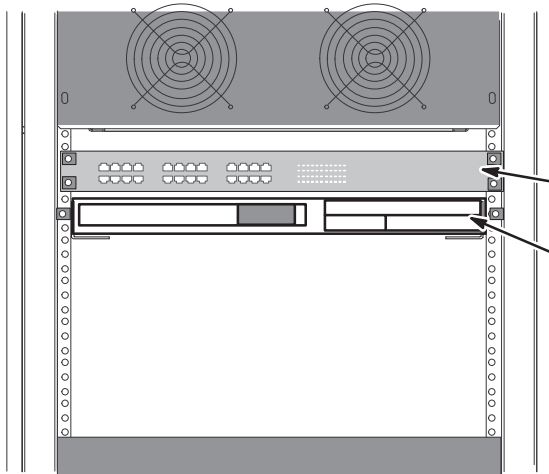
On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.



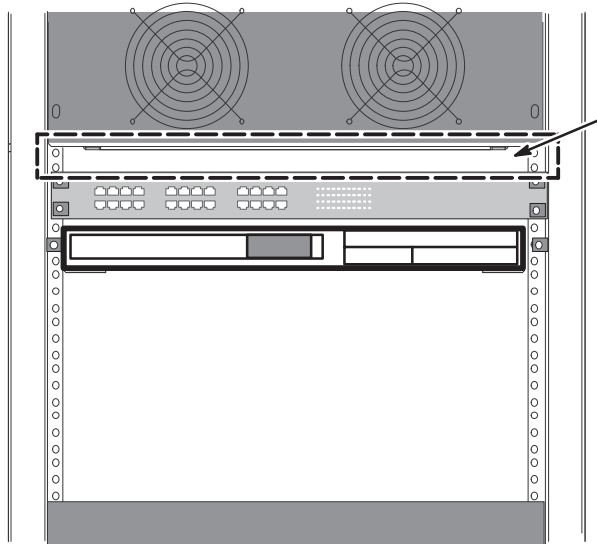
- ① Install 2 (5306485) Brackets on the ICN as shown.



- ② Install Ethernet Switch at this location using M5 Screws (2279742-2).
- ③ Install ICN Module (5127452-3) on Rail at the location shown. Tighten PEM screws on the ICN brackets to secure.

9-8-3 INSTALL VRE CABLES

Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables.



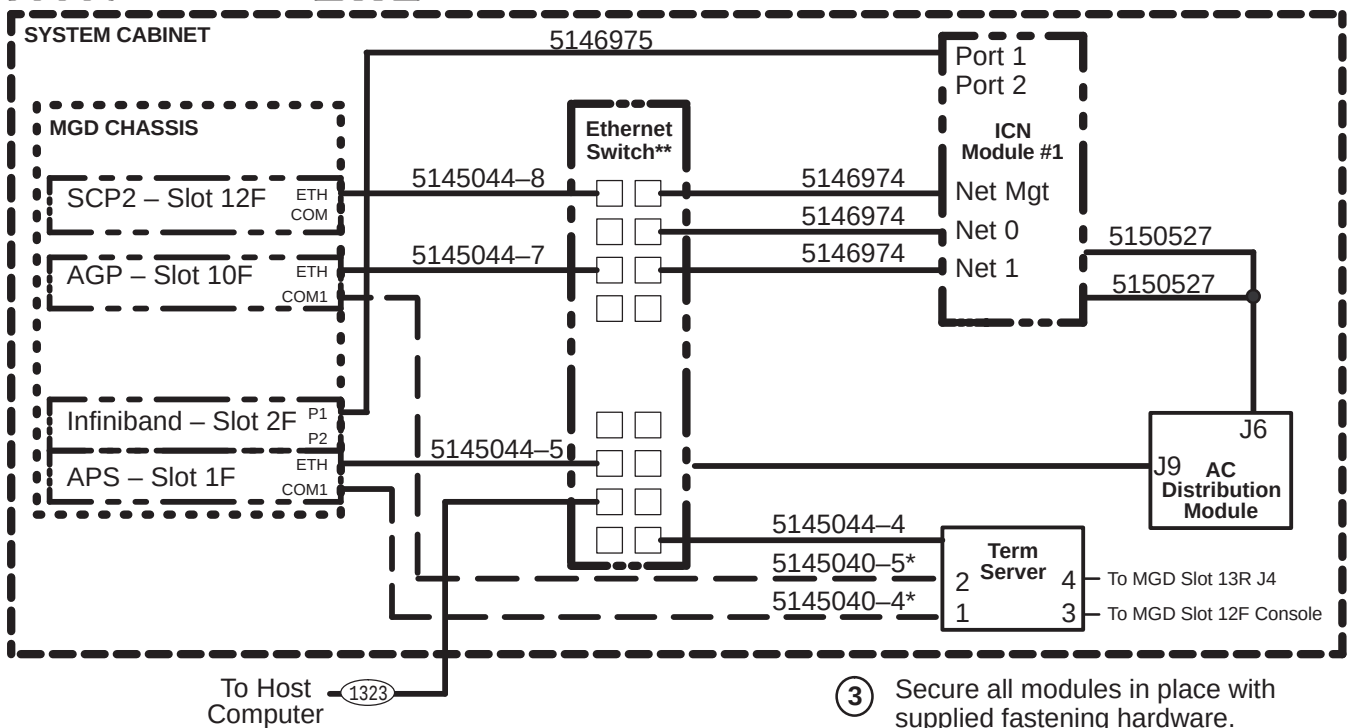
- ① All cables connecting between Ethernet Module and ICN Modules must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

- ② Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables. Make sure 5146975 is connected to P1 (Port 1) and not P2 on the back of the ICN.

- ③ Use power cable supplied with Ethernet switch, connect to J9.

LEGEND:



- ③ Secure all modules in place with supplied fastening hardware.

* = New cable replacing existing cable

** = Ethernet cables routed to Ethernet Switch can be connected to any port.

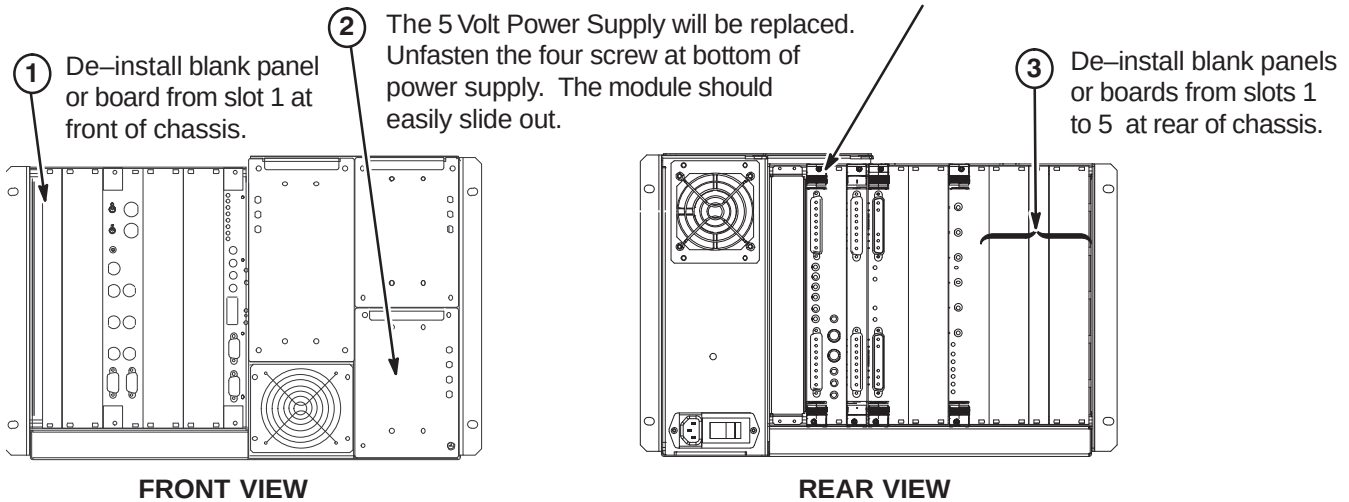
9-9 PROCEDURES FOR UPGRADING TO 16 CHANNEL (M3000PE)

9-9-1 INSTALL REFLEX 800 AP BOARDS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.

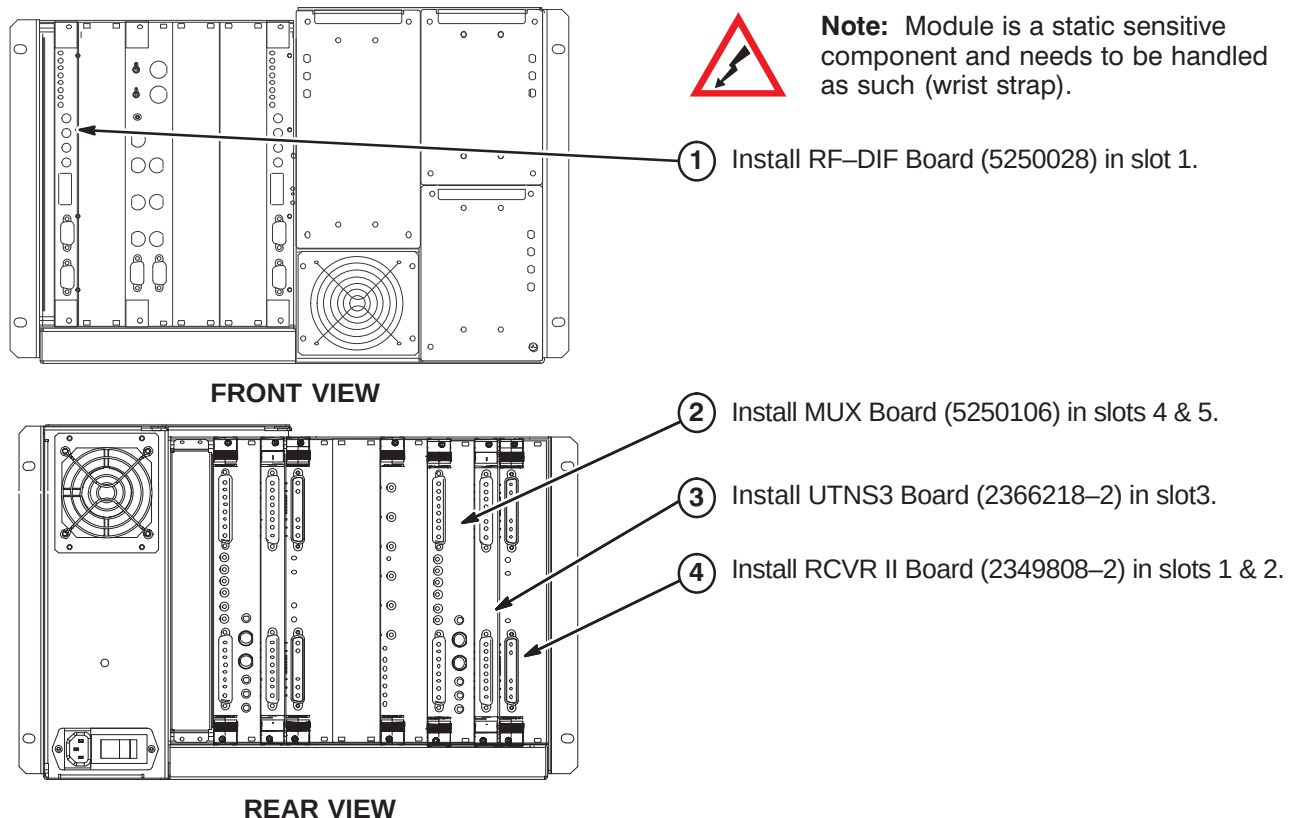


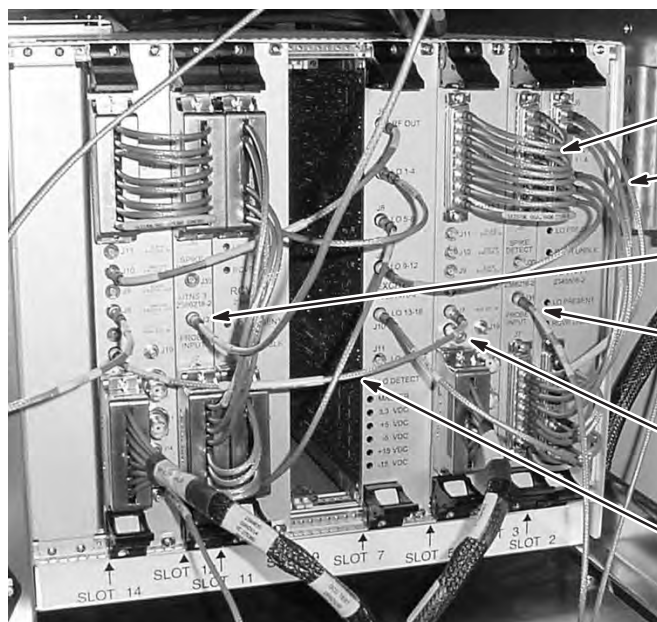
Note: All de-installed boards are to be returned for recycling.

9-9-2 INSTALL NEW MODULES IN RRF CHASSIS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



9-9-3 INSTALL CABLES AT REAR OF RRF CHASSIS

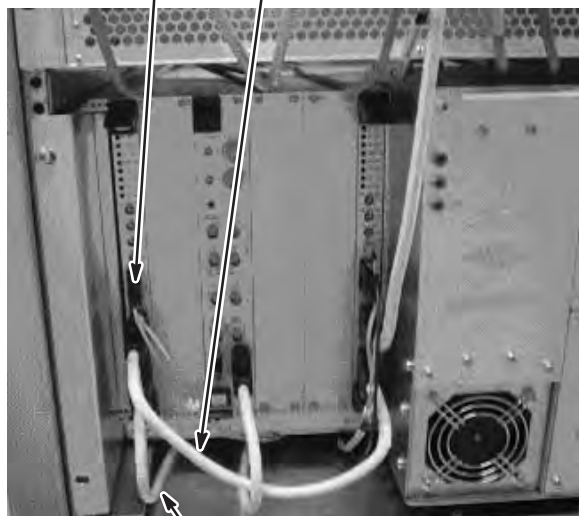
- ① Connect cable 5123706 as marked.
- ② Connect cable 5113193-2 as marked.
- ③ Make sure cable 5115417 slot 12R J31 is connected as marked. it should be routed and connect to Coax Power Splitter "1".
- ④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2". Remove 50 Ohm terminator if installed at "2".
- ⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).
- ⑥ Connect cable 5115630 as marked.

9-9-4 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS

- ③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.

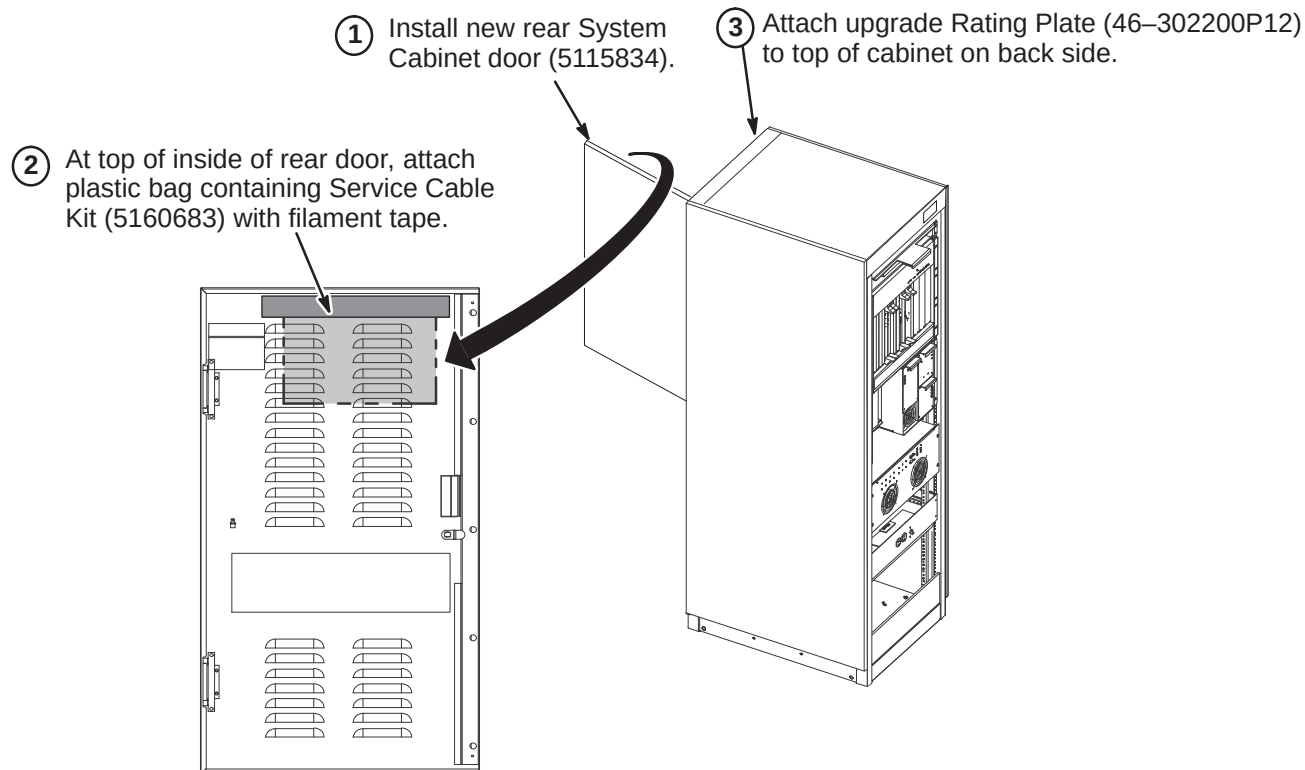
- ② Connect cable 5113182-2 as marked.

- ④ Slide in new 5V Power Supply (2292267-2). Make sure that it is properly seated before tightening the four screws.



- ① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.



9-10 INSTALL NEW REAR DOOR AND RATING PLATE

SECTION 10 – COMPLETION

10-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 10-2, 10-3, and 10-4.

10-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 10-3. Recycling Operation will route them to appropriate location.

10-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

10-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

10-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.